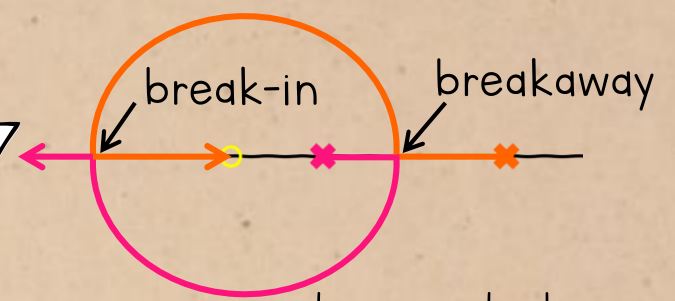


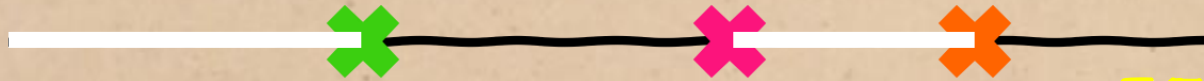
CHAPTER 4:
ROOT LOCUS 3
APPLICATIONS

SUMMARY



1

real axis segments: any section of the real axis with an odd number of real-valued critical frequencies on its right is on the root locus.

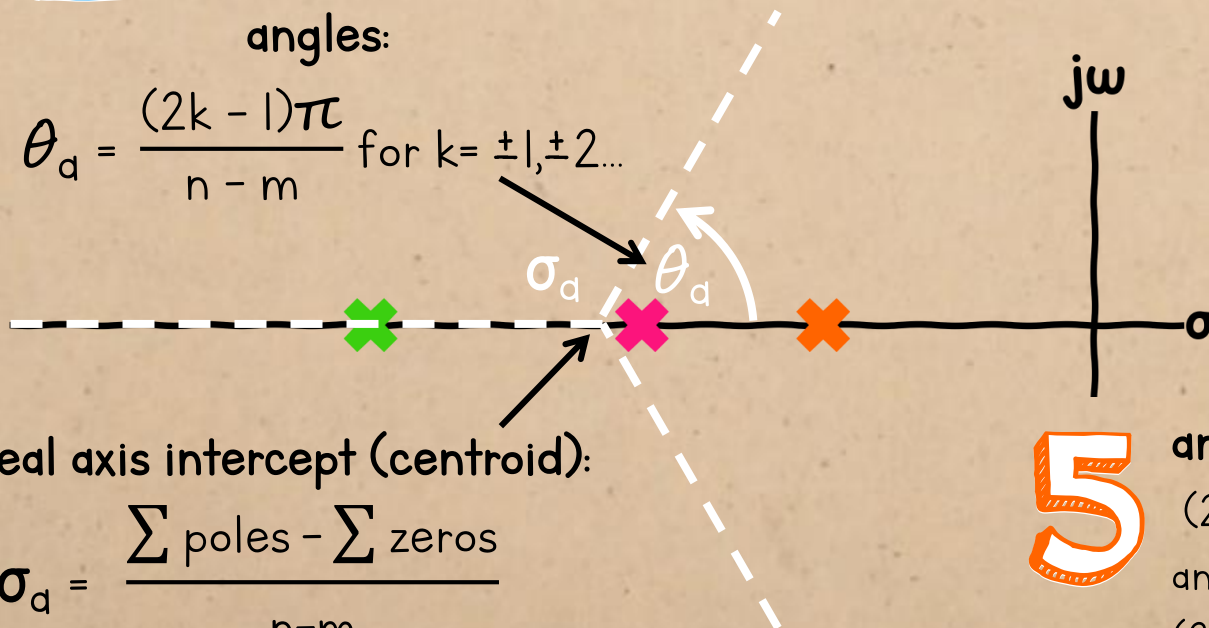


2

asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

angles:

$$\theta_d = \frac{(2k-1)\pi}{n-m} \text{ for } k = \pm 1, \pm 2, \dots$$



real axis intercept (centroid):

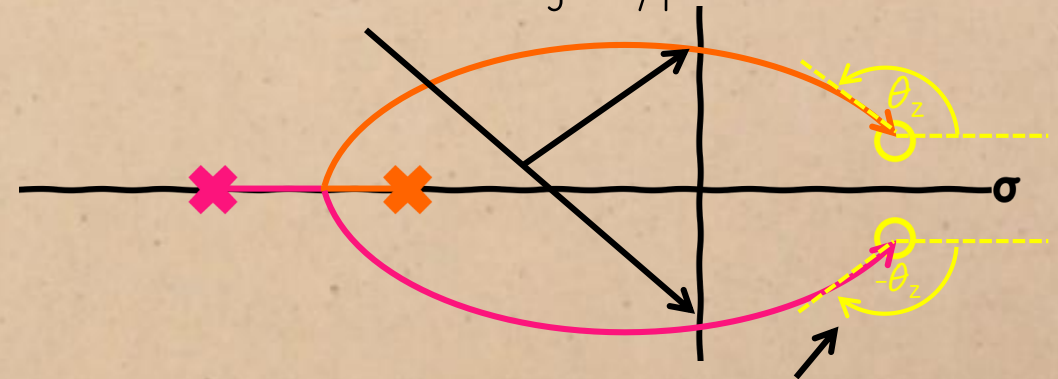
$$\sigma_d = \frac{\sum \text{poles} - \sum \text{zeros}}{n-m}$$

3

breakaway/-in points: where $\frac{d}{ds} K(s) = \frac{d}{ds} \frac{1}{L(s)} = 0$
OR solve for λ where $\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$

4

imaginary axis intercepts: find the maximum gain using the Routh array and then substitute this into the characteristic equation
OR substitute $j\omega$ into the characteristic equation and solve for K and ω such that the real and imaginary parts = 0



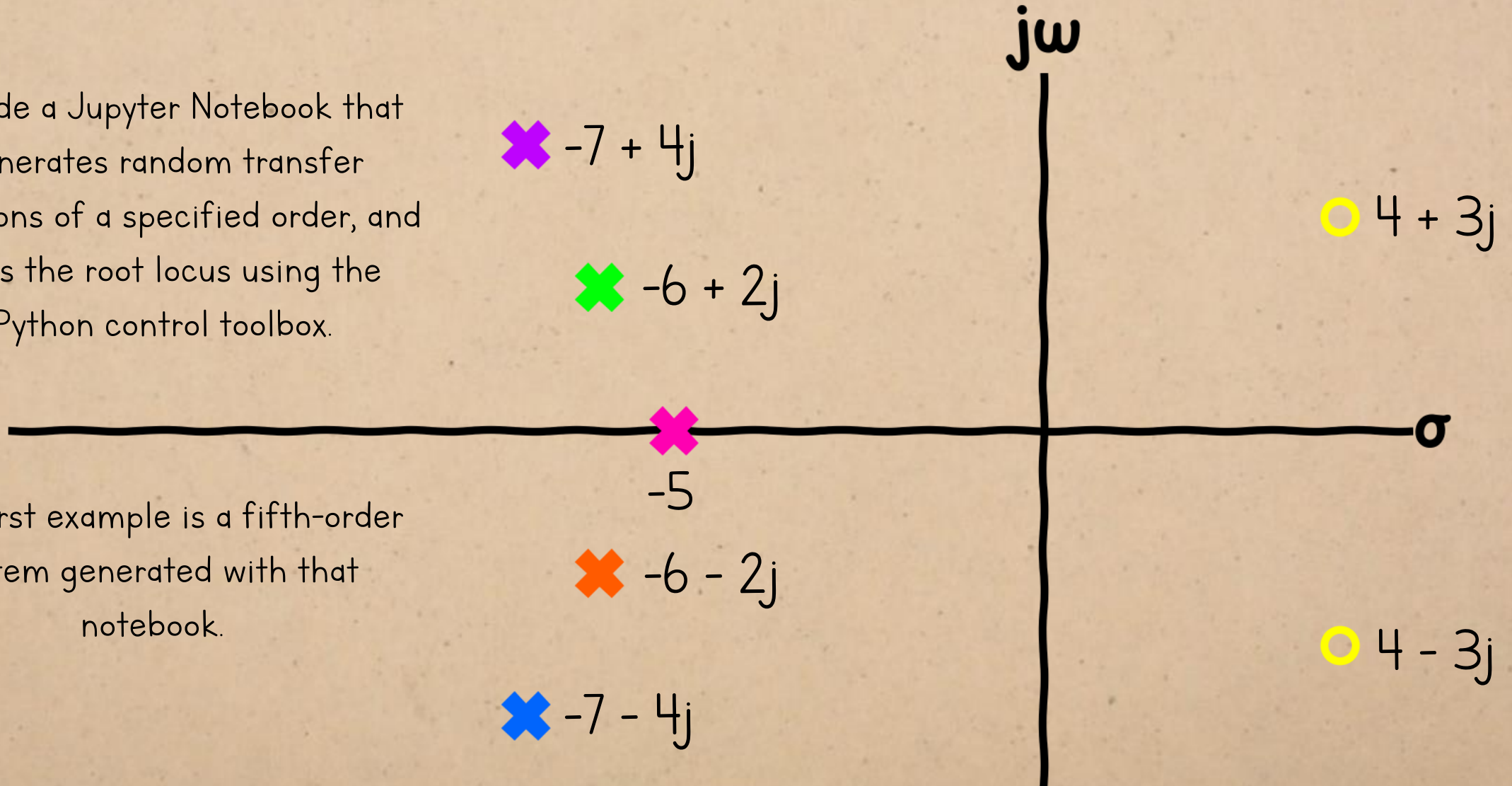
5

angles of arrival/ departure: the angle of arrival at a complex zero:
 $(2k-1)\pi - \sum \text{angles from poles} + \sum \text{angles from other zeros}$
angle of departure from a complex pole:
 $(2k-1)\pi - \sum \text{angles from other poles} + \sum \text{angles from zeros}$

EXAMPLE 1

I made a Jupyter Notebook that generates random transfer functions of a specified order, and plots the root locus using the Python control toolbox.

This first example is a fifth-order system generated with that notebook.



EXAMPLE 1

real axis segments: any section of the real axis with an odd number of real-valued critical frequencies on its right is on the root locus.

✕ $-7 + 4j$

✕ $-6 + 2j$

✕ -5

✕ $-6 - 2j$

✕ $-7 - 4j$

○ $4 + 3j$

○ $4 - 3j$

This is the only part of the real axis with an odd number of **real** critical frequencies to its right.

$j\omega$

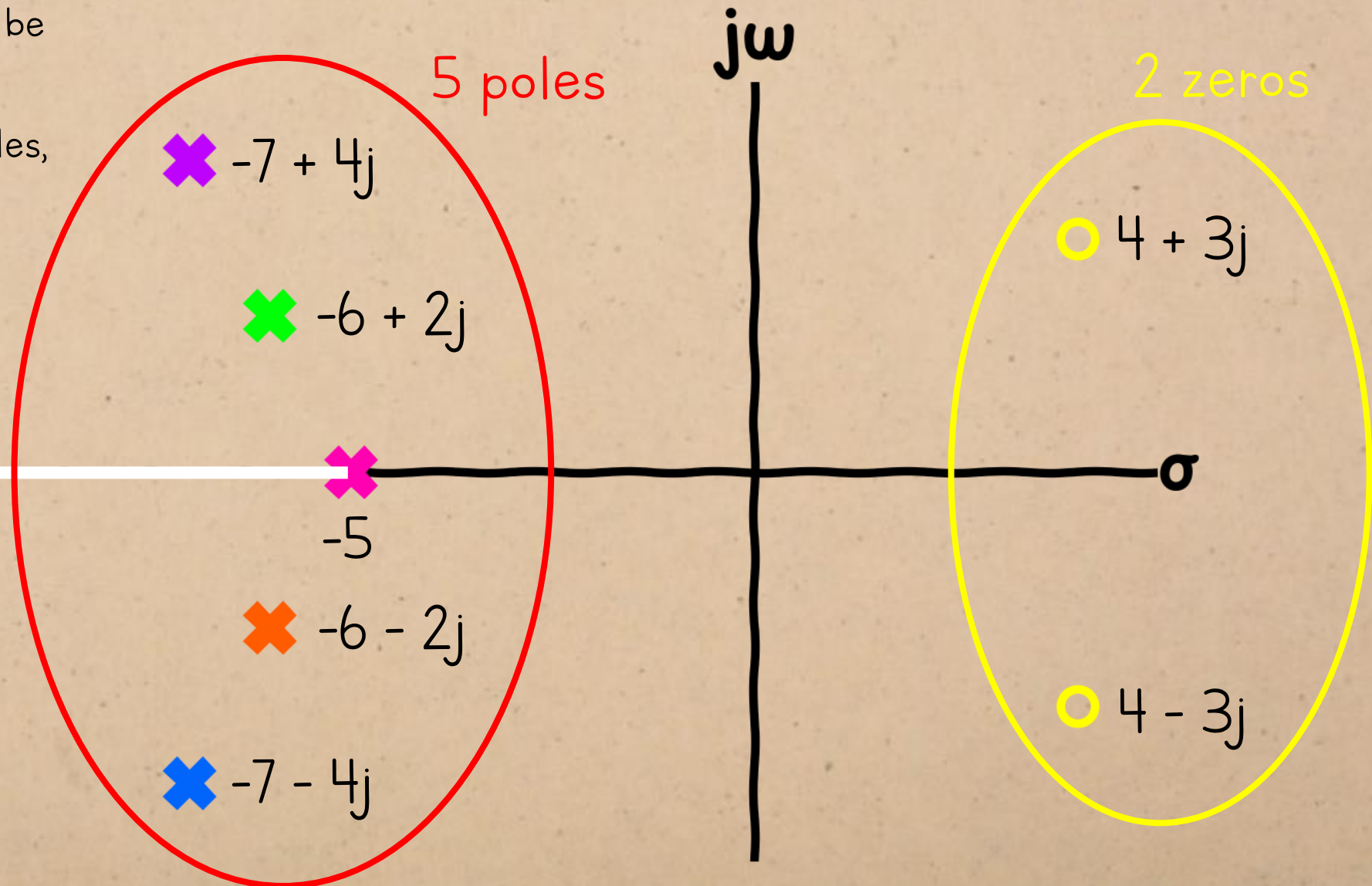
σ

EXAMPLE 1

2

asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

$\therefore$ 3 asymptotes



EXAMPLE 1

2

asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

$\therefore$ 3 asymptotes

real axis intercept (centroid):

$$\sigma_d = \frac{\sum \text{poles} - \sum \text{zeros}}{n-m}$$

✕ $-7 + 4j$

✕ $-6 + 2j$

✕ -5

✕ $-6 - 2j$

✕ $-7 - 4j$

○ $4 + 3j$

○ $4 - 3j$

$j\omega$

σ

EXAMPLE 1

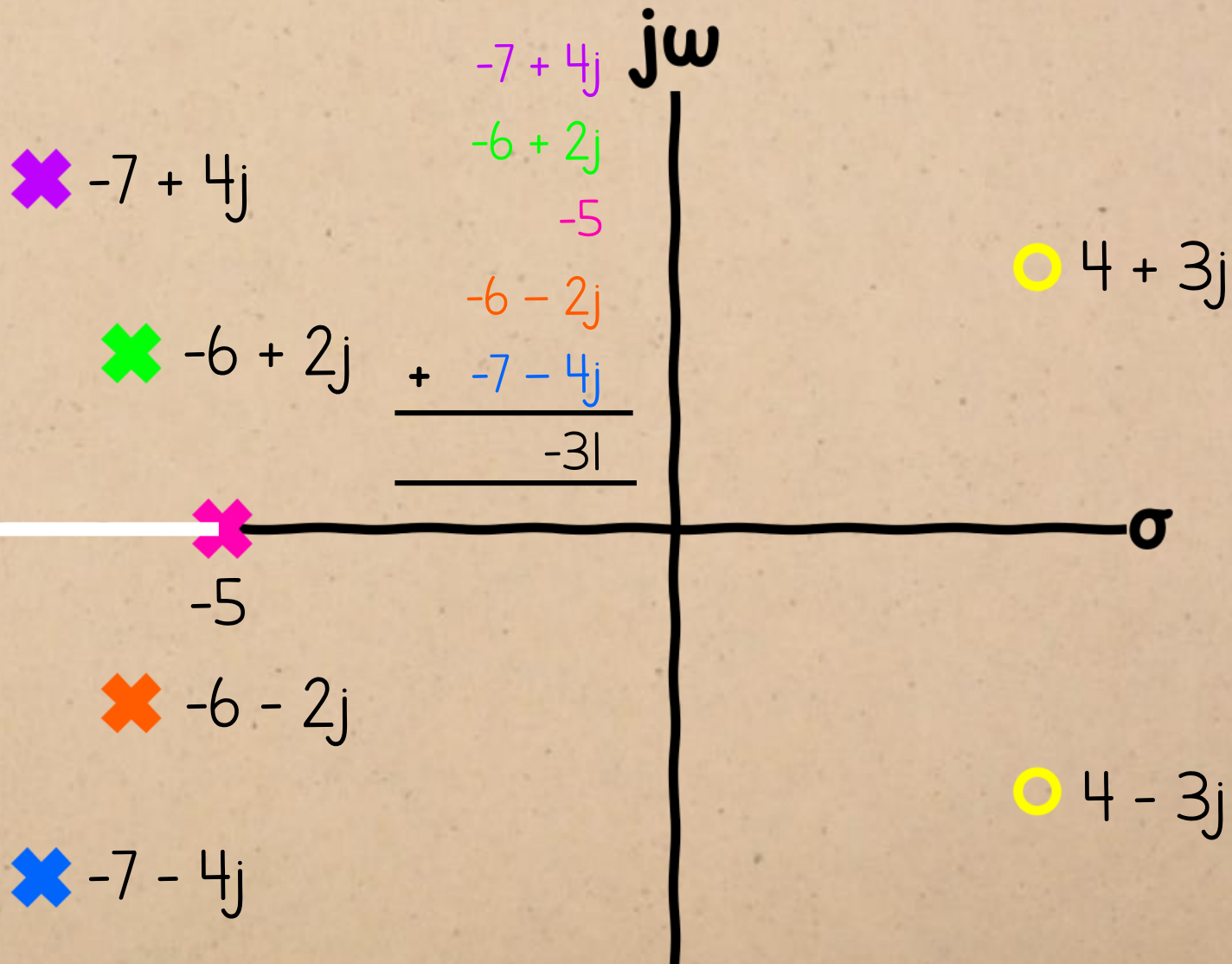


asymptotes: there will be $n-m$ asymptotes,
where n = number of poles,
 m = number of zeros.

$\therefore 3$ asymptotes

real axis intercept (centroid):

$$\sigma_d = \frac{-3I - \sum \text{zeros}}{n-m}$$



EXAMPLE 1

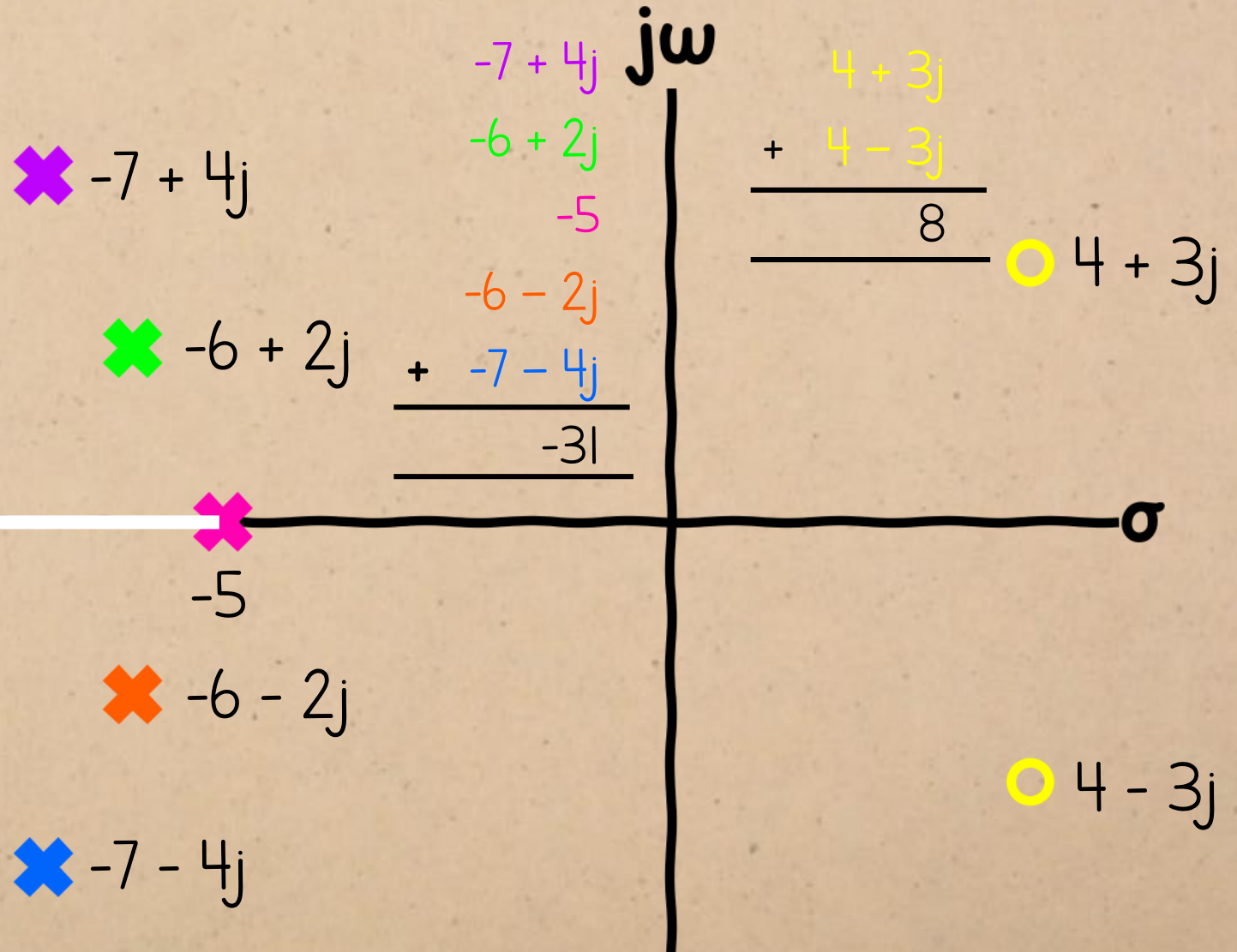
2

asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

$\therefore$ 3 asymptotes

real axis intercept (centroid):

$$\sigma_d = \frac{-31 - 8}{n-m}$$



EXAMPLE 1

2

asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

$\therefore$ 3 asymptotes

$$\times -7 + 4j$$

$$\times -6 + 2j$$

$$\times -5$$

$$\times -6 - 2j$$

$$\times -7 - 4j$$

$$\circ 4 + 3j$$

$$\circ 4 - 3j$$

$$\star -13$$

real axis intercept (centroid):

$$\sigma_d = \frac{-31 - 8}{3} = -13$$

$j\omega$

σ

EXAMPLE 1

2

asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

$\therefore$ 3 asymptotes

★
-13

angles:

$$\theta_d = \frac{(2k-1)\pi}{n-m} \text{ for } k = \pm 1, \pm 2, \dots$$

✕ $-7 + 4j$

✕ $-6 + 2j$

✕ $-6 - 2j$

✕ $-7 - 4j$

○ $4 + 3j$

○ $4 - 3j$

$j\omega$

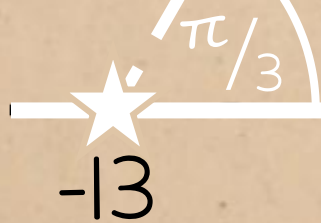
σ

EXAMPLE 1

2

asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

$\therefore 3$ asymptotes



angles:

$$\theta_d = \frac{\pi}{3} (k = 1)$$

✕ $-7 + 4j$

✕ $-6 + 2j$

✕ -5

✕ $-6 - 2j$

✕ $-7 - 4j$

$j\omega$

○ $4 + 3j$

○ $4 - 3j$

σ

EXAMPLE 1

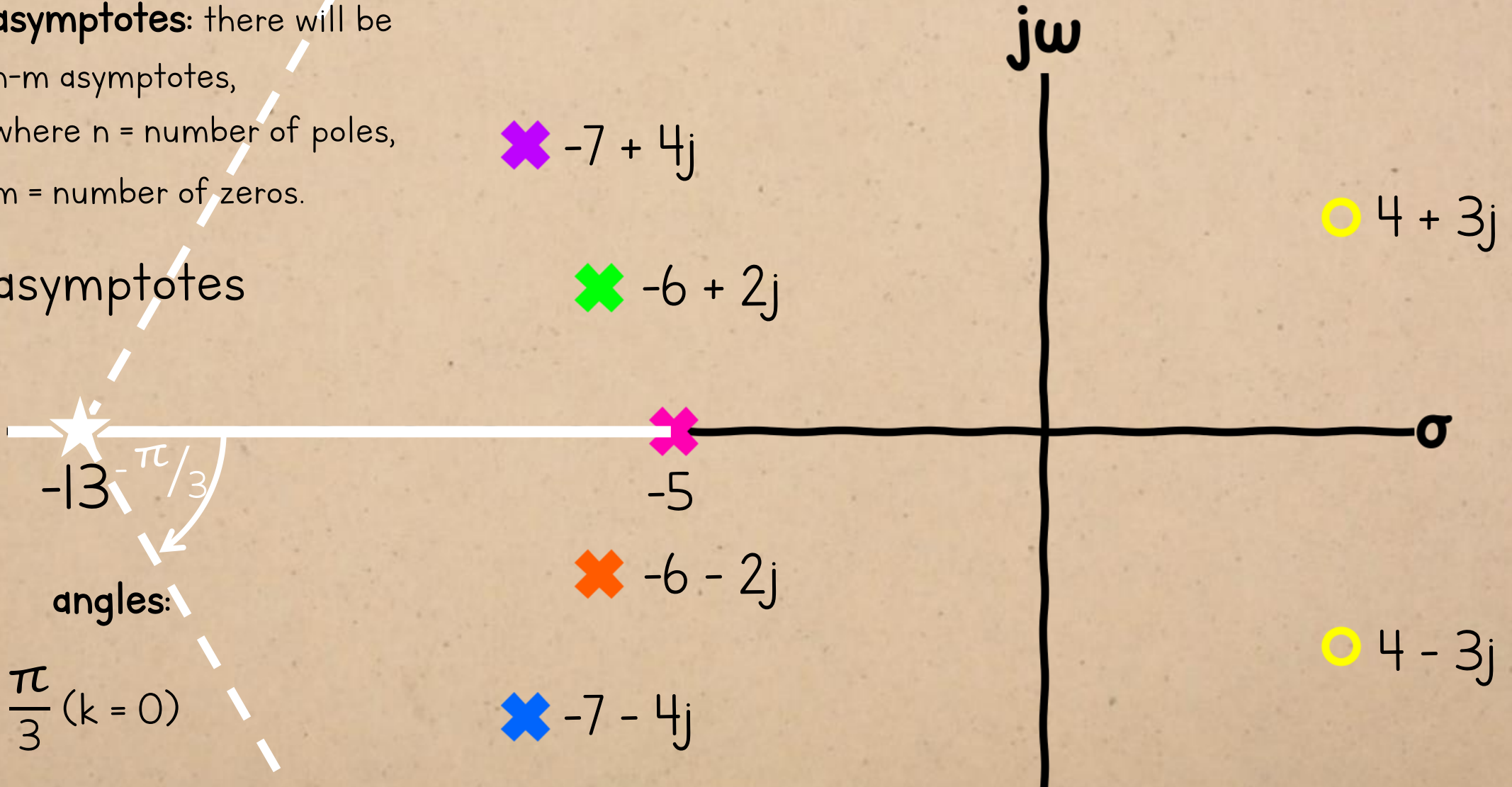
2

asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

$\therefore 3$ asymptotes

$$\theta_d = -\frac{\pi}{3} (k=0)$$

angles:



EXAMPLE 1

2

asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

$\therefore 3$ asymptotes



angles:

$$\theta_d = -\frac{3\pi}{3} \quad (k = 2)$$

✕ $-7 + 4j$

✕ $-6 + 2j$

✕ -5

✕ $-6 - 2j$

✕ $-7 - 4j$

$j\omega$

○ $4 + 3j$

○ $4 - 3j$

σ

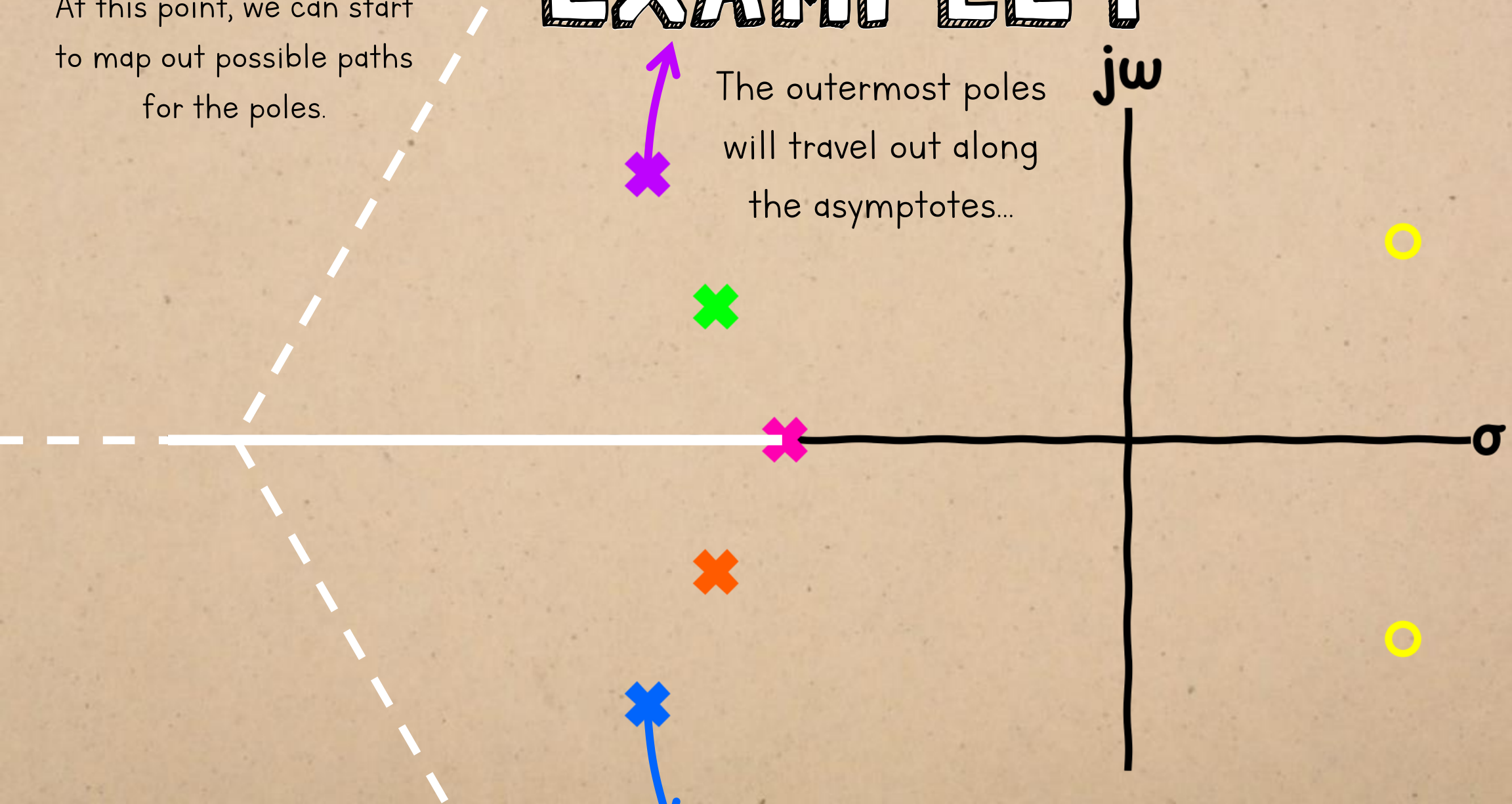
EXAMPLE 1

At this point, we can start to map out possible paths for the poles.

The outermost poles will travel out along the asymptotes...

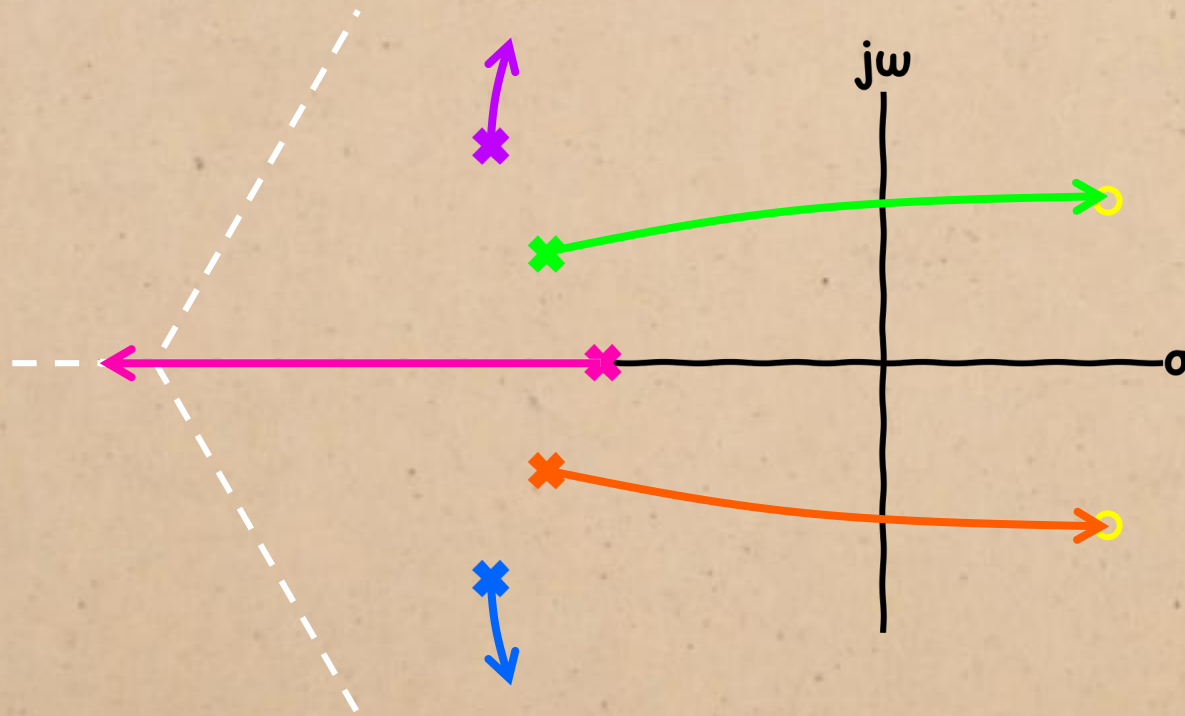
$j\omega$

σ



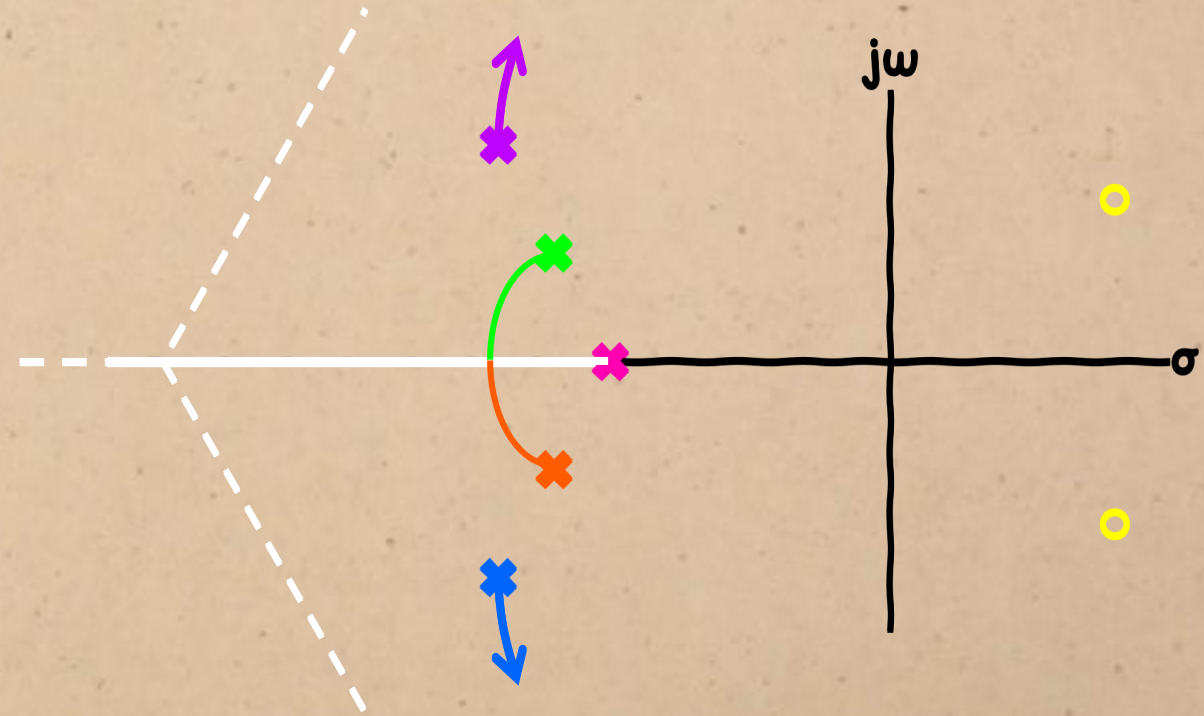
EXAMPLE 1

But there are two possibilities for where the other poles might go.



Option 1

The inner complex poles don't go back to the real axis.



Option 2

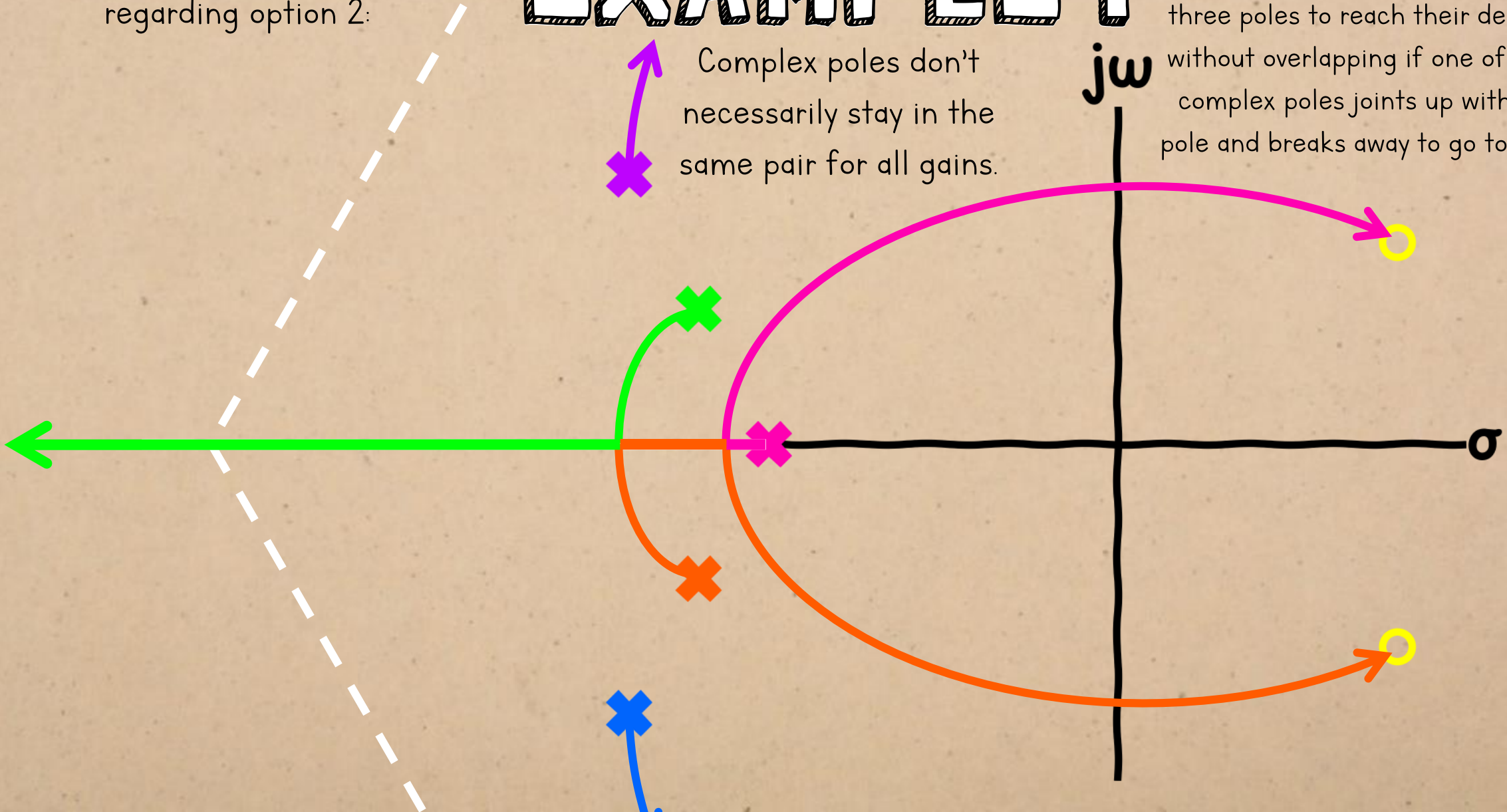
The inner complex poles do go back to the real axis...

Something important to note regarding option 2:

EXAMPLE 1

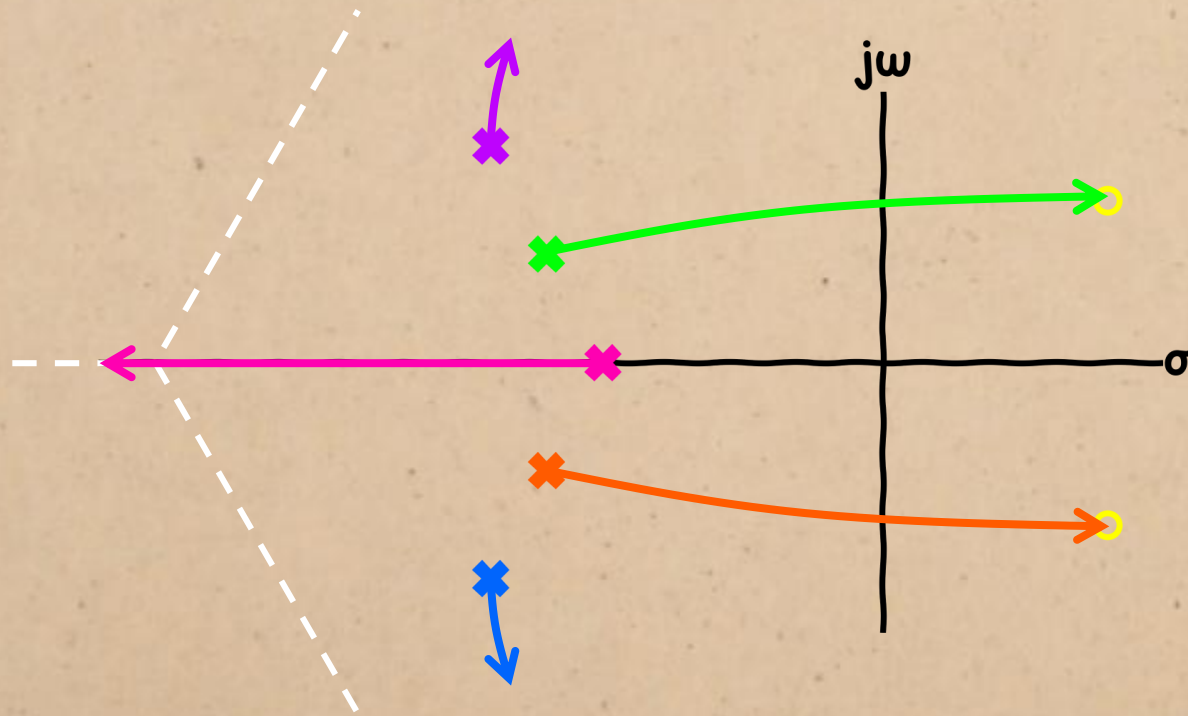
Complex poles don't necessarily stay in the same pair for all gains.

It would only be possible for these three poles to reach their destinations without overlapping if one of the inner complex poles joins up with the real pole and breaks away to go to the zeros.

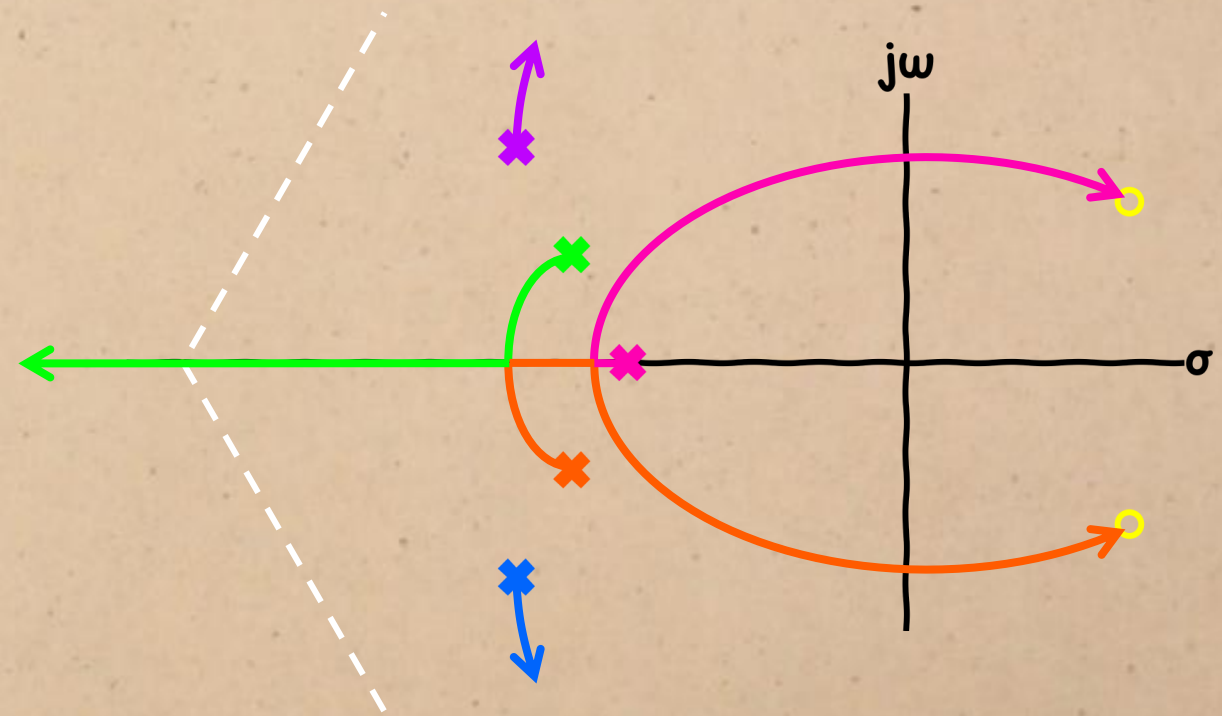


EXAMPLE 1

But there are two possibilities for where the other poles might go.



Option 1



Option 2

EXAMPLE 13

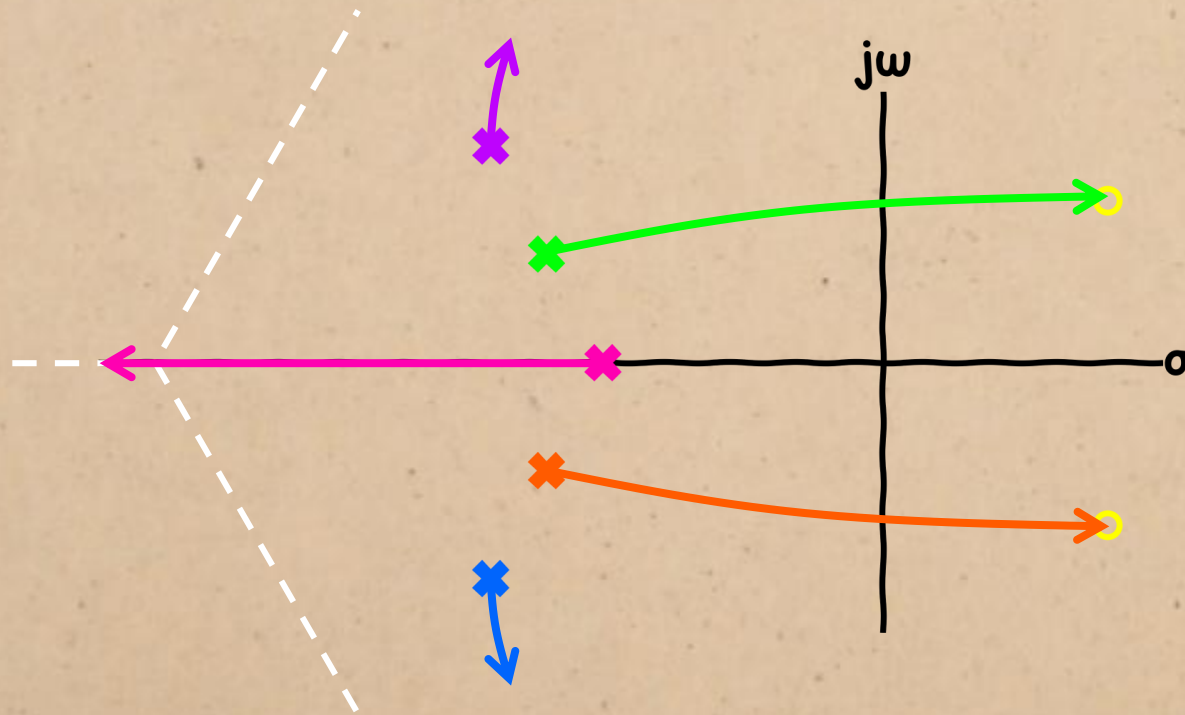
breakaway/-in points:

$$\text{where } \frac{d}{ds} K(s) = \frac{d}{ds} \frac{1}{L(s)} = 0$$

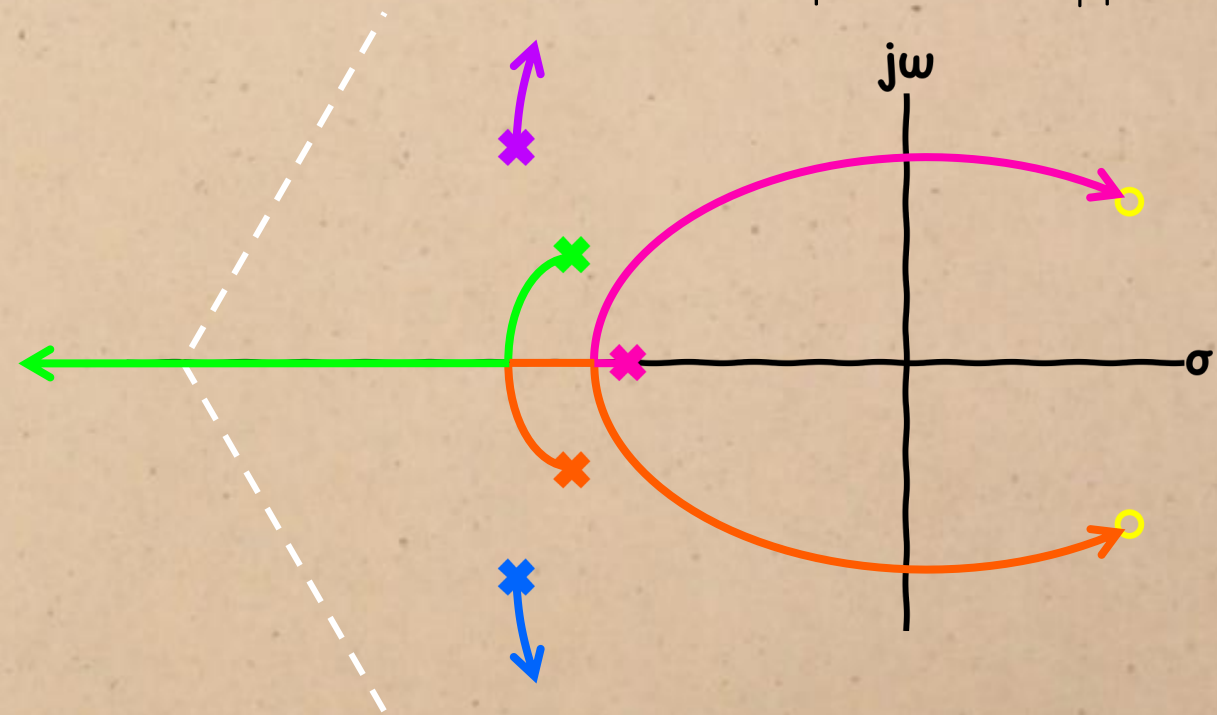
Finding the breakaway/ -in points will tell us which option is correct.

OR solve for λ where

$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$



Option 1



Option 2

...but doing it by hand with a complicated transfer function like this is going to suck no matter how we do it.

EXAMPLE 13

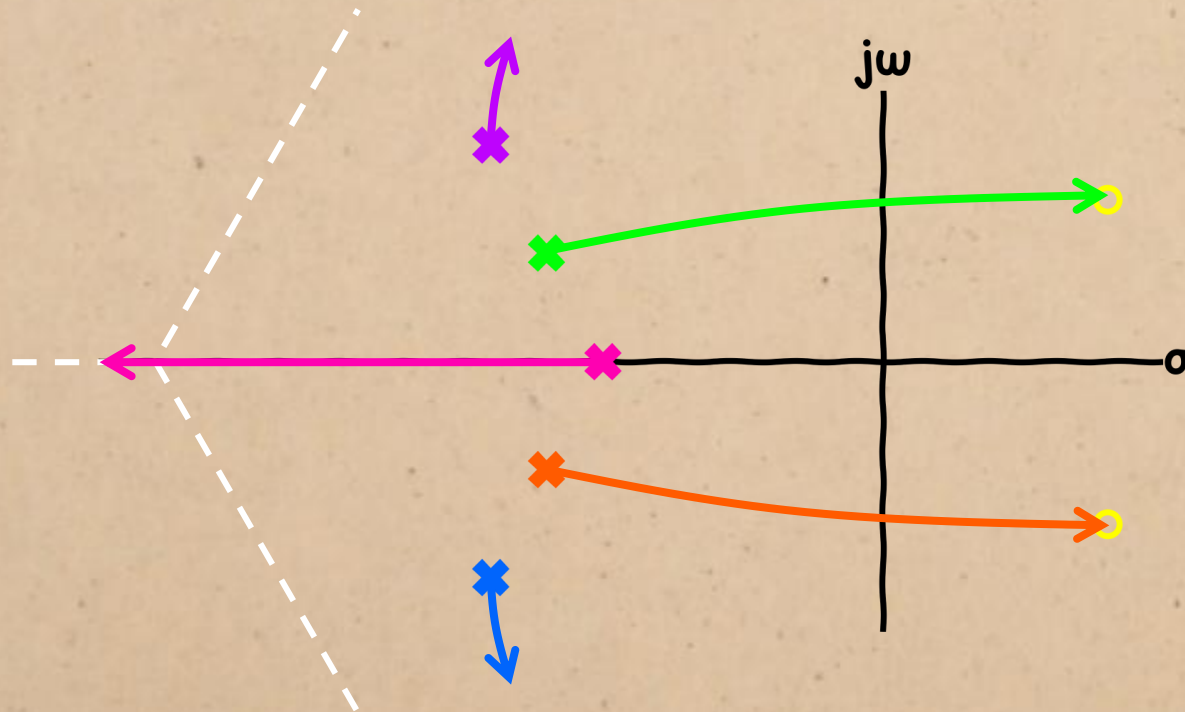
breakaway/-in points:

$$\text{where } \frac{d}{ds} K(s) = \frac{d}{ds} \frac{1}{L(s)} = 0$$

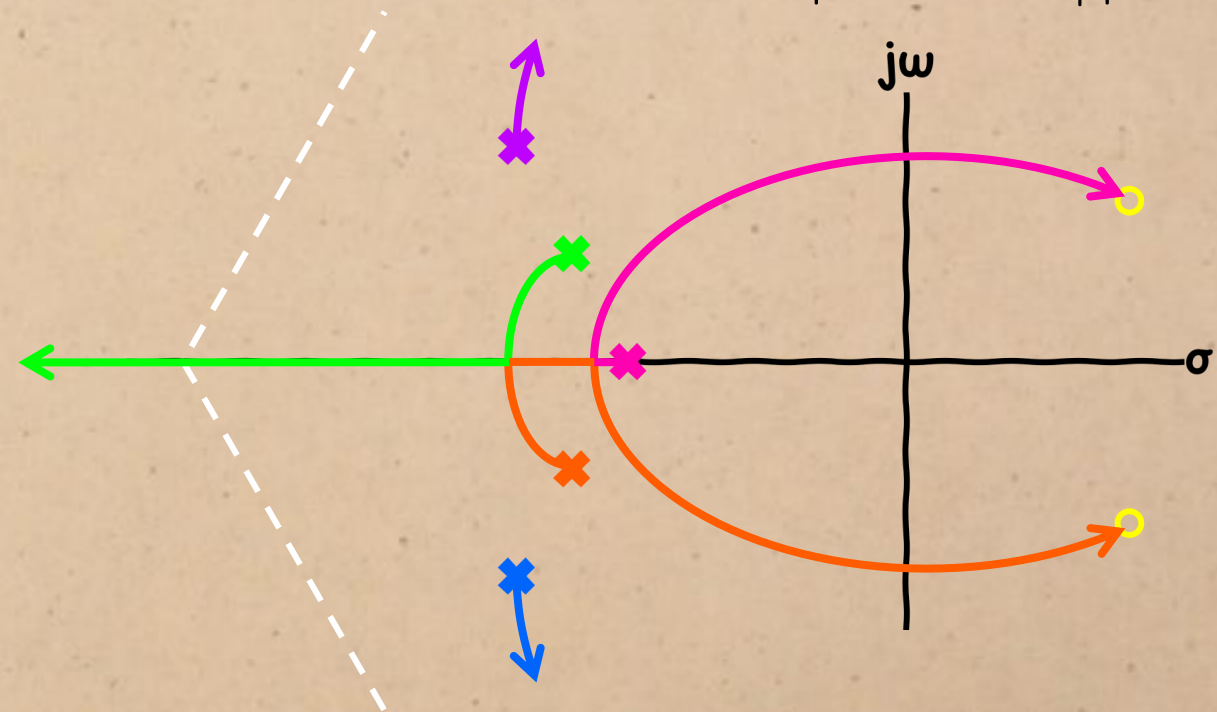
OR solve for λ where

$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

$$L(s) = \frac{s^2 - 8s + 25}{s^5 + 31s^4 + 403s^3 + 2705s^2 + 9300s + 13000}$$



Option 1



Option 2

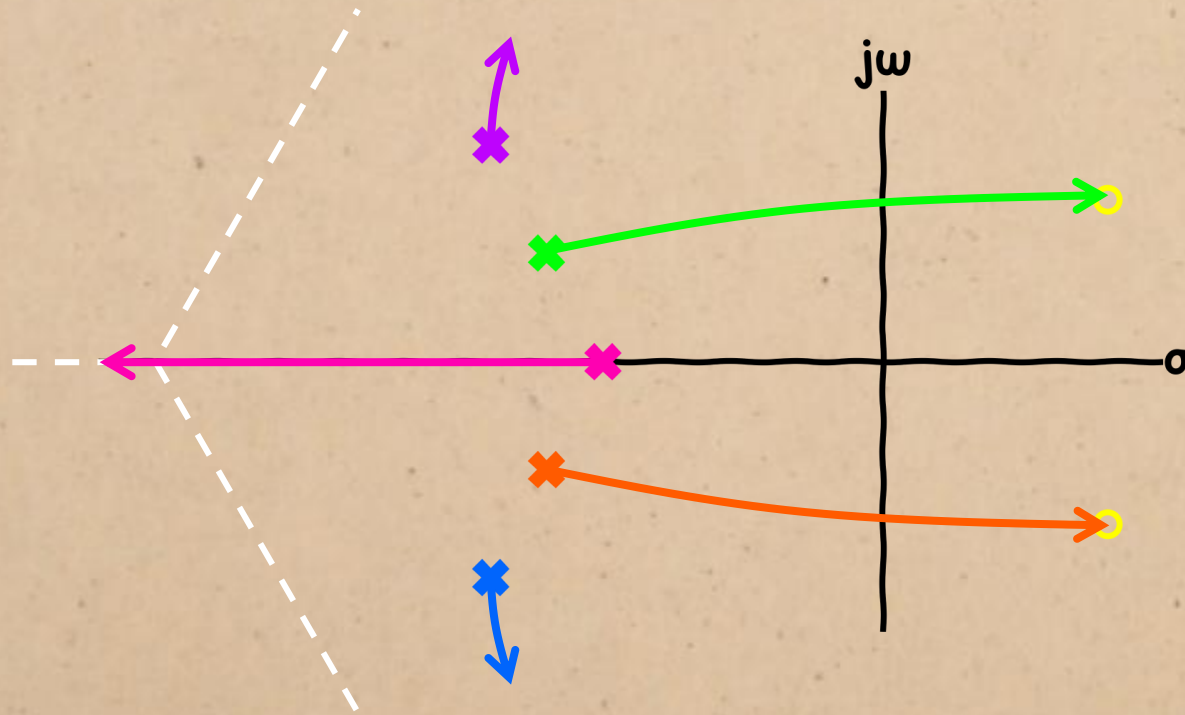
I used Sympy (the Python symbolic toolbox) to find the derivative of the inverse of $L(s)$ and solve for its zeros.

EXAMPLE 13

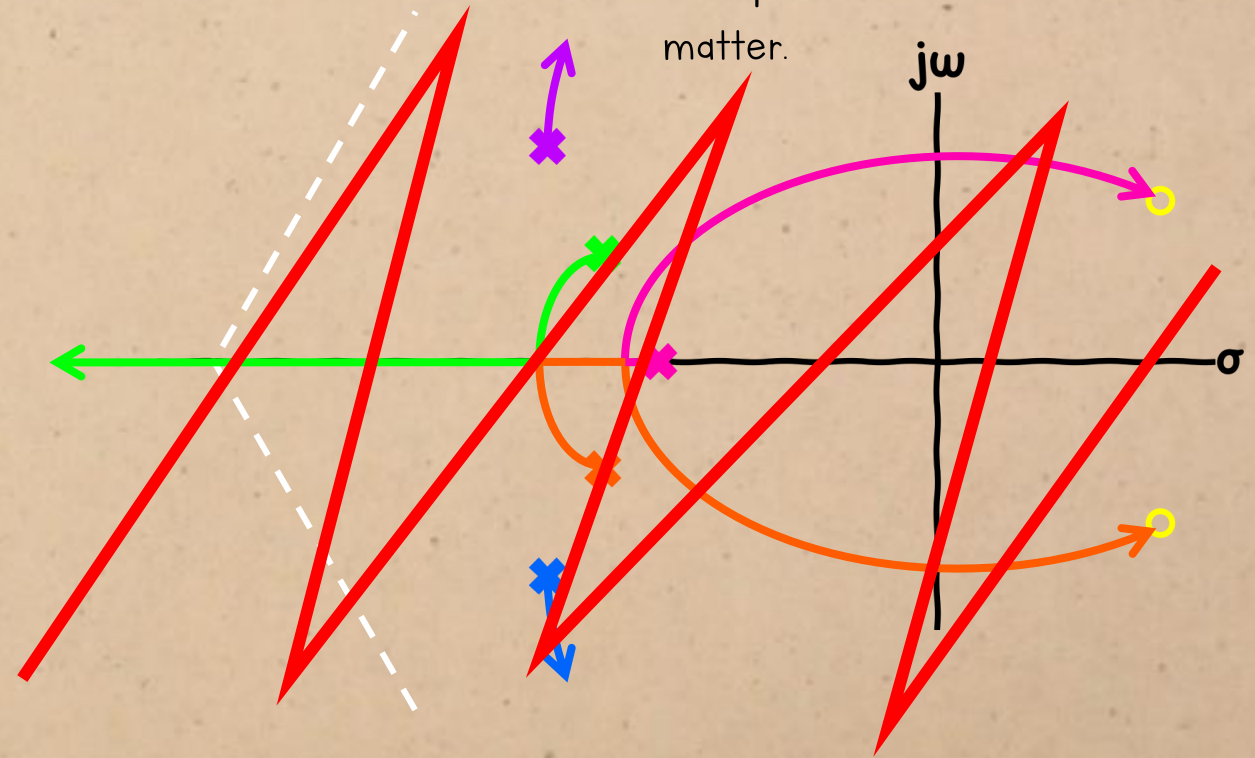
breakaway/-in points:

where $\frac{d}{ds} K(s) = \frac{d}{ds} \frac{1}{L(s)} = 0$

$$\frac{d}{ds} \frac{1}{L(s)} = \frac{3s^6 + 30s^5 - 216s^4 - 3348s^3 - 715s^2 + 109250 + 335600}{s^4 - 16s^3 + 114s^2 - 400s + 625} = 0 \text{ for } s = \text{a bunch of complex values that don't matter.}$$



Option 1



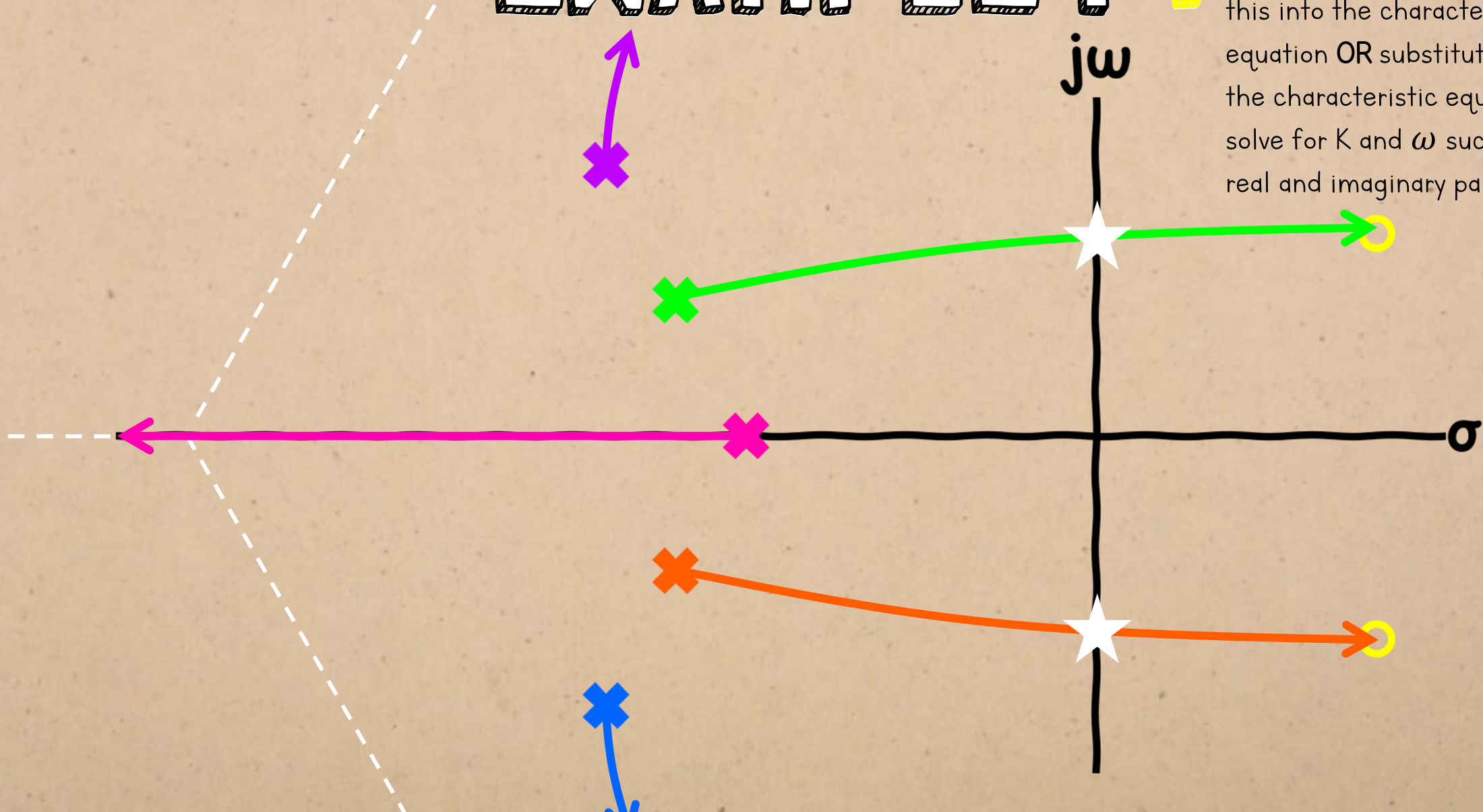
Option 2

$\therefore$ no breakaway/-in points

EXAMPLE 14

imaginary axis intercepts:

find the maximum gain using the Routh array and then substitute this into the characteristic equation OR substitute $j\omega$ into the characteristic equation and solve for K and ω such that the real and imaginary parts = 0

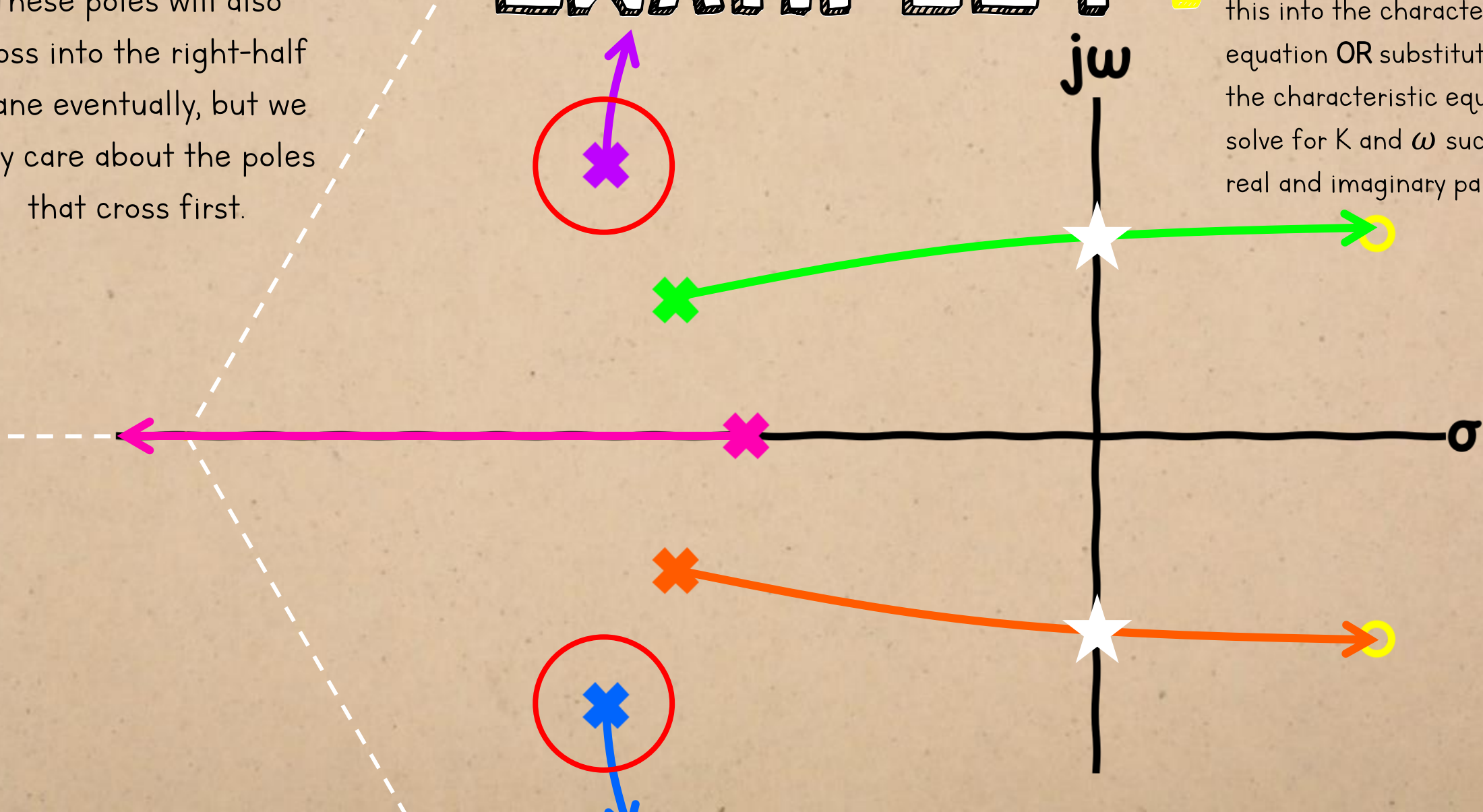


EXAMPLE 14

imaginary axis intercepts:

find the maximum gain using the Routh array and then substitute this into the characteristic equation OR substitute $j\omega$ into the characteristic equation and solve for K and ω such that the real and imaginary parts = 0

These poles will also cross into the right-half plane eventually, but we only care about the poles that cross first.



Finding these also sucks either way,
but the Routh-Hurwitz criterion is
probably the easier choice for
dealing with high-order transfer
functions "by hand"

EXAMPLE 14

imaginary axis intercepts:
find the maximum gain using the
Routh array and then substitute
this into the characteristic
equation

$$L(s) = \frac{s^2 - 8s + 25}{s^5 + 31s^4 + 403s^3 + 2705s^2 + 9300s + 13000}$$

characteristic equation:

$$s^5 + 31s^4 + 403s^3 + (K + 2705)s^2 + (9300 - 8K)s + (25K + 13000) = 0$$

(I also used Sympy to find the
zeros of the ugly stuff
happening in the first column)

$$K \leq 9788$$

$$K \leq 16625$$

$$\therefore K \leq 683$$

$$K \leq 683$$

1	403	-8K + 9300
31	K + 2705	25K + 13000
$\frac{9788 - K}{31}$	$\frac{-273K + 275300}{31}$	0
$\frac{K^2 - 15546K - 17942240}{K - 9788}$	25K + 13000	0
$\frac{-8K^3 + 130418K^2 + 89001720K - 119162400000}{K^2 - 15546 - 17942240}$	0	
25K + 13000		

EXAMPLE 14

imaginary axis intercepts:

find the maximum gain using the Routh array and then substitute this into the characteristic equation

$$L(s) = \frac{s^2 - 8s + 25}{s^5 + 31s^4 + 403s^3 + 2705s^2 + 9300s + 13000}$$

characteristic equation:

$$s^5 + 31s^4 + 403s^3 + (683 + 2705)s^2 + (9300 - 8 \times 683)s + (25 \times 683K + 13000) = 0$$

$$\text{for } s = \pm 3.12j$$

(and some other values, but these are the ones that matter)

$$K \leq 9788$$

$$K \leq 16625$$

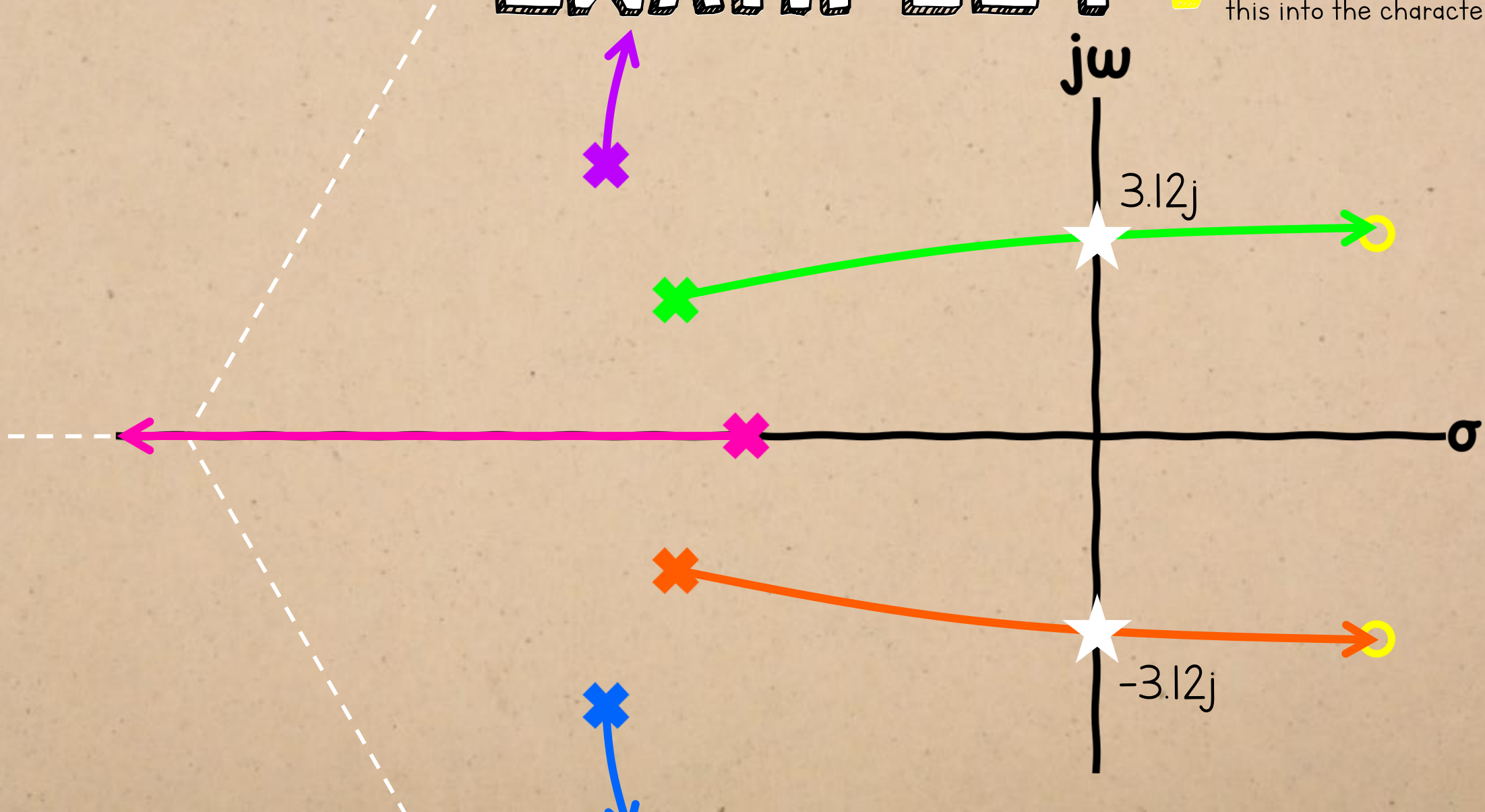
$$K \leq 683$$

$$\therefore K \leq 683$$

	1	403	-8K + 9300
	31	K + 2705	25K + 13000
	$\frac{9788 - K}{31}$	$\frac{-273K + 275300}{31}$	0
	$\frac{K^2 - 15546K - 17942240}{K - 9788}$	25K + 13000	0
	$\frac{-8K^3 + 130418K^2 + 89001720K - 119162400000}{K^2 - 15546 - 17942240}$	0	
	25K + 13000		

EXAMPLE 14

imaginary axis intercepts:
find the maximum gain using the
Routh array and then substitute
this into the characteristic

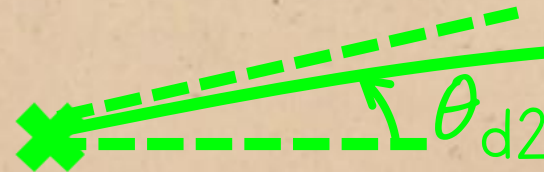
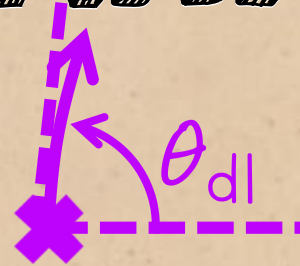


5 angles of arrival/ departure:

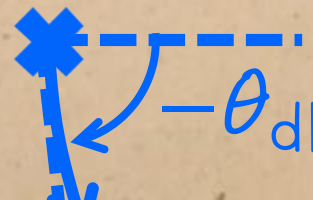
EXAMPLE 1

angle of arrival at complex zero:
 $(2k-1)\pi - \sum \text{angles from poles}$
 $+ \sum \text{angles from other zeros}$

$j\omega$



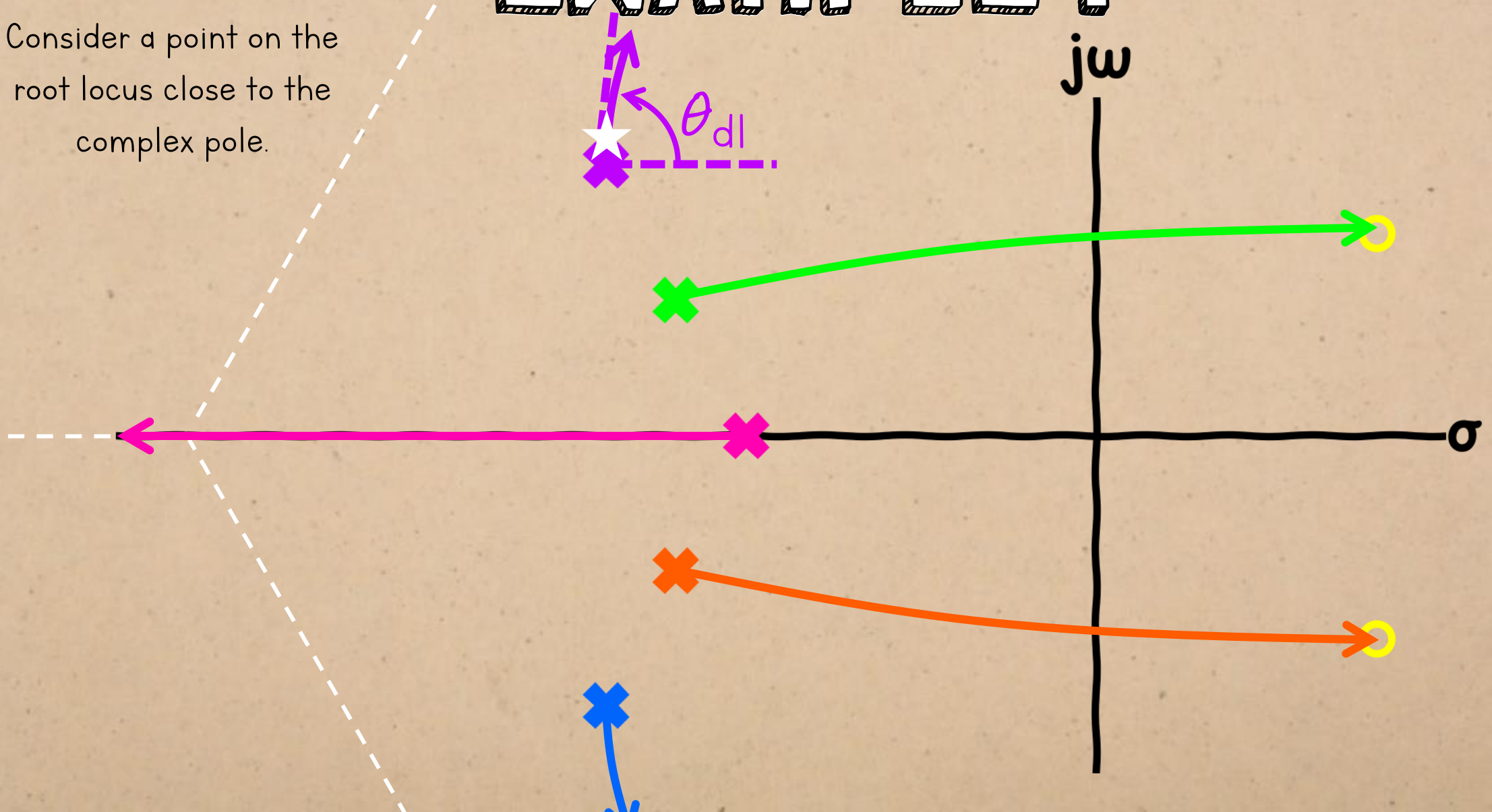
angle of departure from complex pole:
 $(2k-1)\pi - \sum \text{angles from other poles}$
 $+ \sum \text{angles from zeros}$



5 angles of
arrival/ departure:

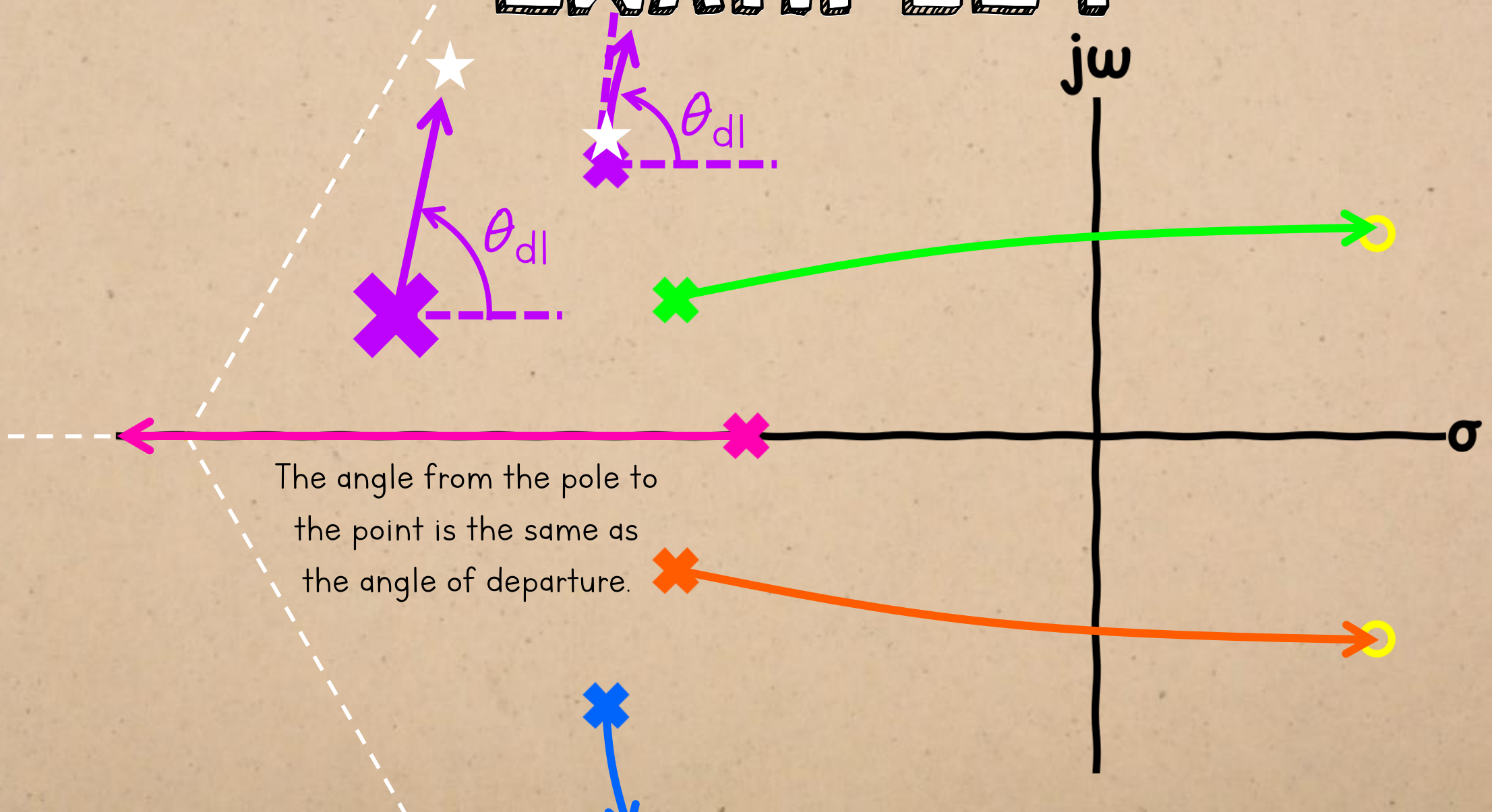
Consider a point on the
root locus close to the
complex pole.

EXAMPLE 1



5 angles of
arrival/ departure:

EXAMPLE 1



5 angles of
arrival/ departure:

EXAMPLE 1

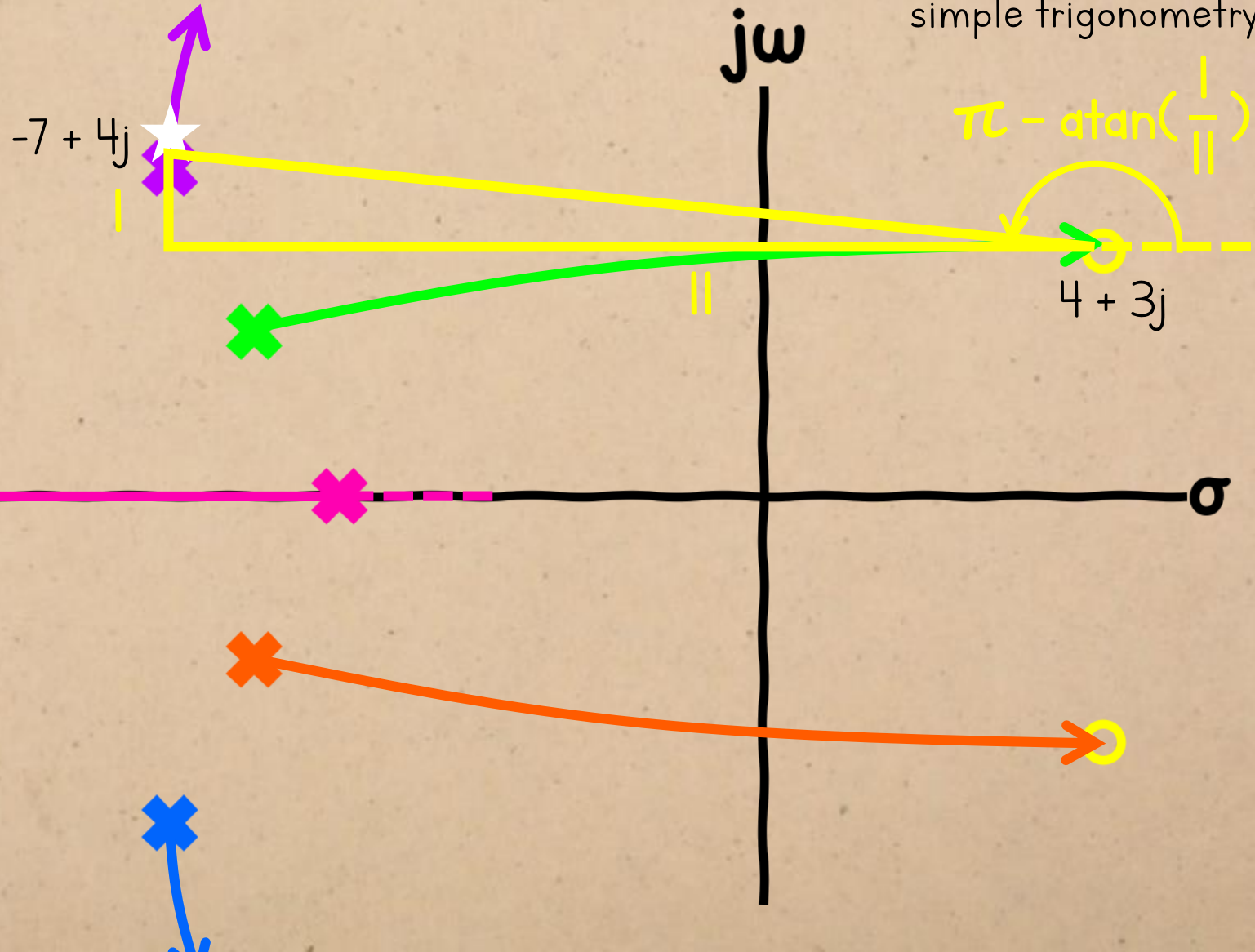
The angles from all the other
critical frequencies to the point
will be the same as their angles
to the pole.



5 angles of arrival/ departure:

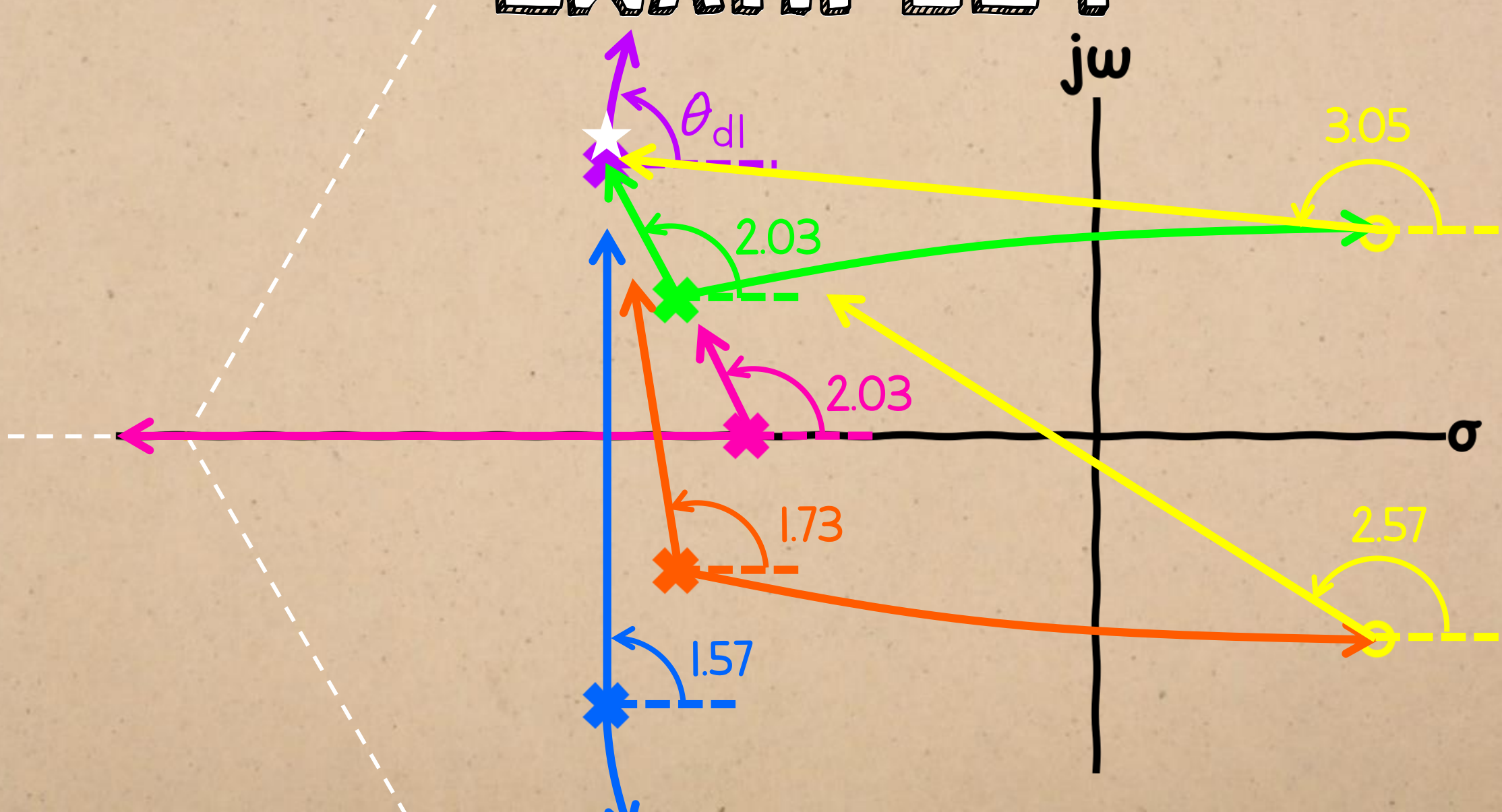
EXAMPLE 1

The positions of the critical frequencies are known, so we can work these angles out with simple trigonometry.



5 angles of
arrival/ departure:

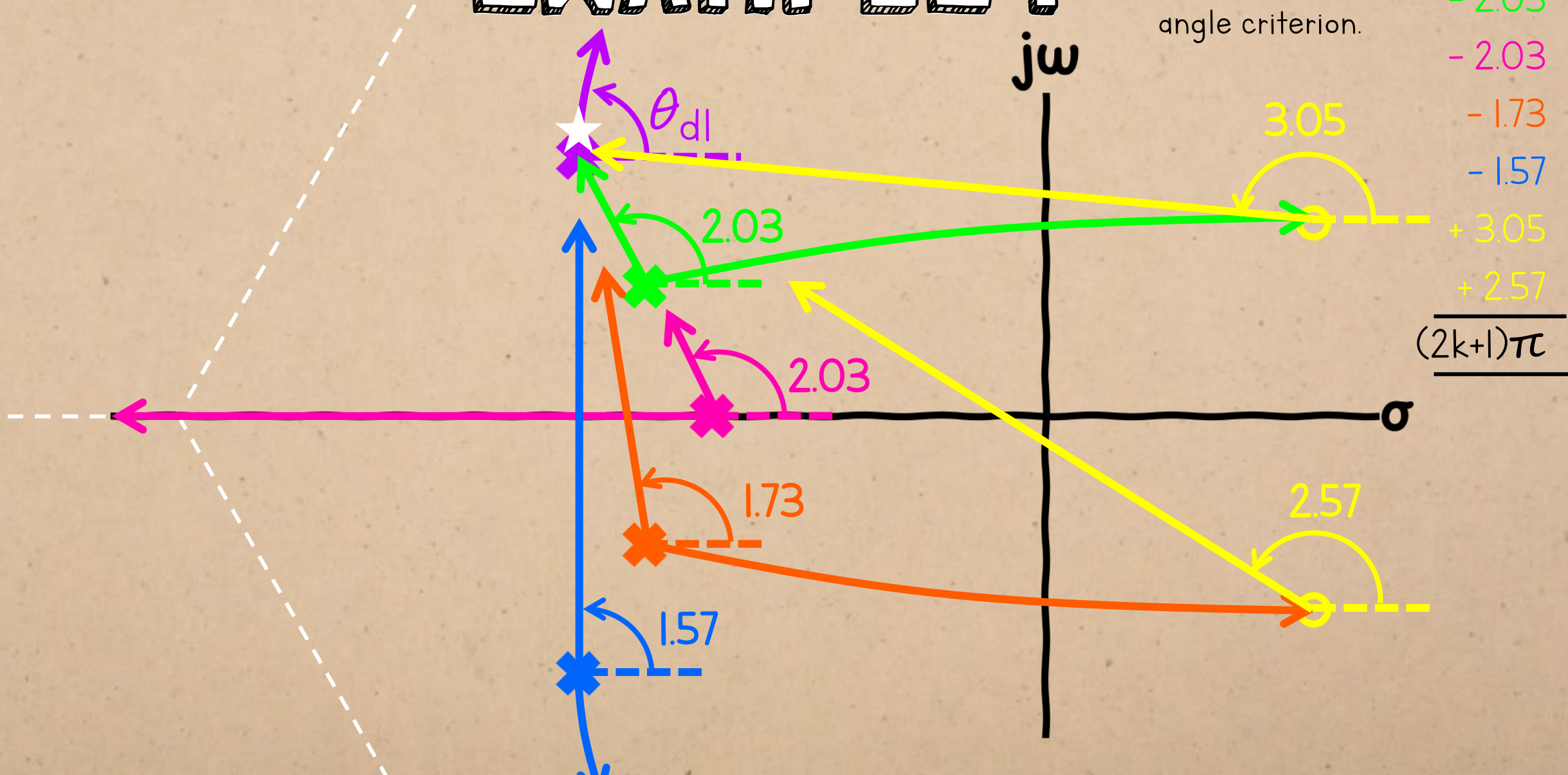
EXAMPLE 1



5 angles of arrival/ departure:

EXAMPLE 1

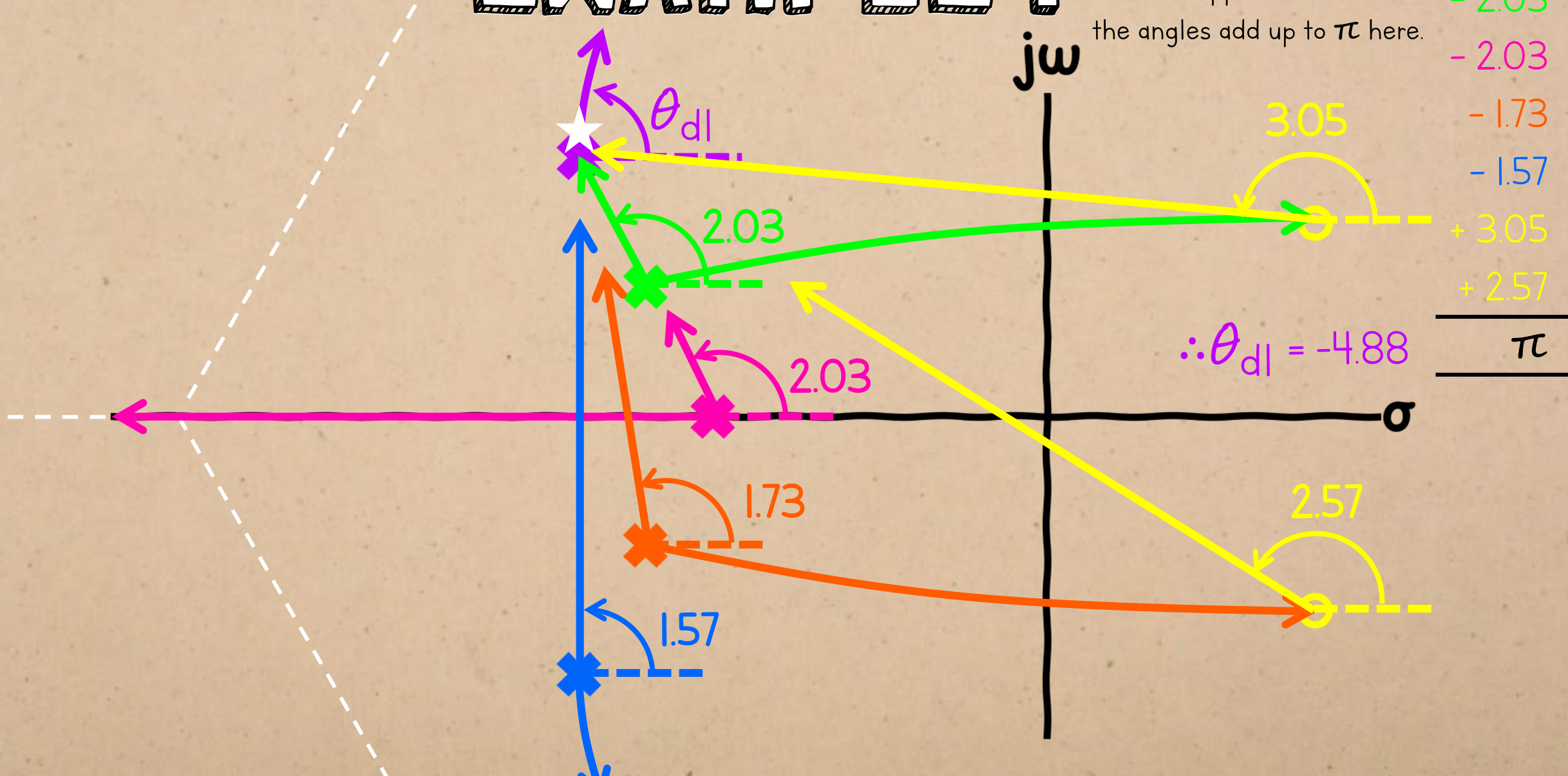
The point must satisfy the angle criterion.



5 angles of arrival/ departure:

EXAMPLE 1

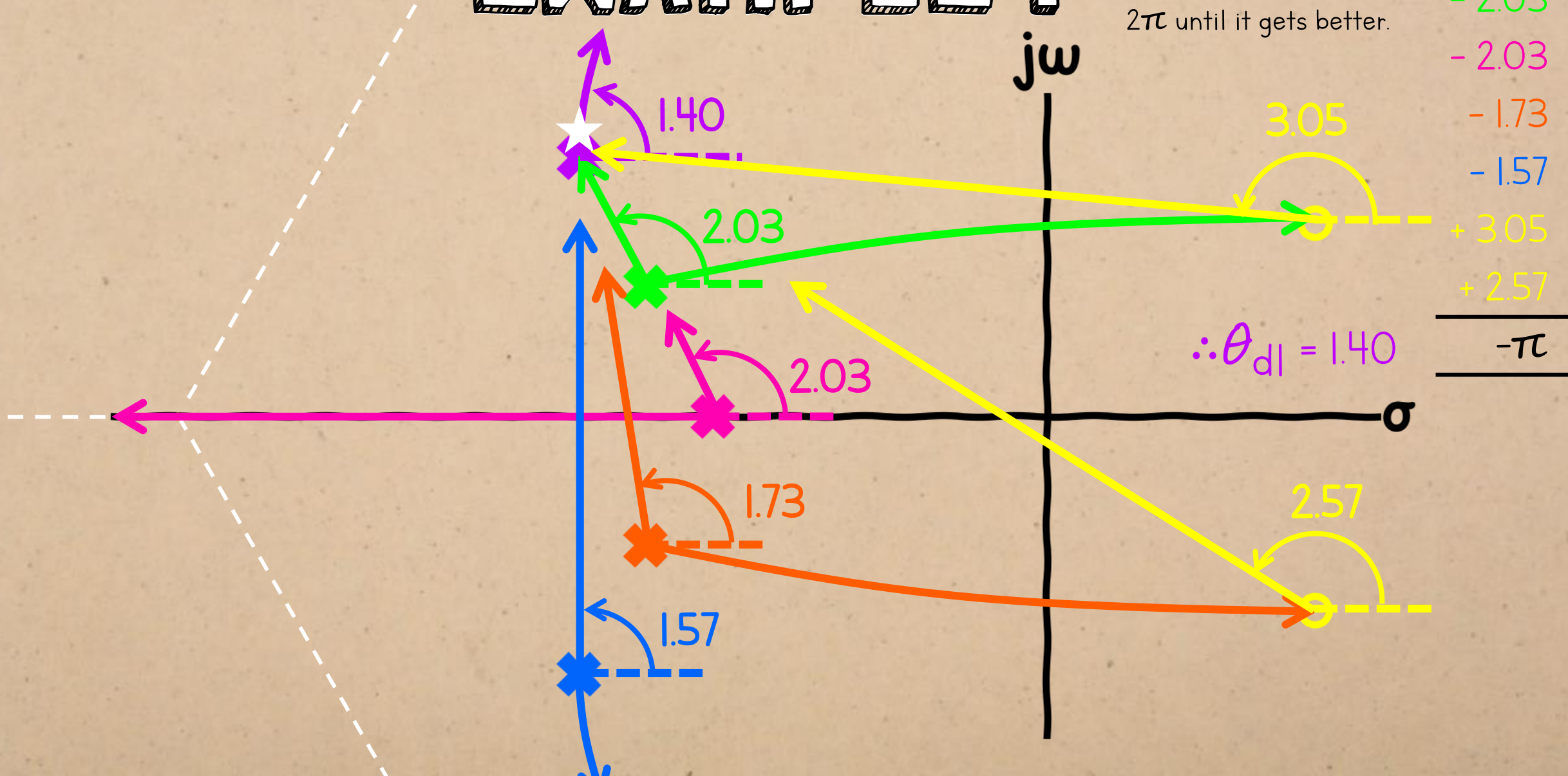
I like the answer to fall between $-\pi$ and π , which doesn't happen if we assume the angles add up to π here.



5 angles of arrival/ departure:

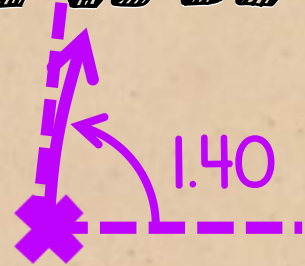
EXAMPLE 1

...but if you get a value you don't like, you can just add or subtract 2π until it gets better.



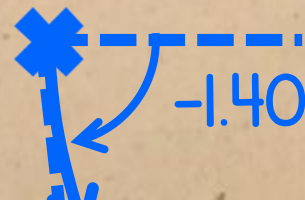
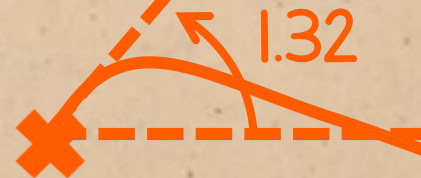
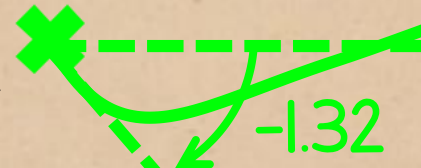
5 angles of arrival/ departure:

EXAMPLE 1



And repeat...

Knowing this departure angle helped me refine the shape of my initial rough sketch.



$j\omega$

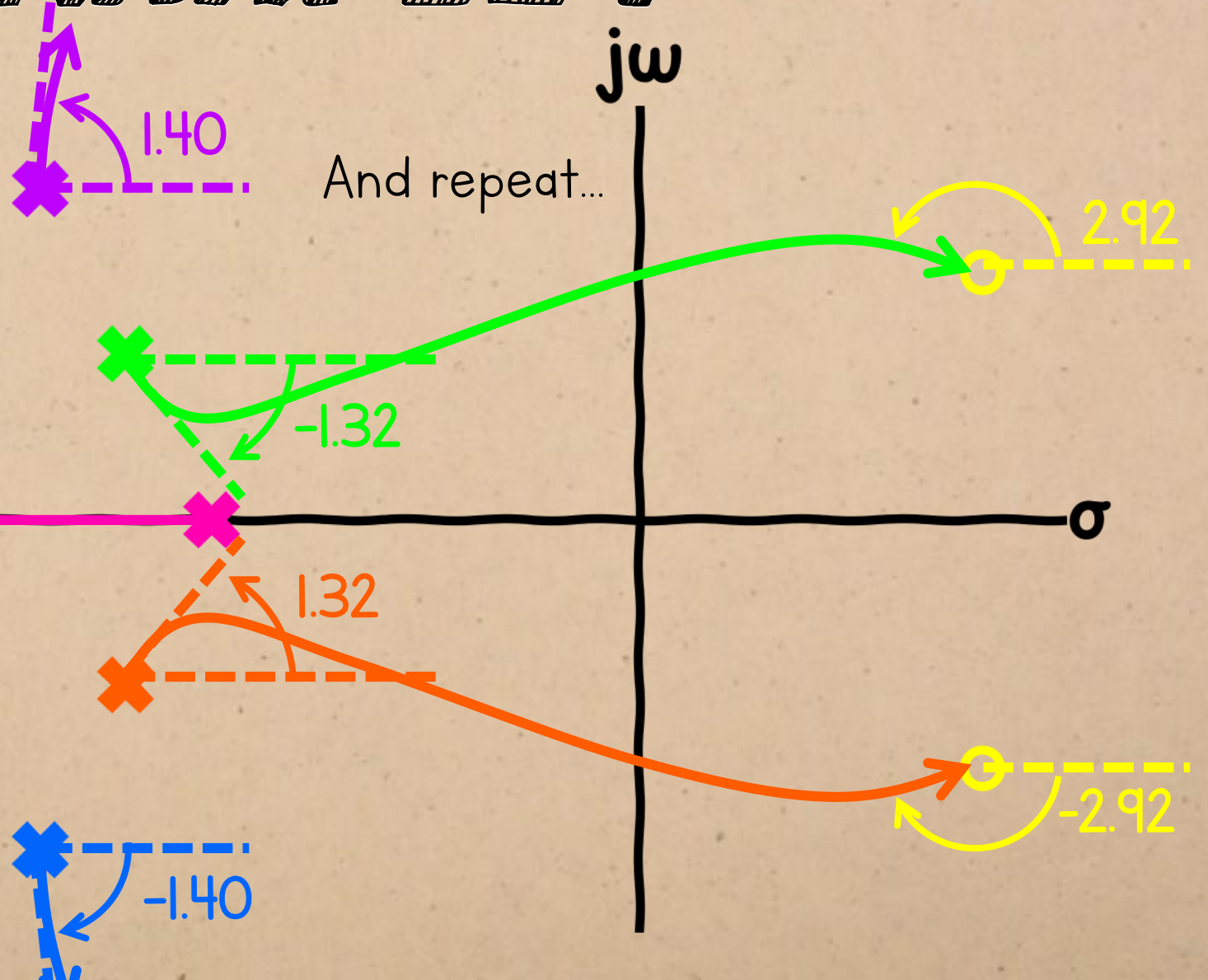
θ_{d1}

$-\theta_{d1}$

σ

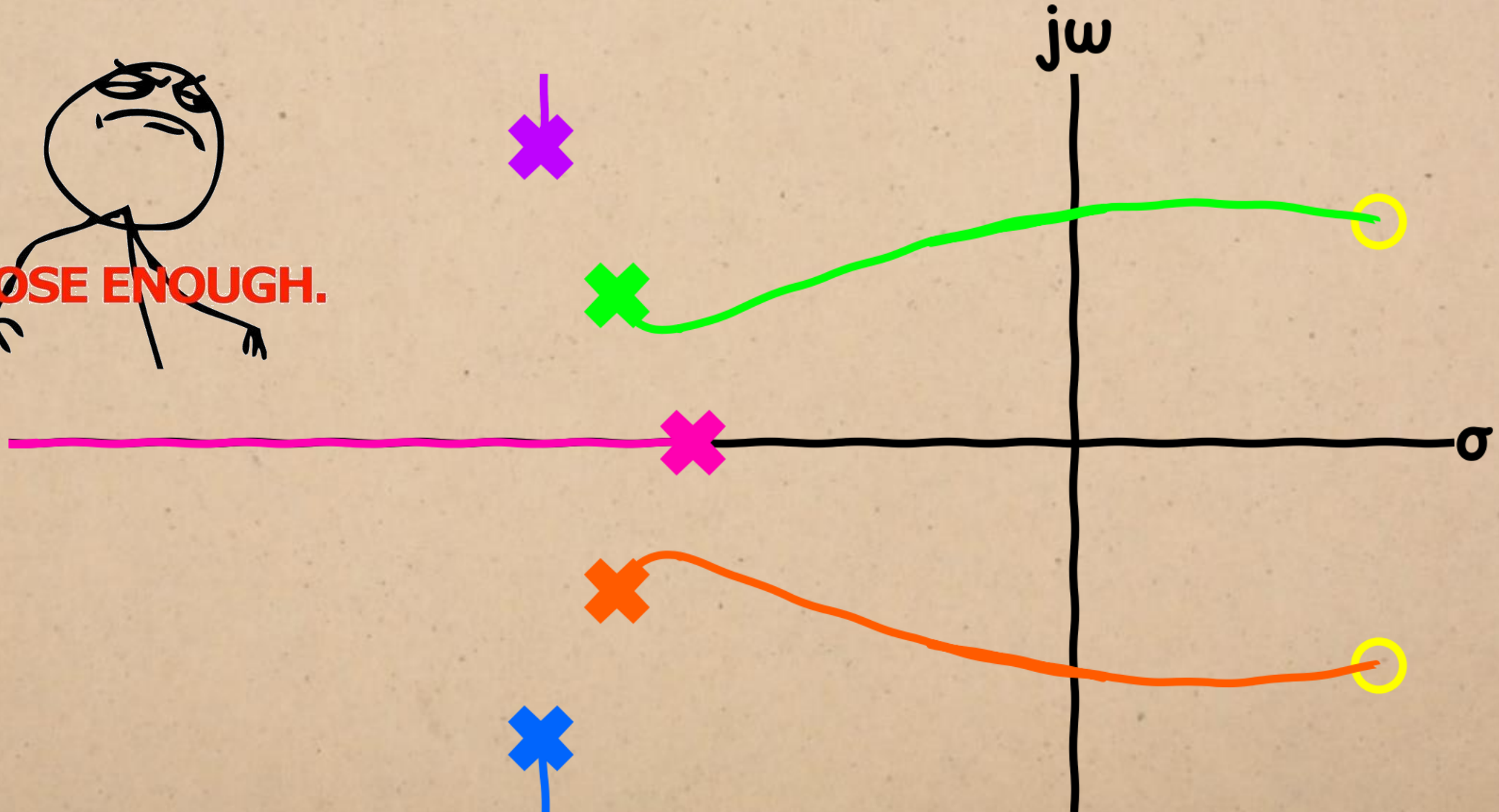
5 angles of
arrival/ departure:

EXAMPLE 1



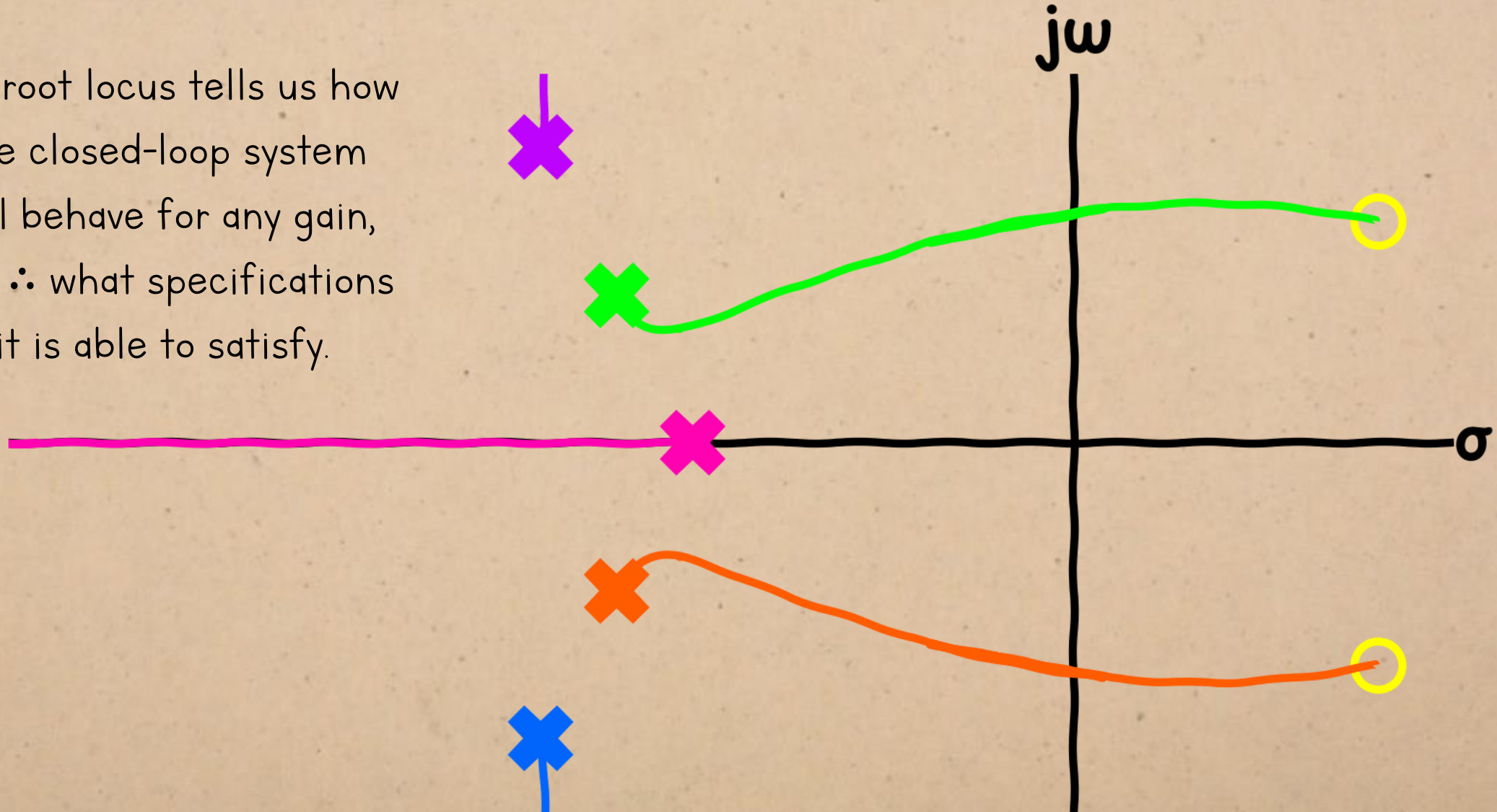
The true solution:

EXAMPLE 1



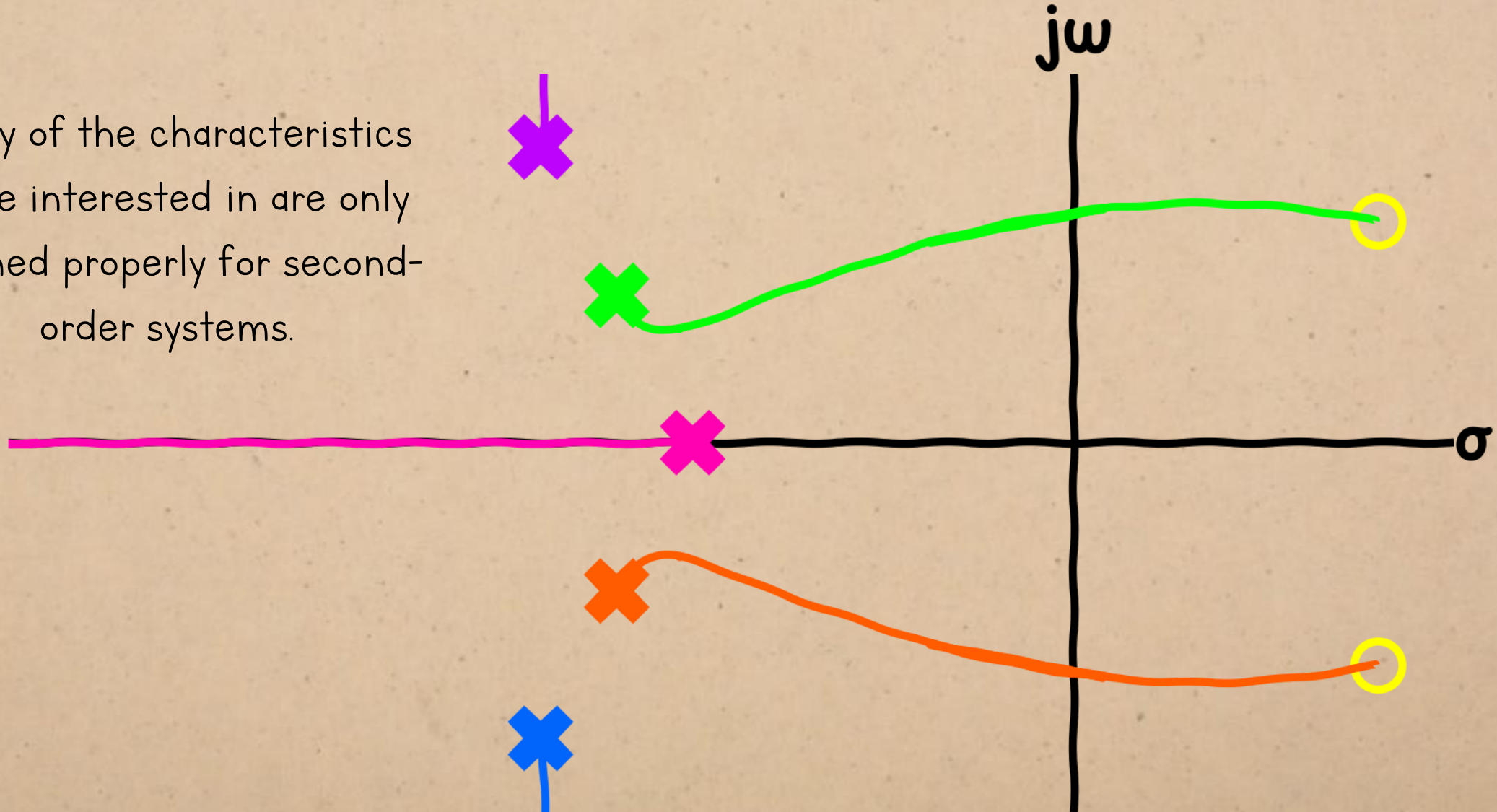
INTERPRETATION

The root locus tells us how the closed-loop system will behave for any gain, and $\therefore$ what specifications it is able to satisfy.



INTERPRETATION

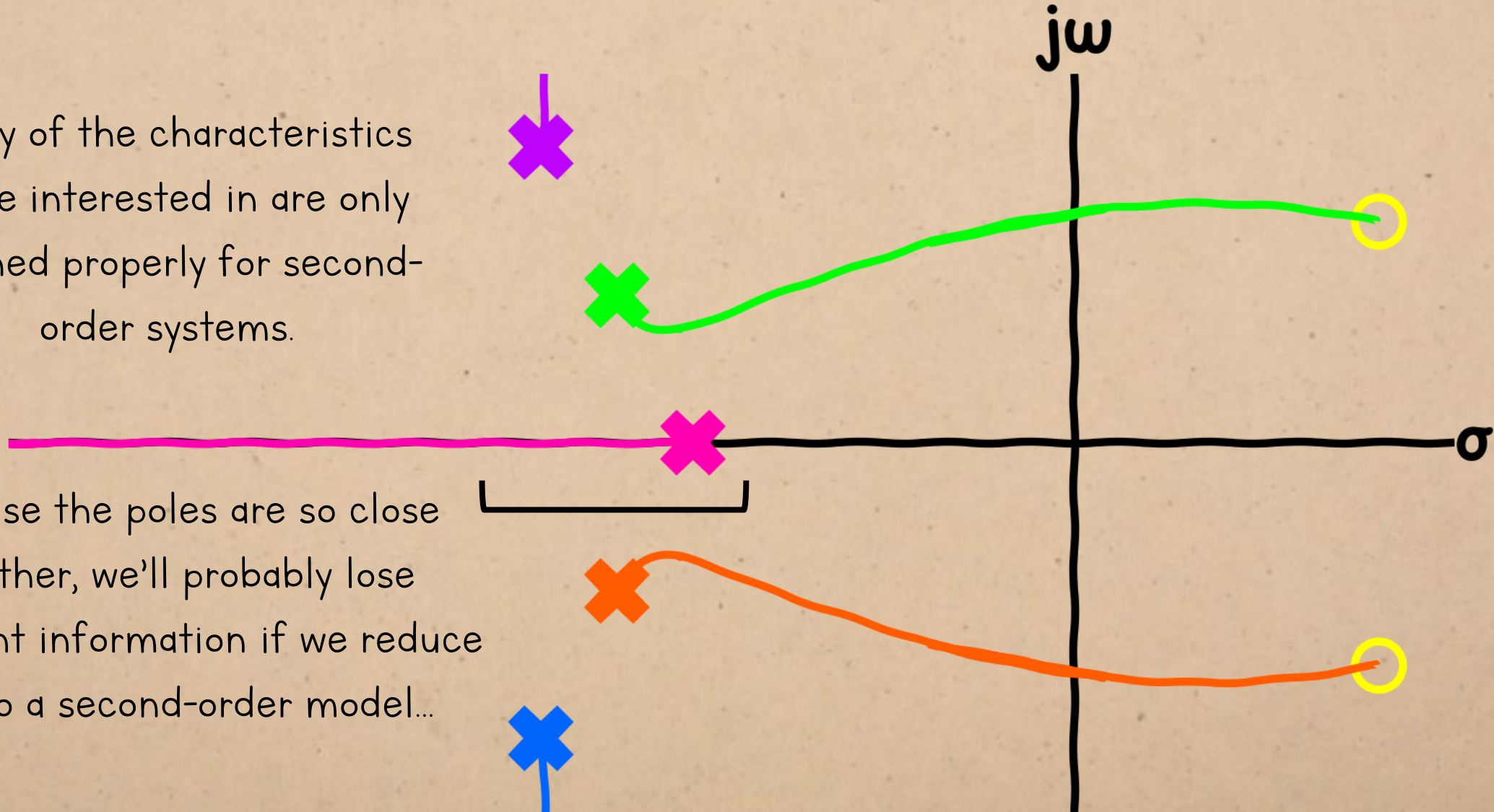
Many of the characteristics we're interested in are only defined properly for second-order systems.



INTERPRETATION

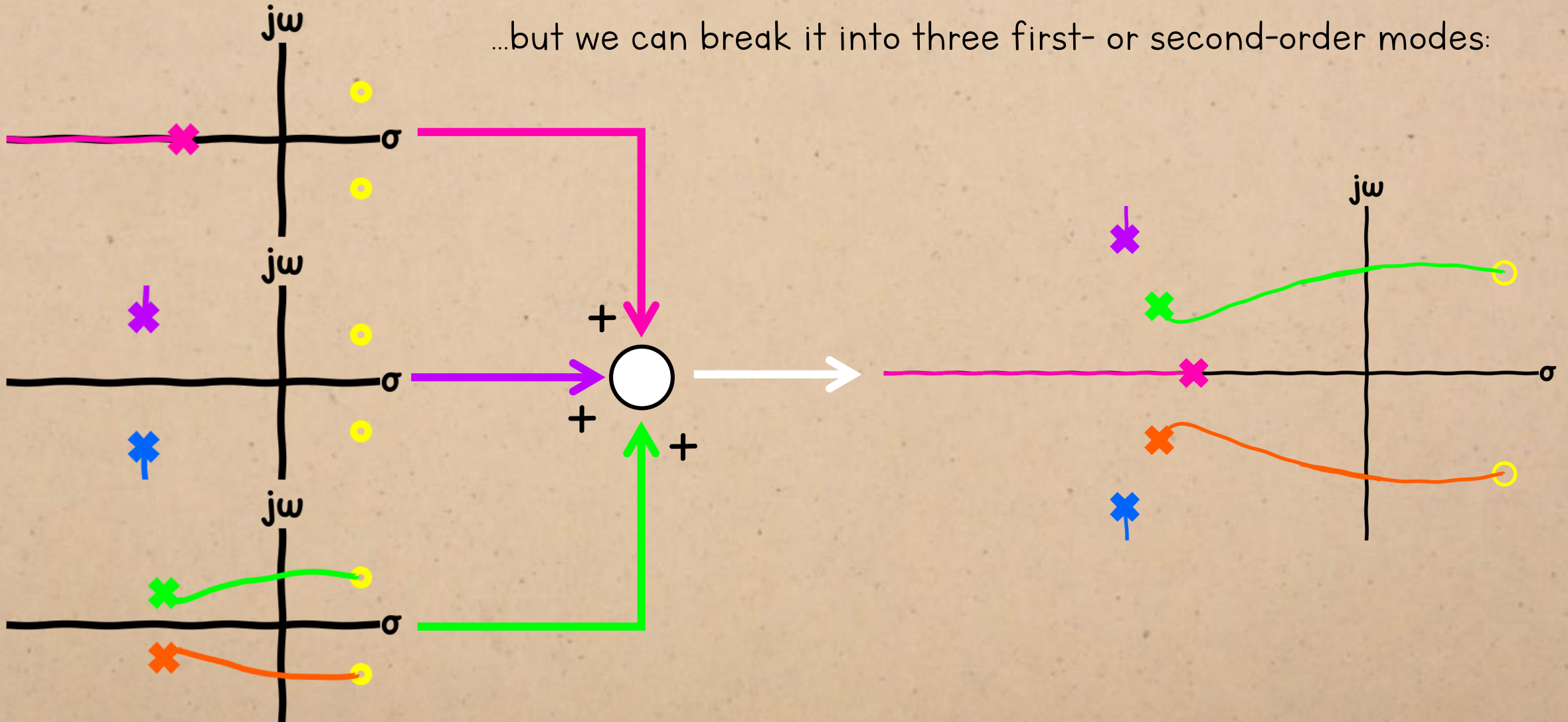
Many of the characteristics we're interested in are only defined properly for second-order systems.

Because the poles are so close together, we'll probably lose important information if we reduce this to a second-order model...

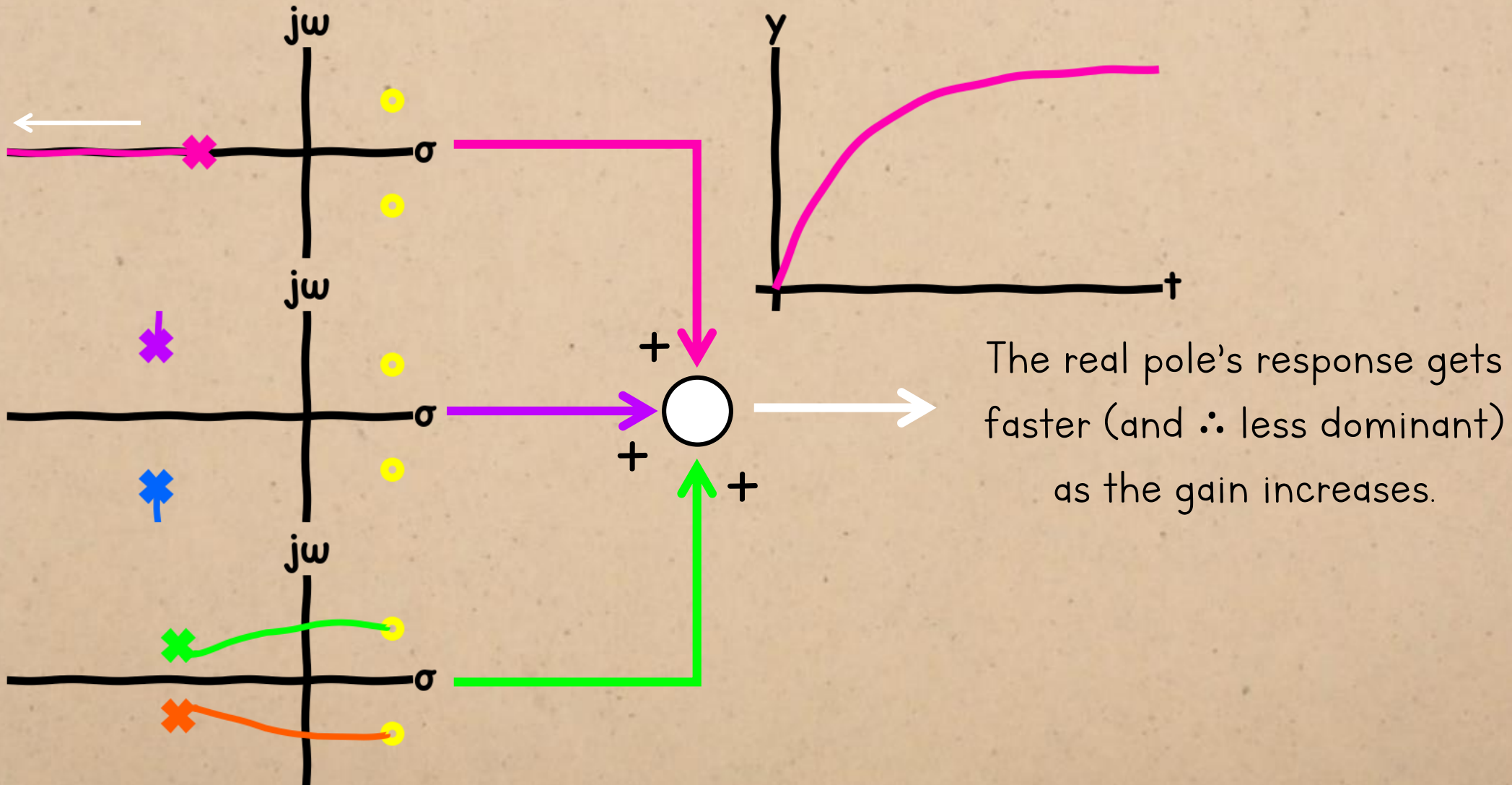


INTERPRETATION

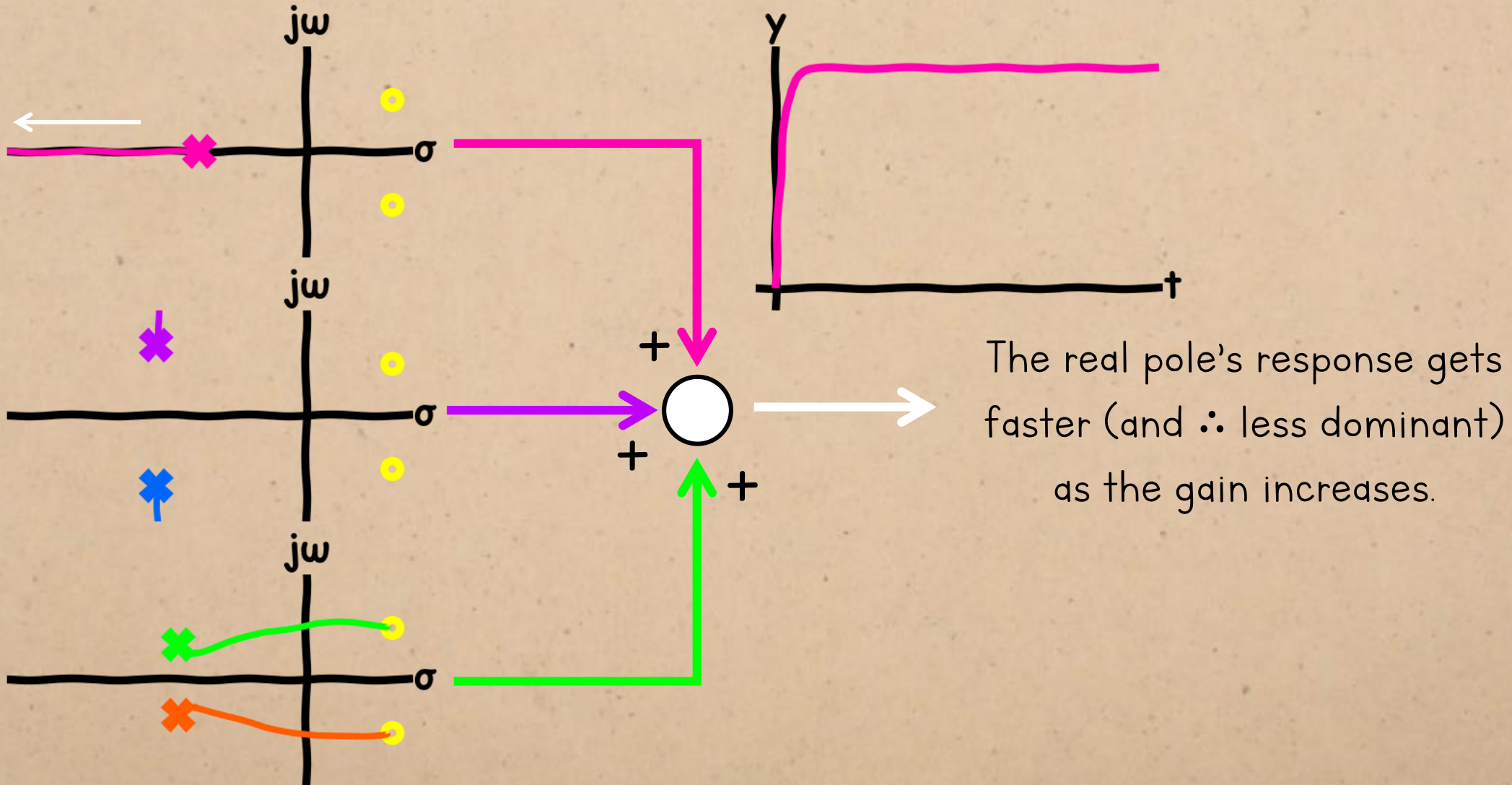
...but we can break it into three first- or second-order modes:



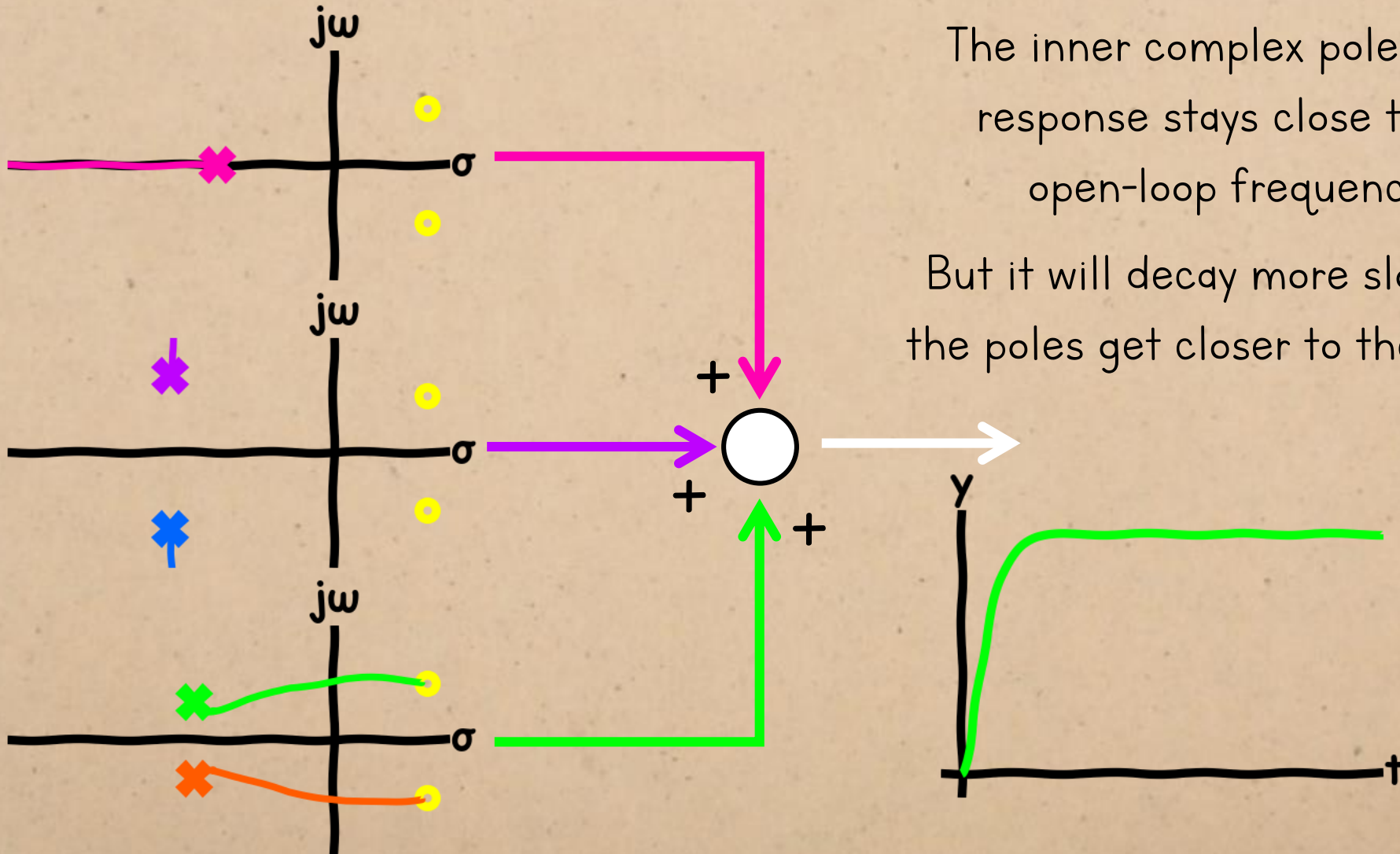
INTERPRETATION



INTERPRETATION



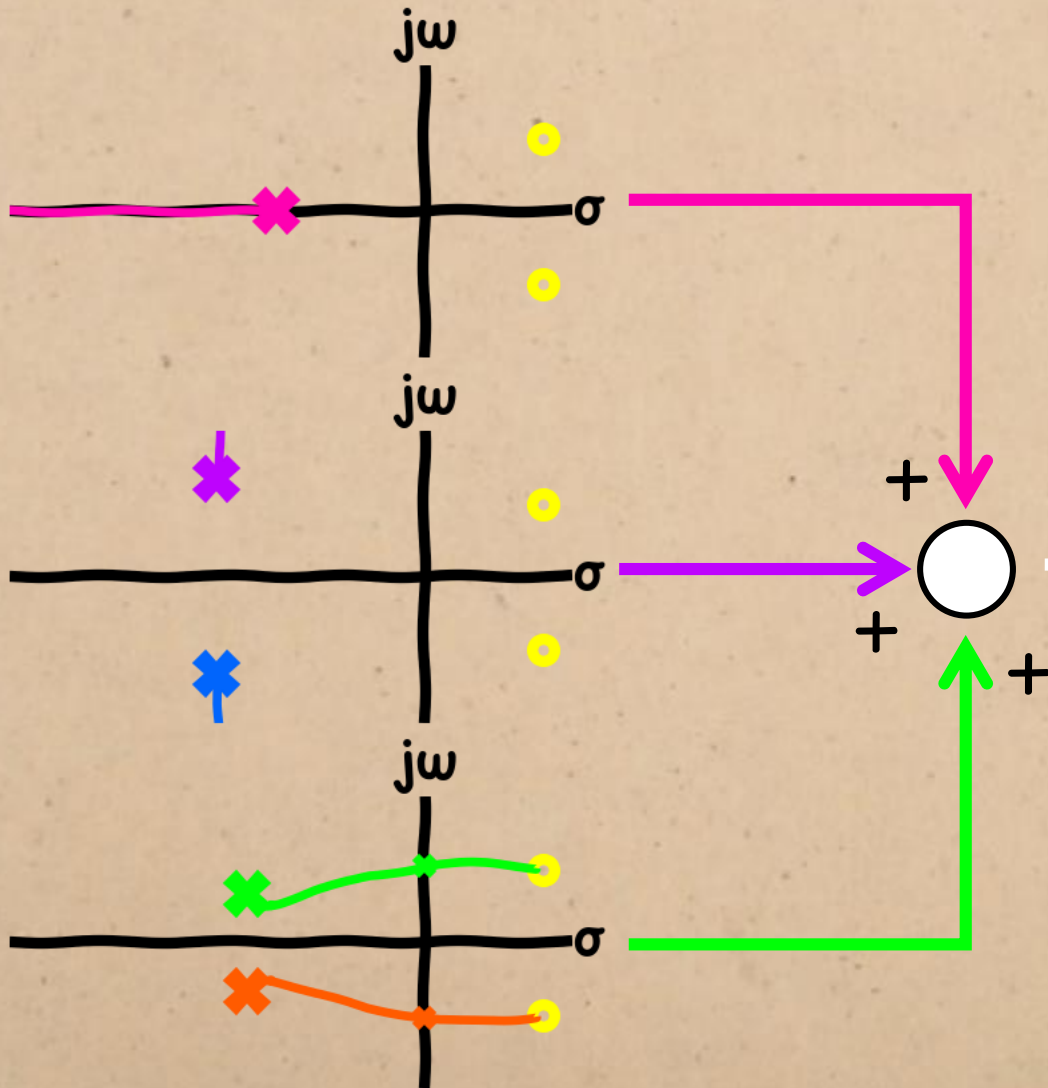
INTERPRETATION



The inner complex pole pair's response stays close to its open-loop frequency.

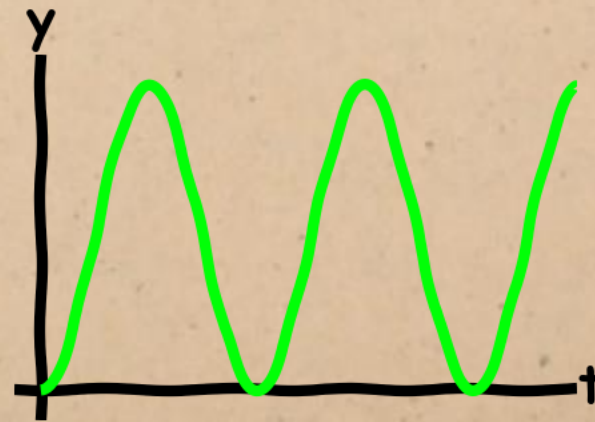
But it will decay more slowly as the poles get closer to the origin.

INTERPRETATION

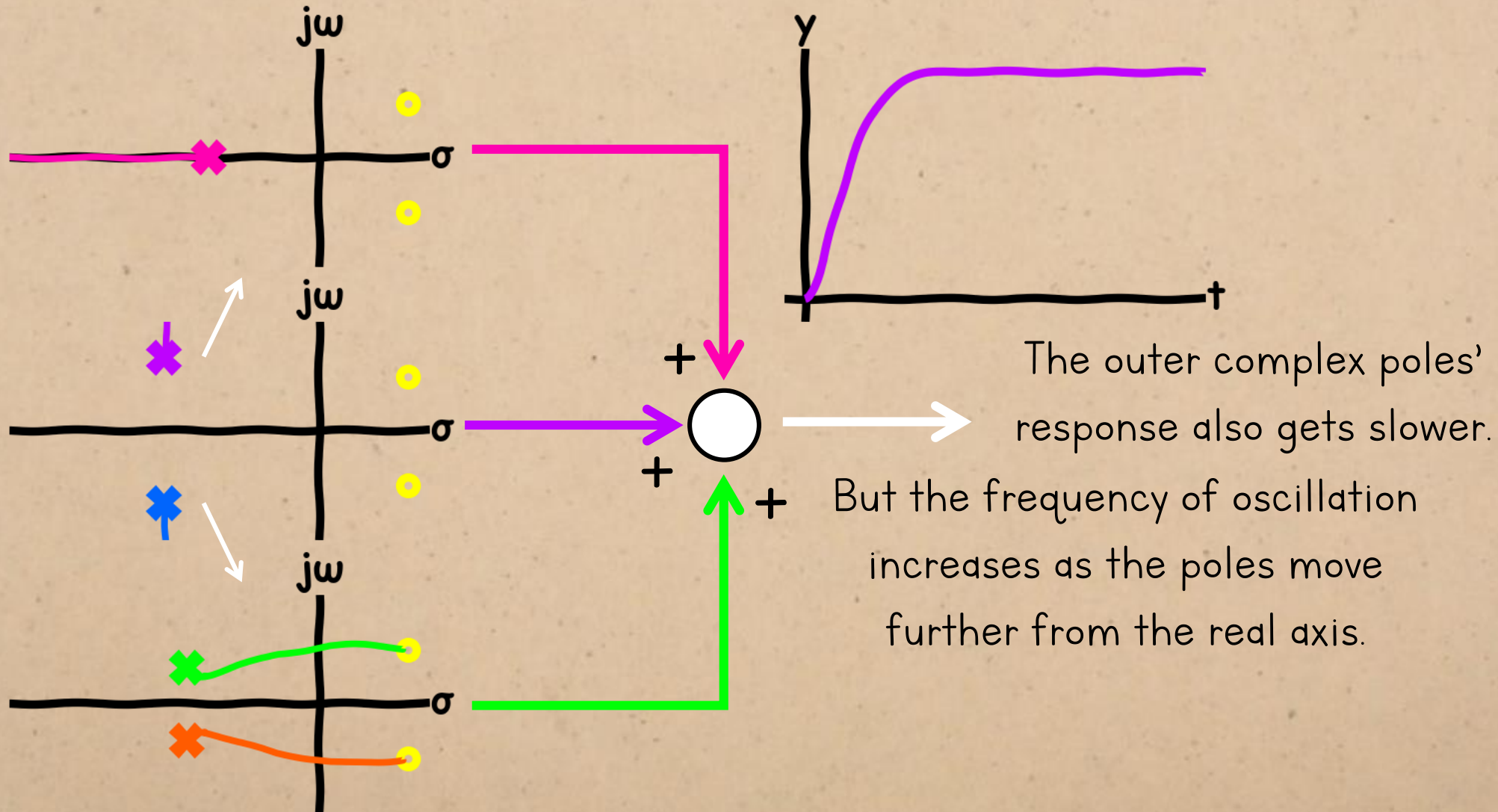


The inner complex pole pair's response stays close to its open-loop frequency.

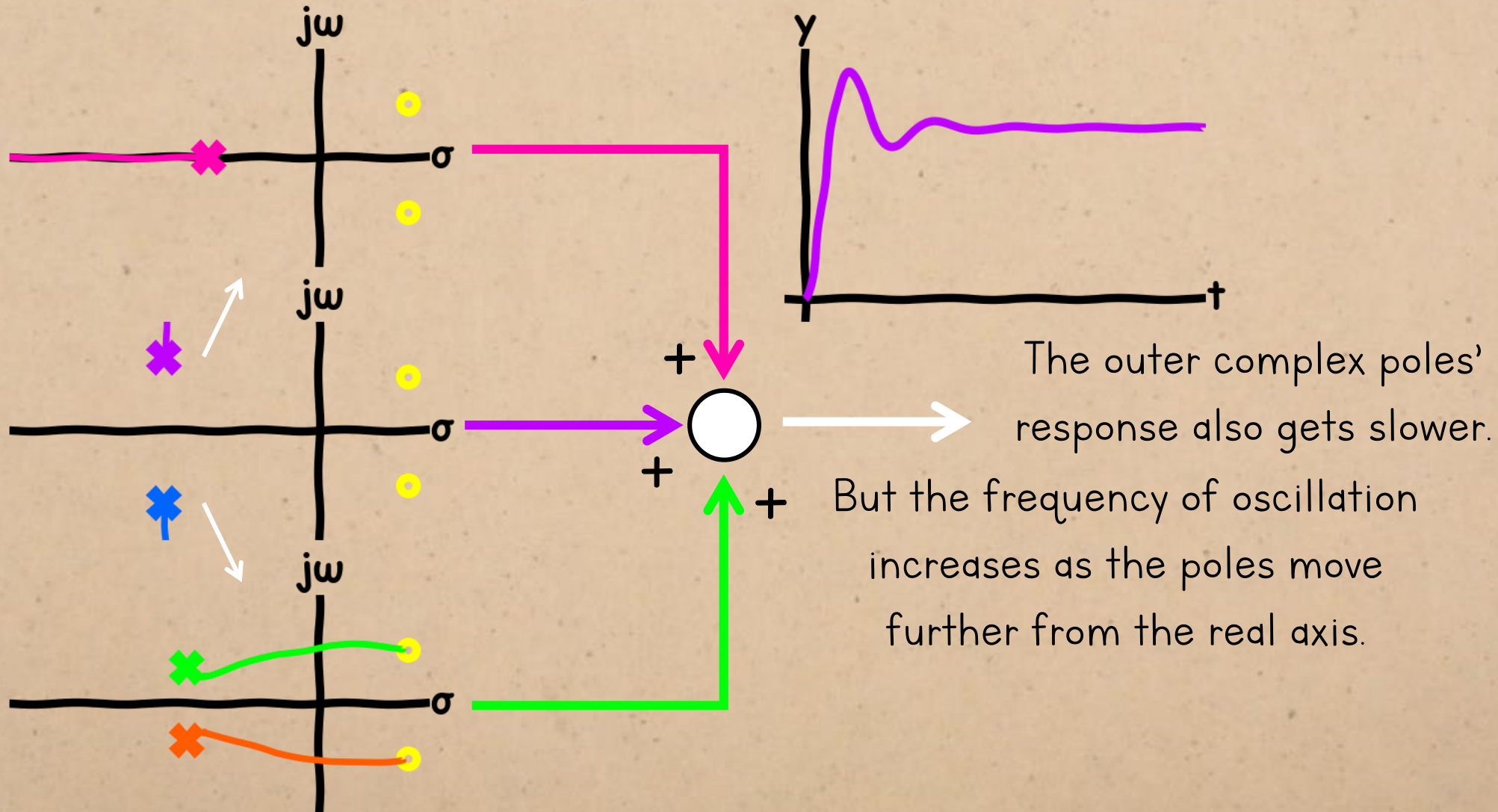
But it will decay more slowly as the poles get closer to the origin.



INTERPRETATION

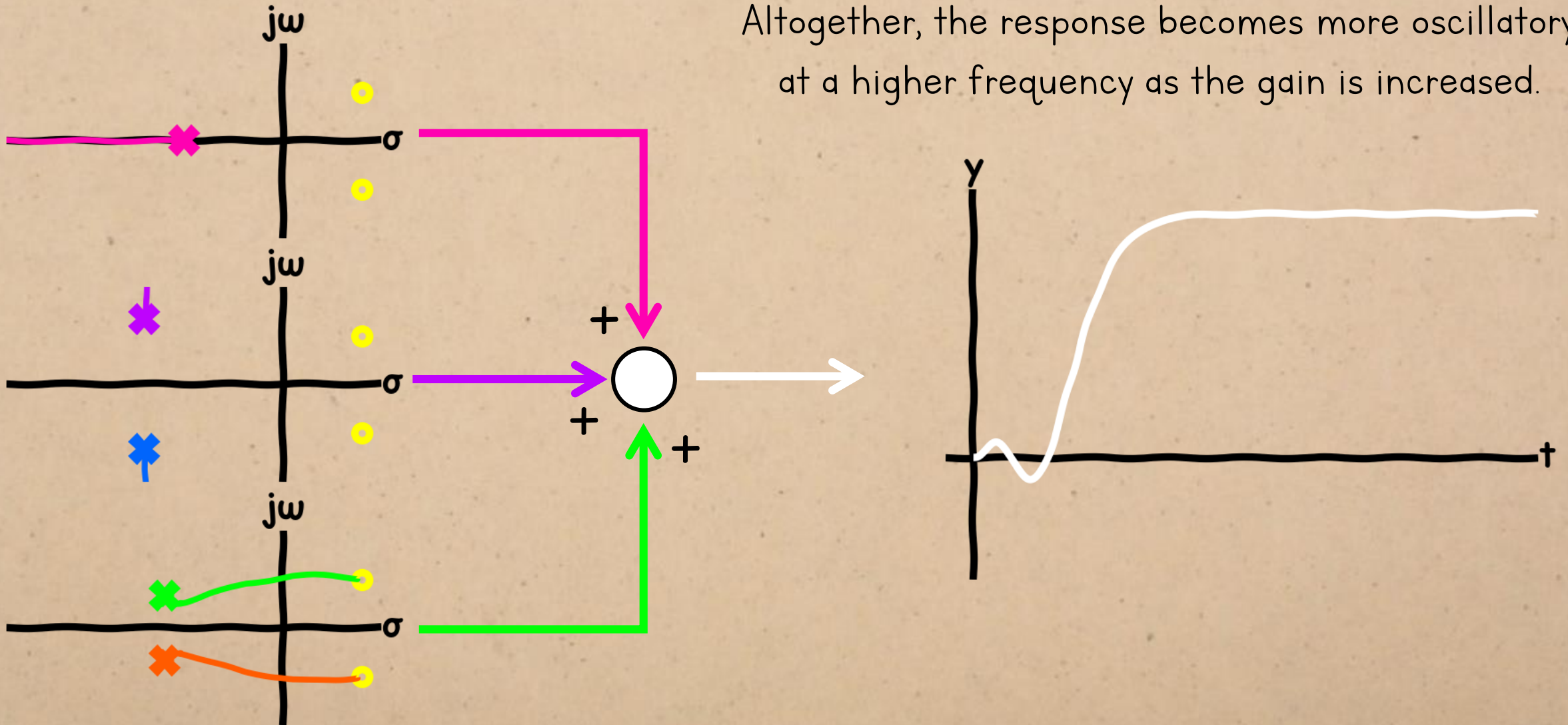


INTERPRETATION



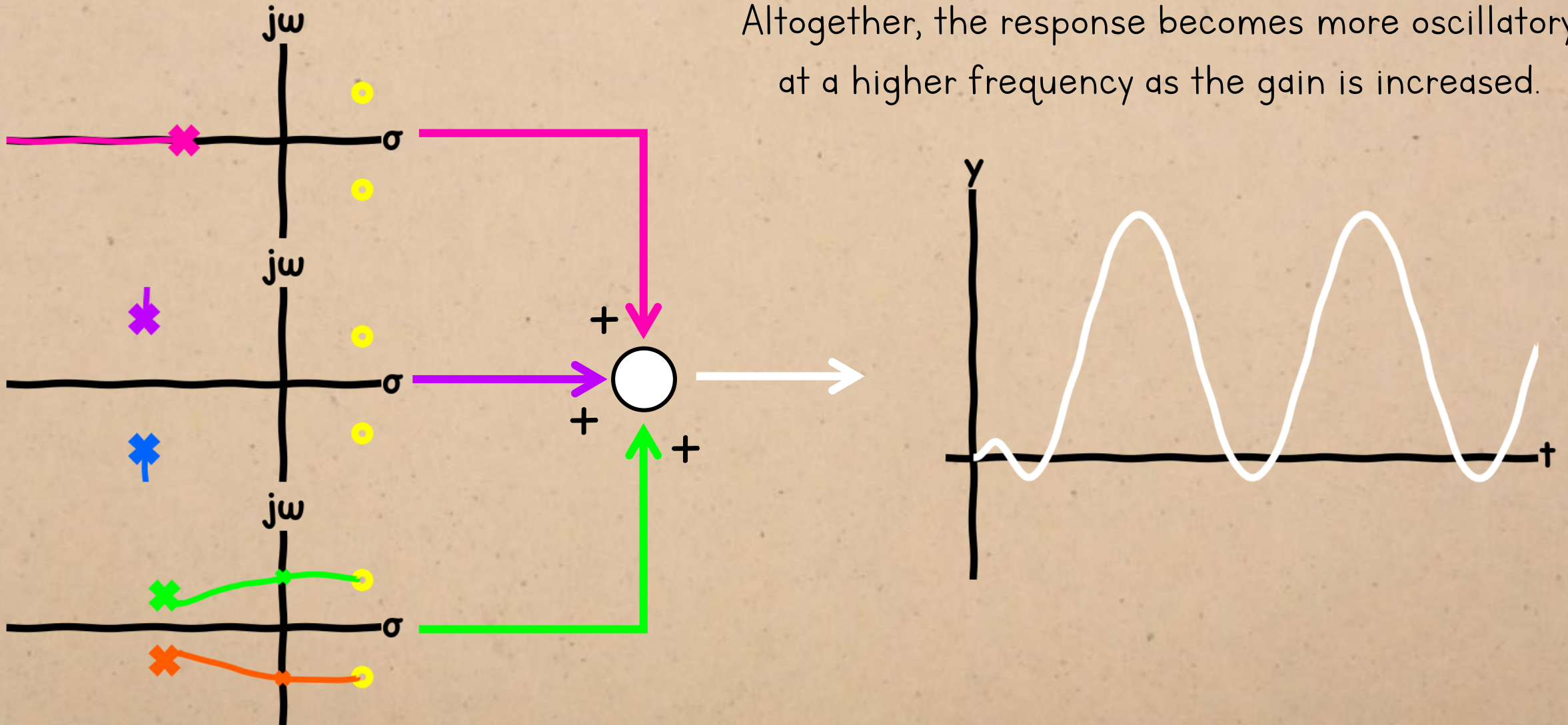
INTERPRETATION

Altogether, the response becomes more oscillatory at a higher frequency as the gain is increased.



INTERPRETATION

Altogether, the response becomes more oscillatory at a higher frequency as the gain is increased.



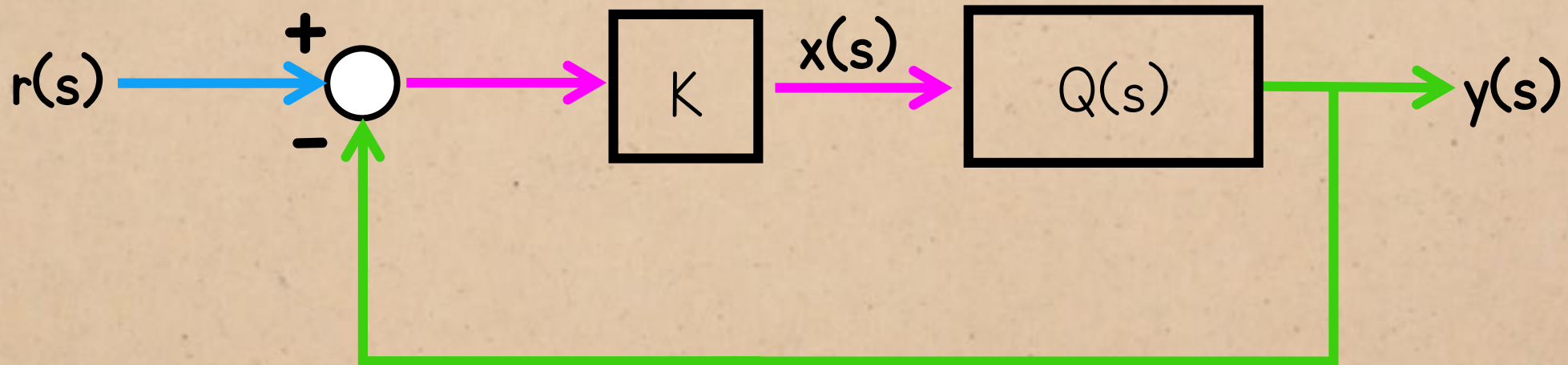
INTERPRETATION

So if your job is to make the response less oscillatory or decrease its frequency...

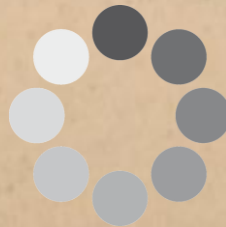


INTERPRETATION

...A proportional controller won't help you.
Increasing the gain will just make the system's behaviour worse.



So what should you do, instead?



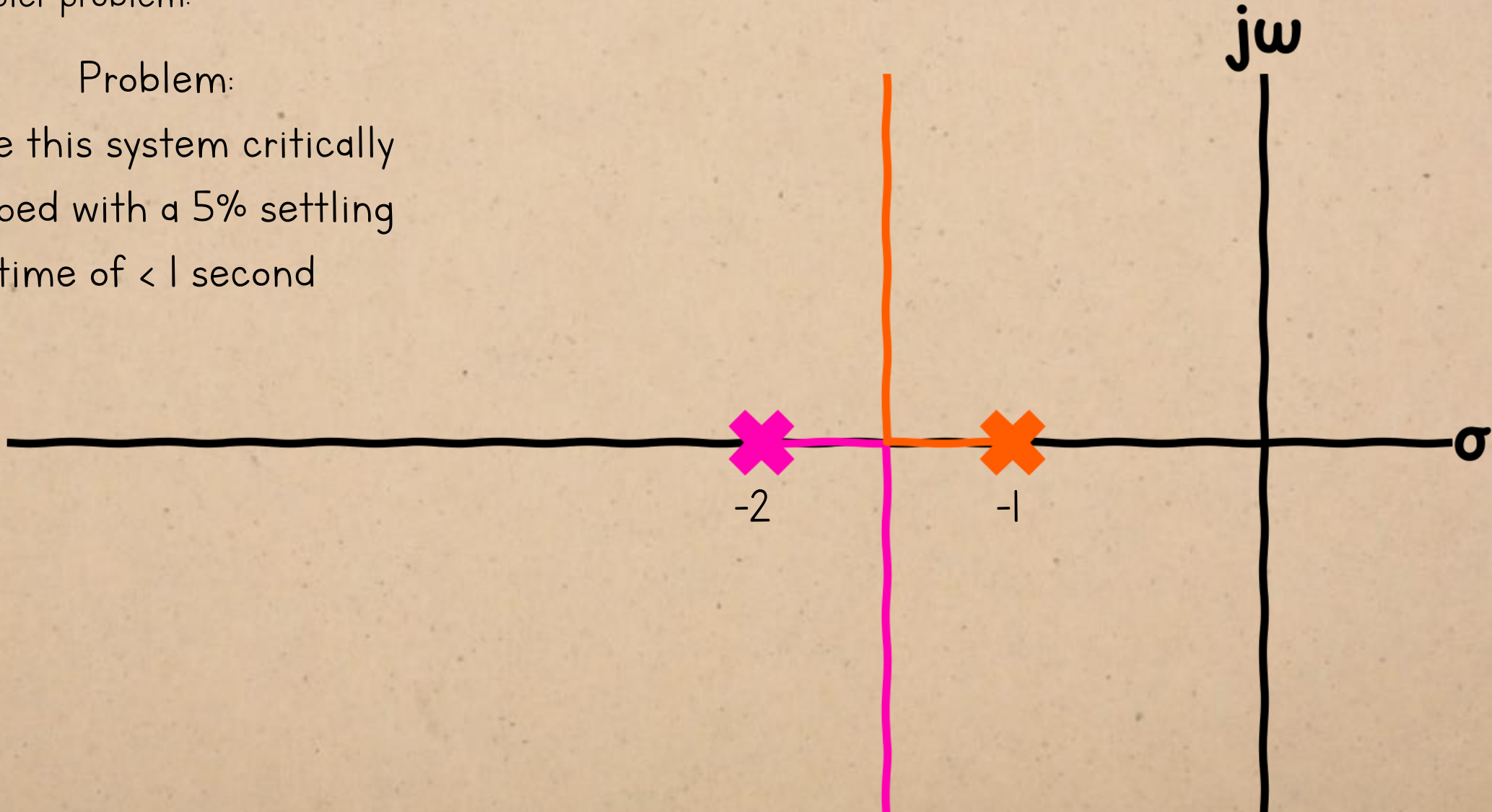
We'll answer this properly in a few weeks' time, when we get to the compensation section, but we already know enough about how the poles move to make some educated guesses.

...Or handle a similar,
but simpler problem:

INTERPRETATION

Problem:

make this system critically
damped with a 5% settling
time of < 1 second

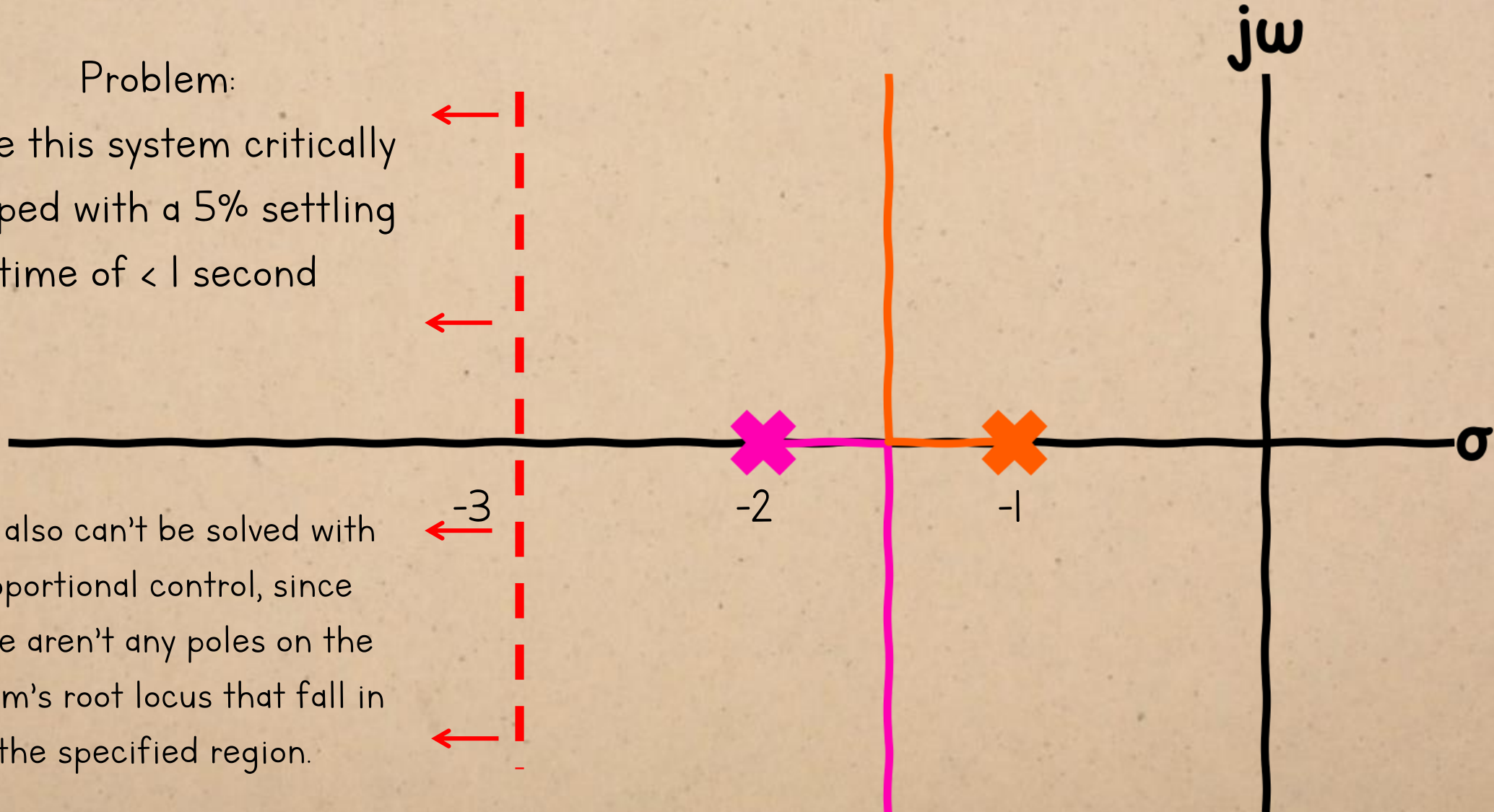


INTERPRETATION

Problem:

make this system critically damped with a 5% settling time of < 1 second

This also can't be solved with proportional control, since there aren't any poles on the system's root locus that fall in the specified region.

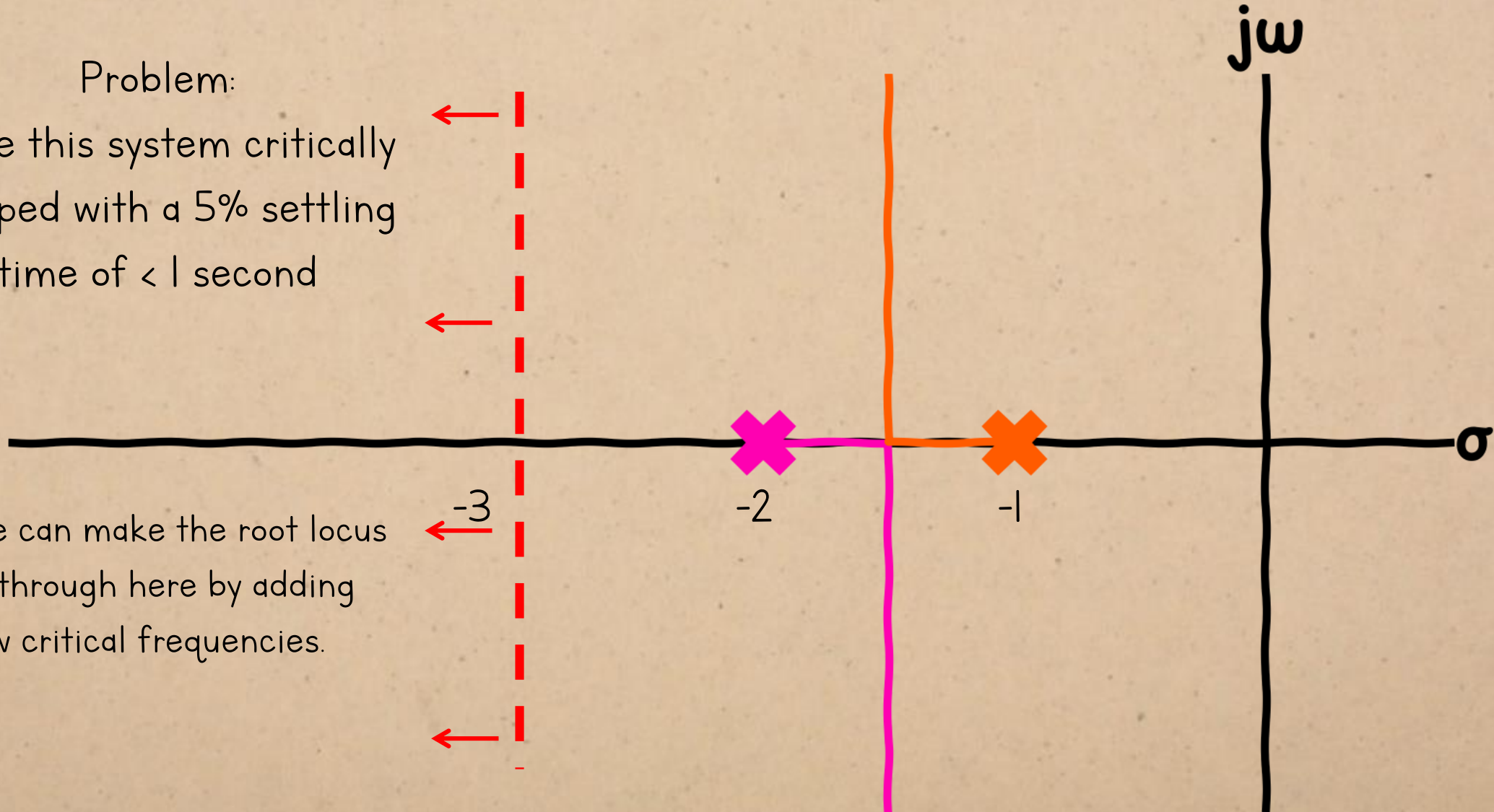


INTERPRETATION

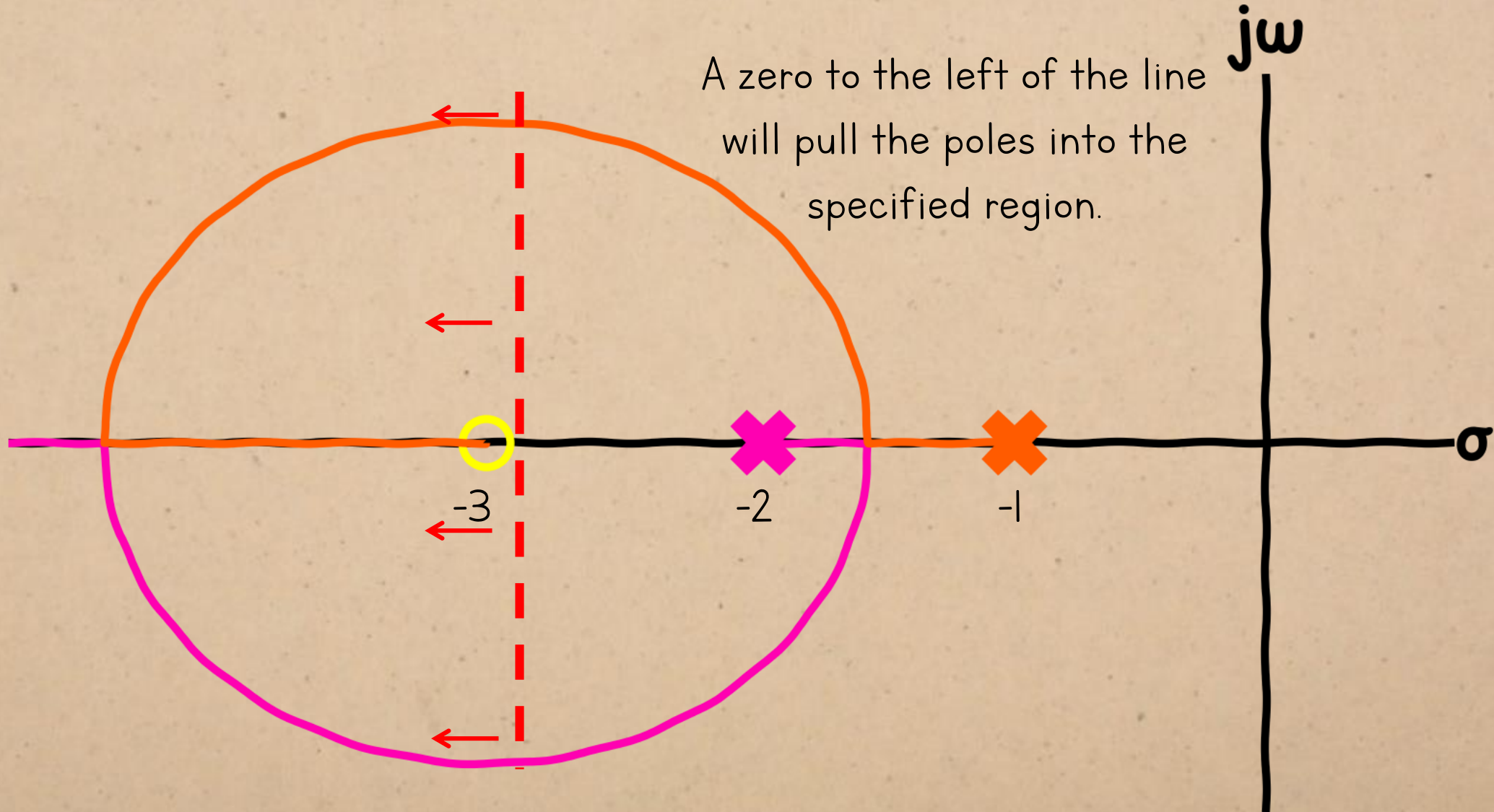
Problem:

make this system critically
damped with a 5% settling
time of < 1 second

...but we can make the root locus
pass through here by adding
new critical frequencies.

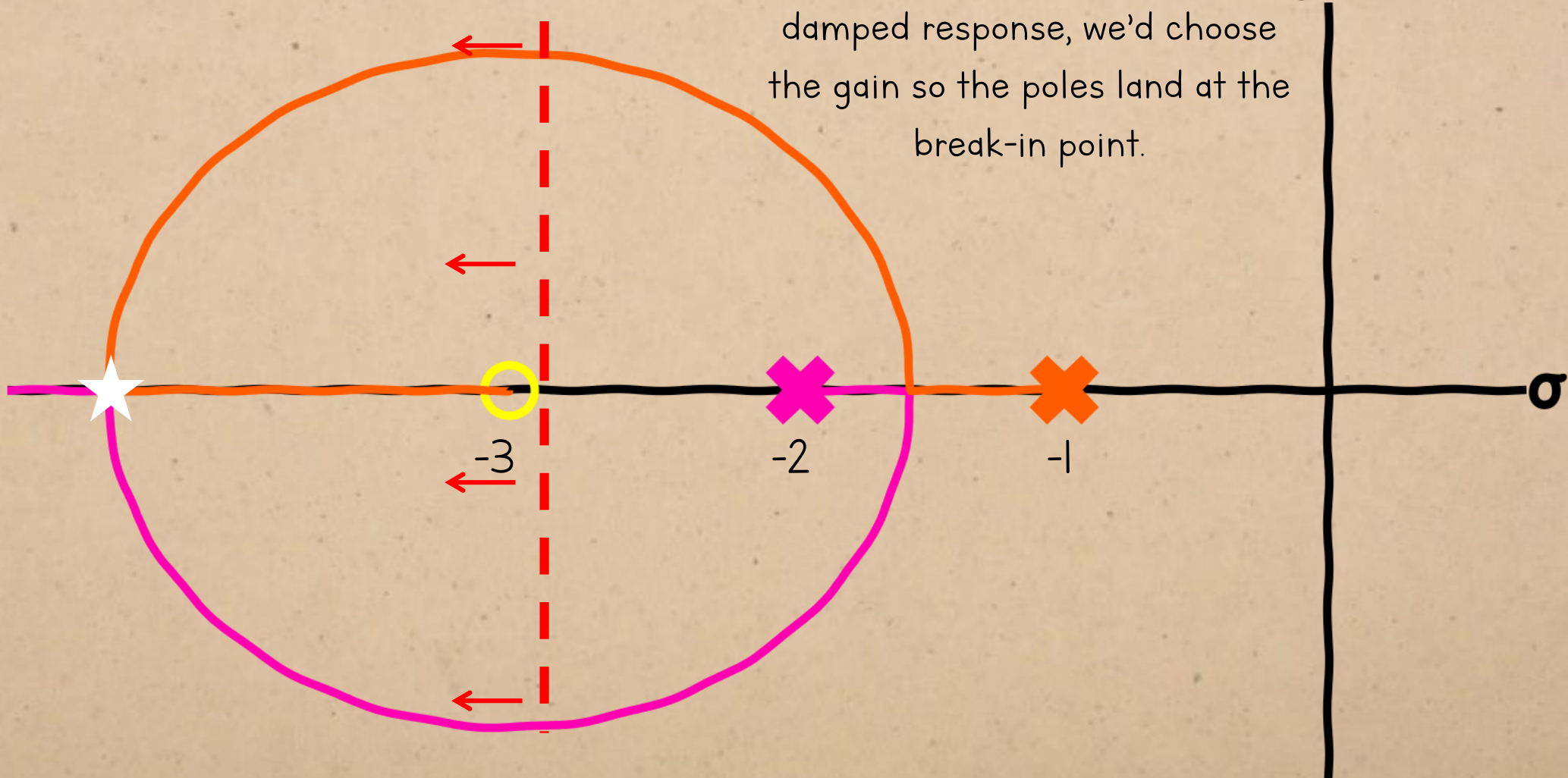


INTERPRETATION



INTERPRETATION

To get the desired critically-damped response, we'd choose the gain so the poles land at the break-in point.

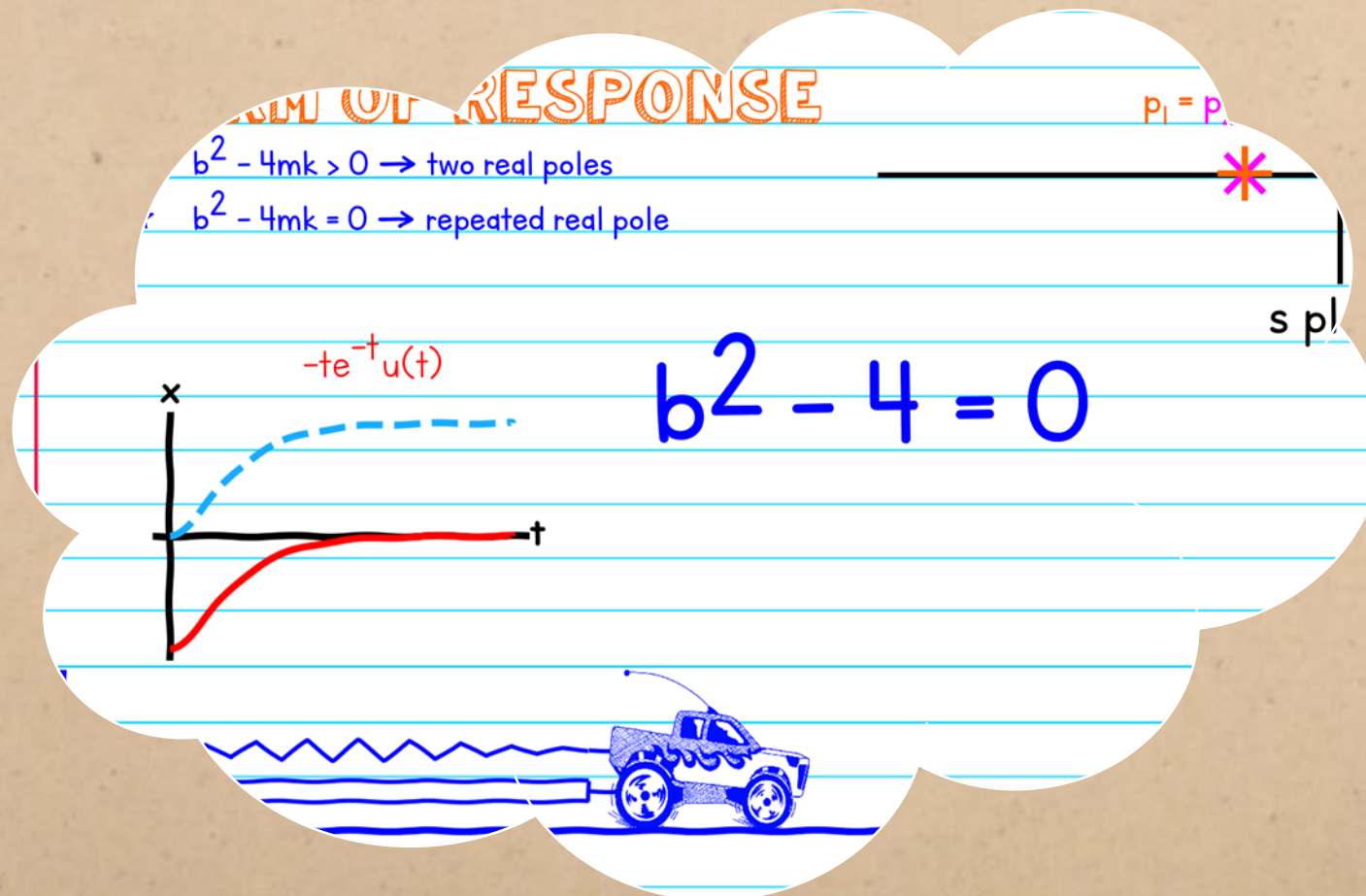


EXAMPLE 2

We can also use the root locus to track the system's behaviour as other parameters are varied:

Example:
the damping coefficient

We're going to repeat the investigation we did in chapter 2 into the effect of the damping coefficient on the transient response, but use the root locus method to do it.



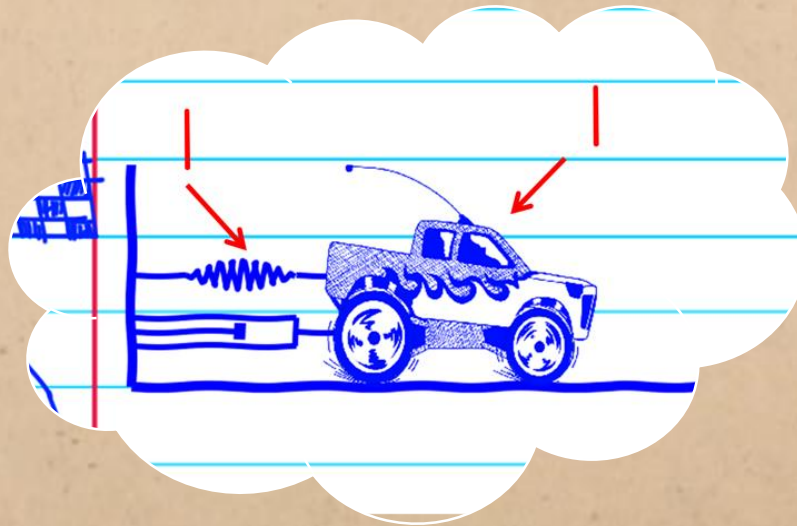
EXAMPLE 2

transfer function:

$$\frac{1}{s^2 + bs + 1}$$

characteristic equation:

$$s^2 + bs + 1 = 0$$



EXAMPLE 2

characteristic equation:

$$s^2 + bs + 1 = 0$$

To construct the root locus, we need to write this in the form:

$$1 + [\text{parameter of interest}][\text{transfer function}] = 0$$

so we group everything that isn't multiplied by the damping coefficient...

$$(s^2 + 1) + bs = 0$$

EXAMPLE 2

characteristic equation:

$$s^2 + bs + 1 = 0$$

To construct the root locus, we need to write this in the form:

$$1 + [\text{parameter of interest}][\text{transfer function}] = 0$$

...and divide the equation by this group:

$$\frac{(s^2 + 1)}{(s^2 + 1)} + \frac{bs}{(s^2 + 1)} = 0$$

EXAMPLE 2

characteristic equation:

$$s^2 + bs + 1 = 0$$

To construct the root locus, we need to write this in the form:

$$1 + [\text{parameter of interest}][\text{transfer function}] = 0$$

...and divide the equation by this group:

$$1 + b \frac{s}{(s^2 + 1)} = 0$$

EXAMPLE 2

characteristic equation:

$$s^2 + bs + 1 = 0$$

To construct the root locus, we need to write this in the form:

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$$1 + b \frac{s}{(s^2 + 1)} = 0$$

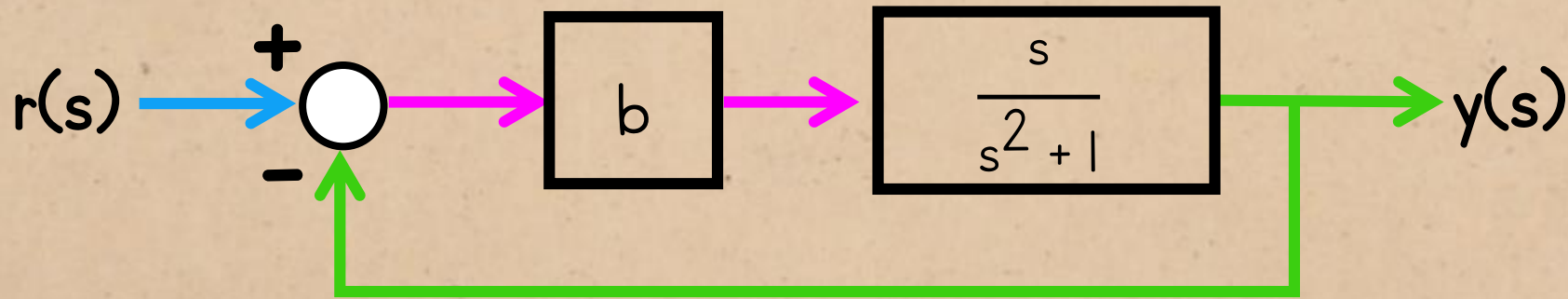
auxiliary transfer function



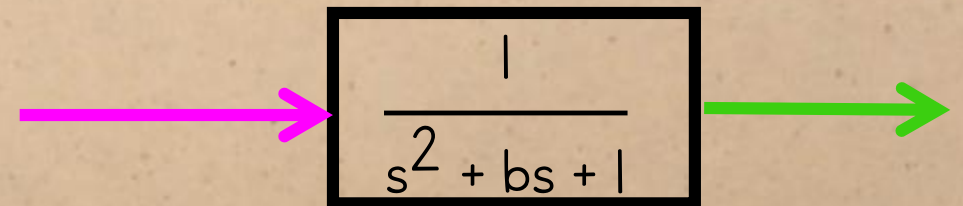
EXAMPLE 2

$$1 + b \frac{s}{(s^2 + 1)} = 0$$

When we put this auxiliary transfer function in closed loop and vary its gain,



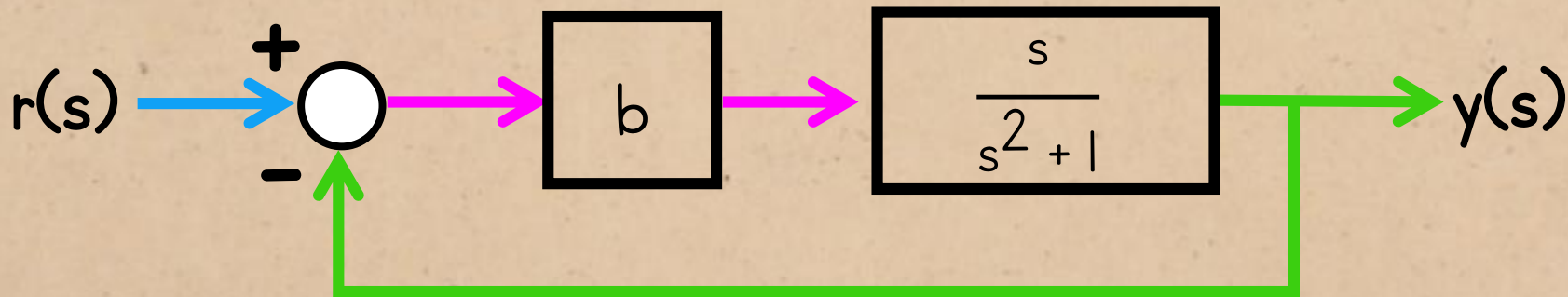
its closed-loop poles will move in the same way the poles of our original system do as the damping coefficient is varied.



EXAMPLE 2

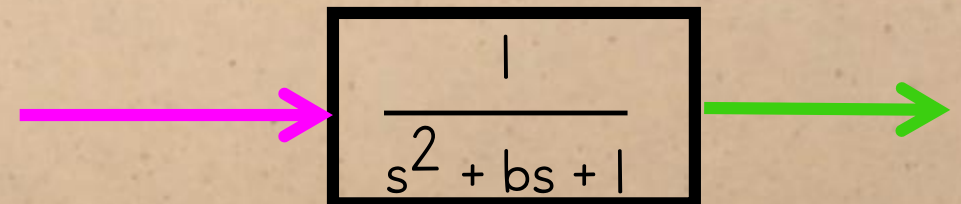
$$1 + b \frac{s}{(s^2 + 1)} = 0$$

So plotting the root locus for this:



(We aren't actually putting our system in closed loop, though, so instead of open-loop/ closed-loop poles, we should talk about undamped/ damped poles in this example.)

actually tells us about this:

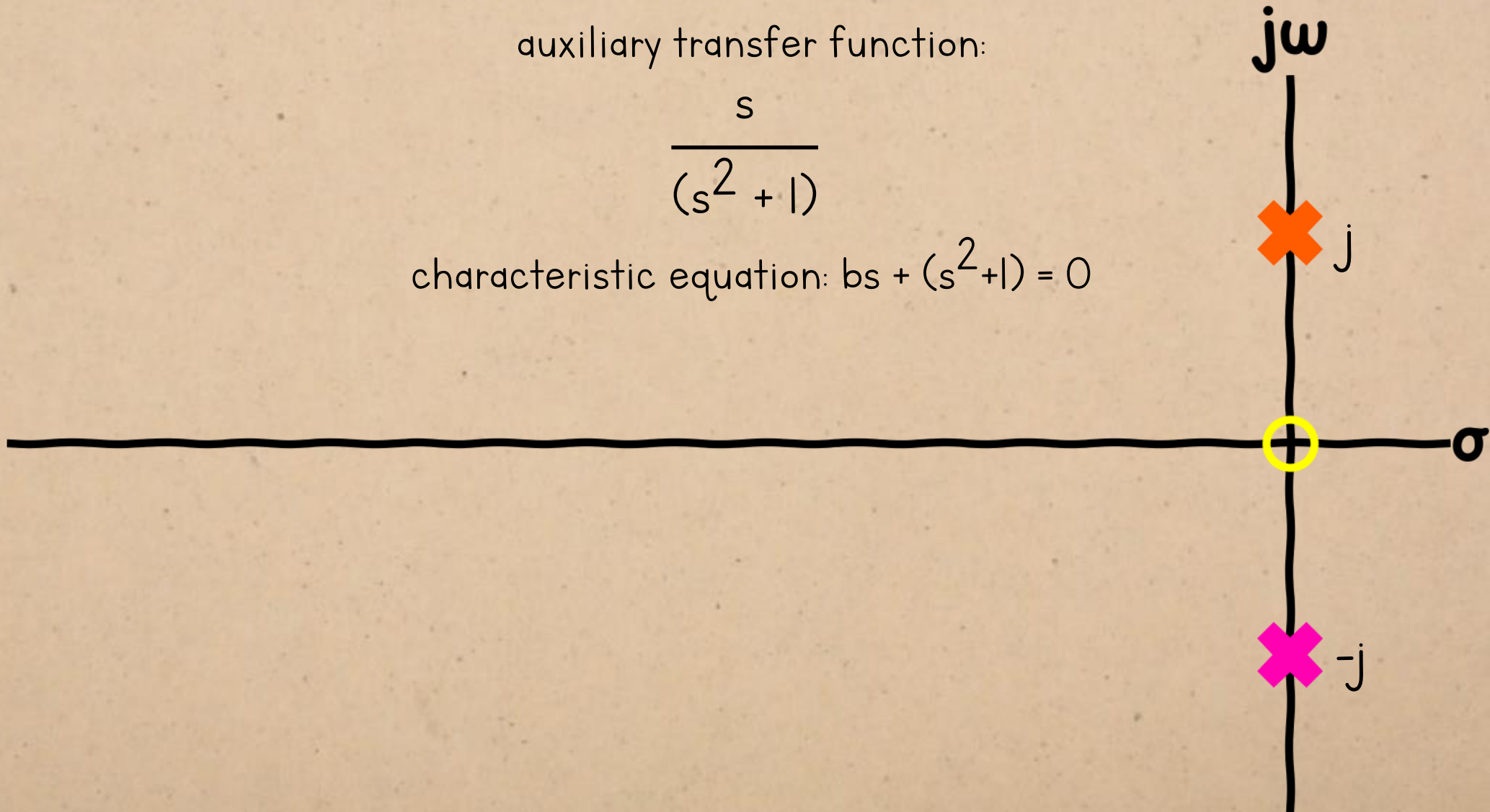


EXAMPLE 2

auxiliary transfer function:

$$\frac{s}{(s^2 + 1)}$$

characteristic equation: $bs + (s^2 + 1) = 0$



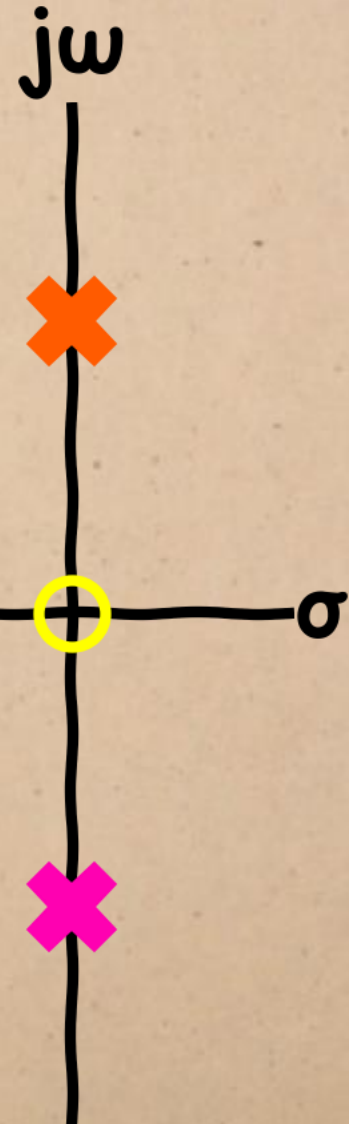
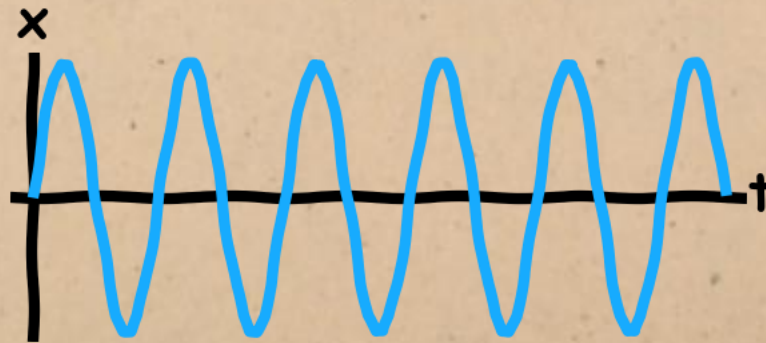
EXAMPLE 2

auxiliary transfer function:

$$\frac{s}{(s^2 + 1)}$$

characteristic equation: $bs + (s^2 + 1) = 0$

It makes sense that the undamped poles are on the imaginary axis, since the undamped system response is a sinusoid.



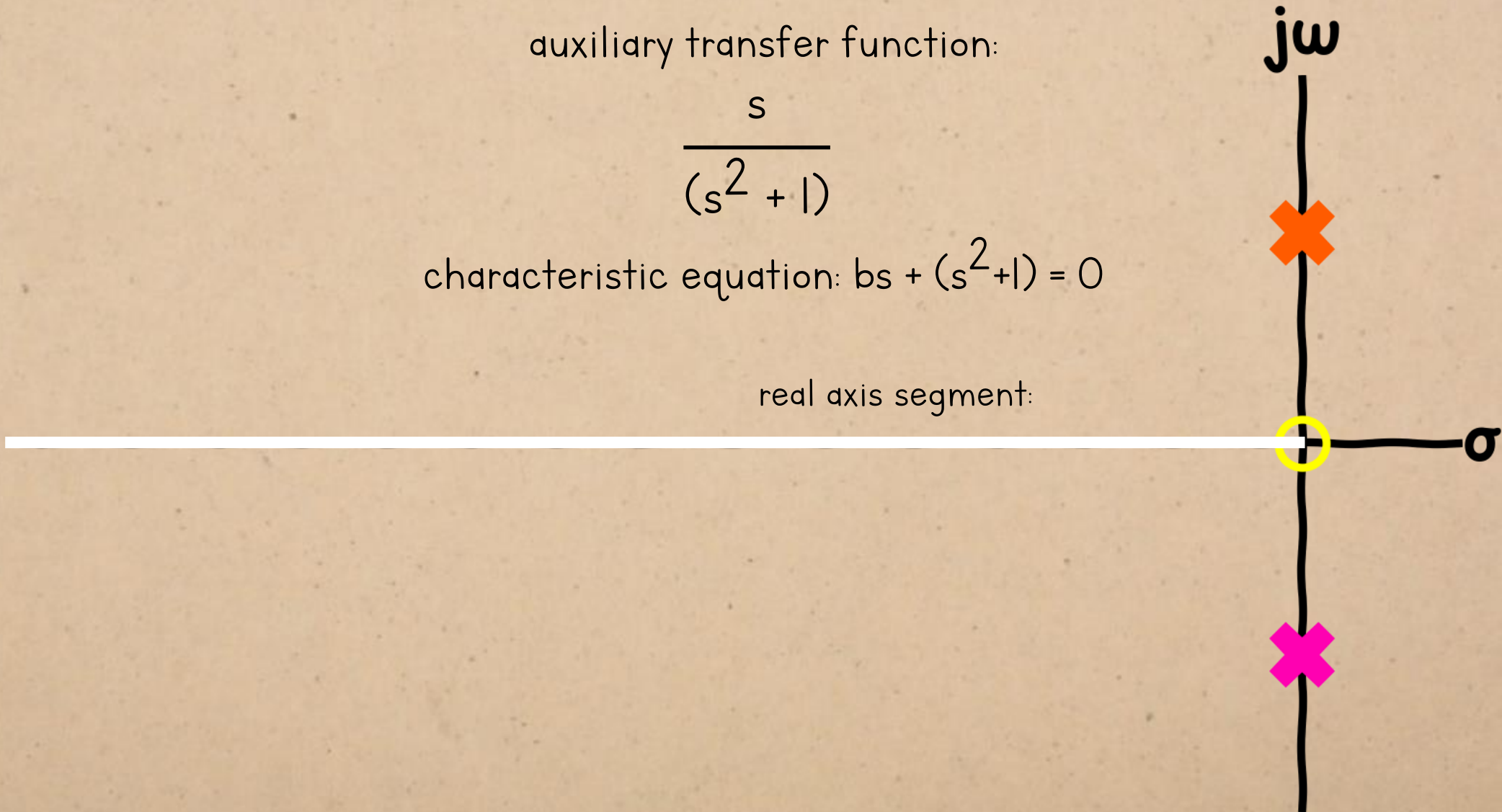
EXAMPLE 2

auxiliary transfer function:

$$\frac{s}{(s^2 + 1)}$$

characteristic equation: $bs + (s^2 + 1) = 0$

real axis segment:



EXAMPLE 2

auxiliary transfer function:

$$\frac{s}{(s^2 + 1)}$$

characteristic equation: $bs + (s^2 + 1) = 0$

asymptote:



$j\omega$



σ

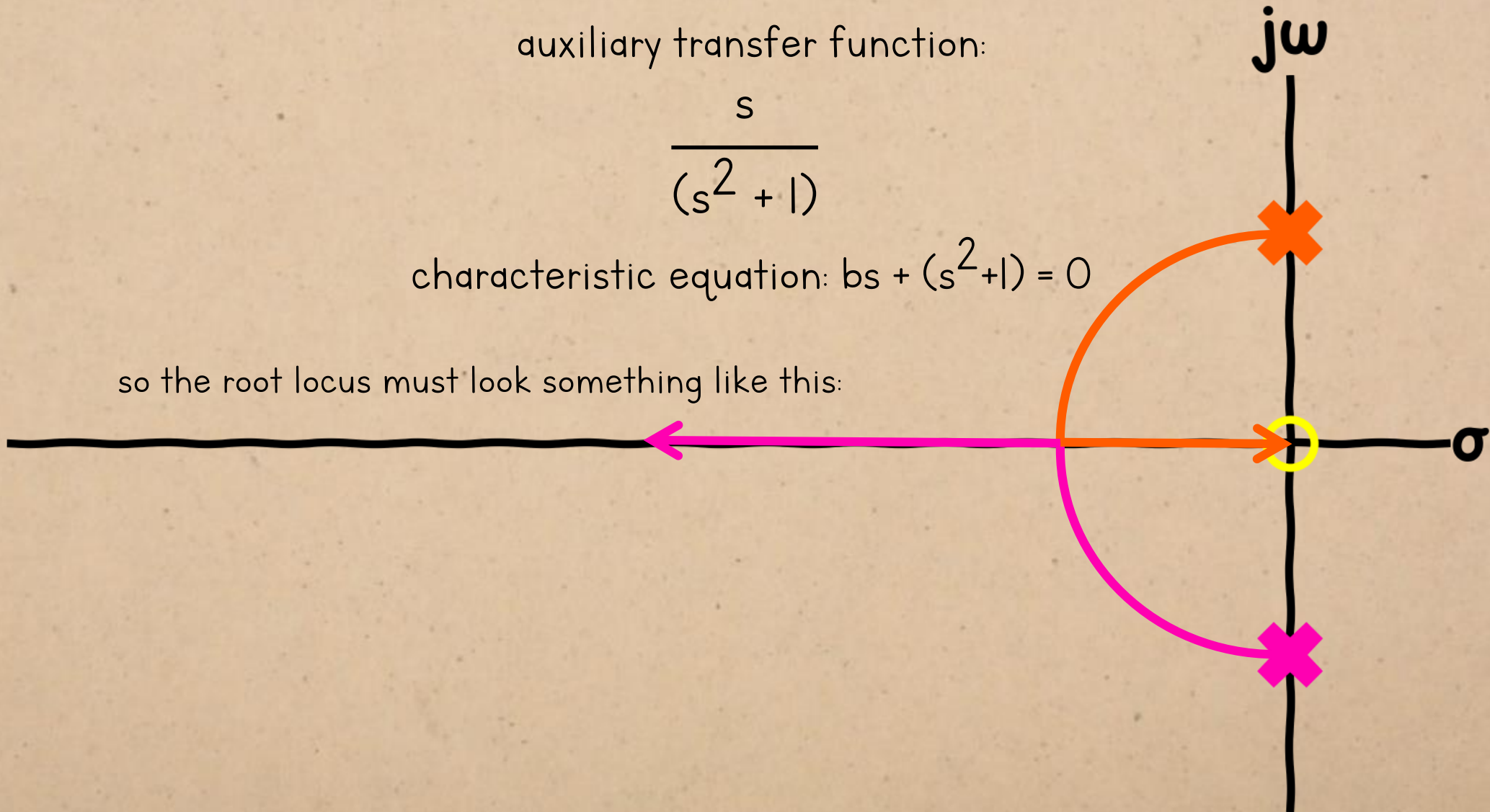
EXAMPLE 2

auxiliary transfer function:

$$\frac{s}{(s^2 + 1)}$$

characteristic equation: $bs + (s^2 + 1) = 0$

so the root locus must look something like this:

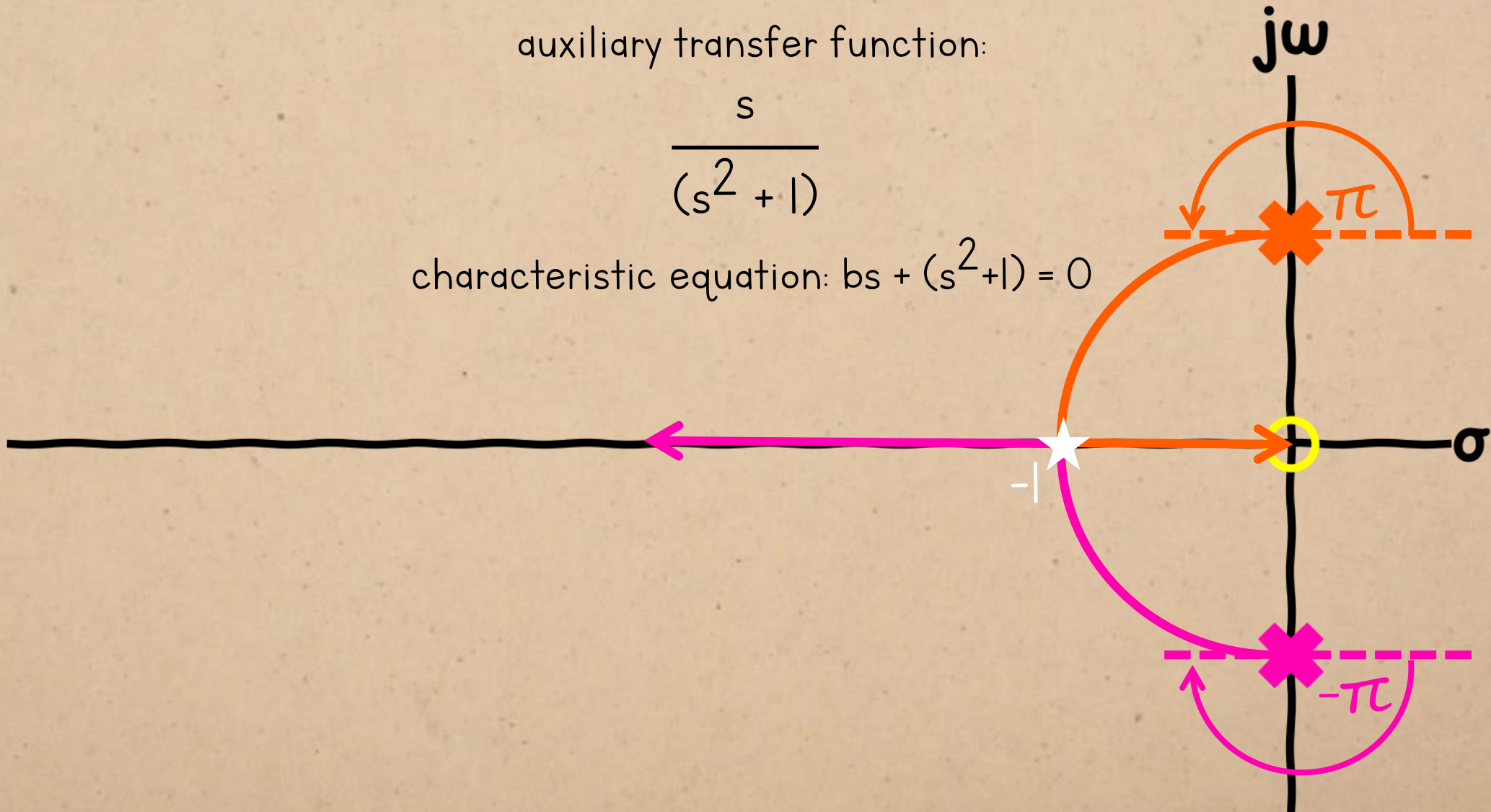


EXAMPLE 2

auxiliary transfer function:

$$\frac{s}{(s^2 + 1)}$$

characteristic equation: $bs + (s^2 + 1) = 0$



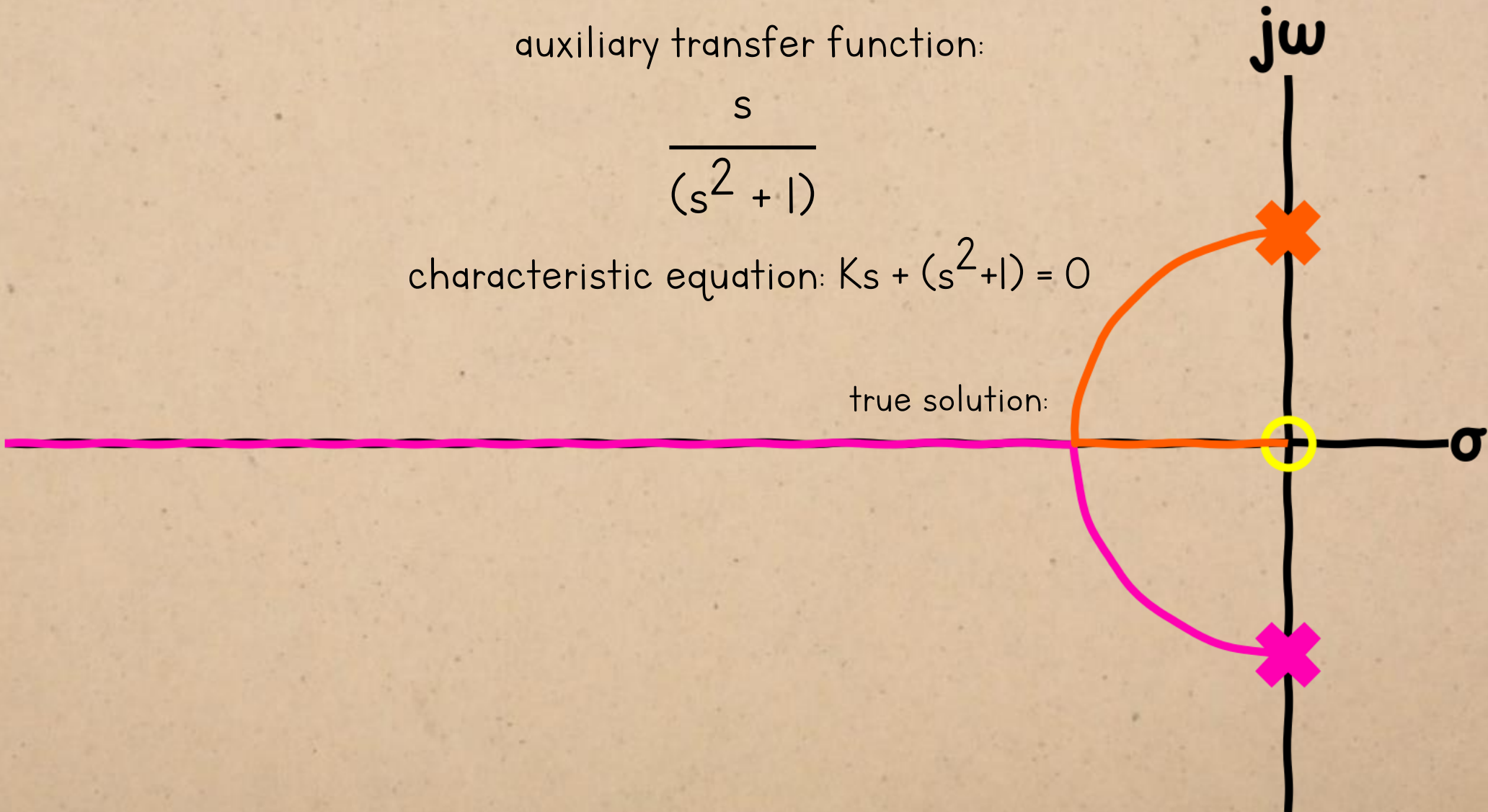
EXAMPLE 2

auxiliary transfer function:

$$\frac{s}{(s^2 + 1)}$$

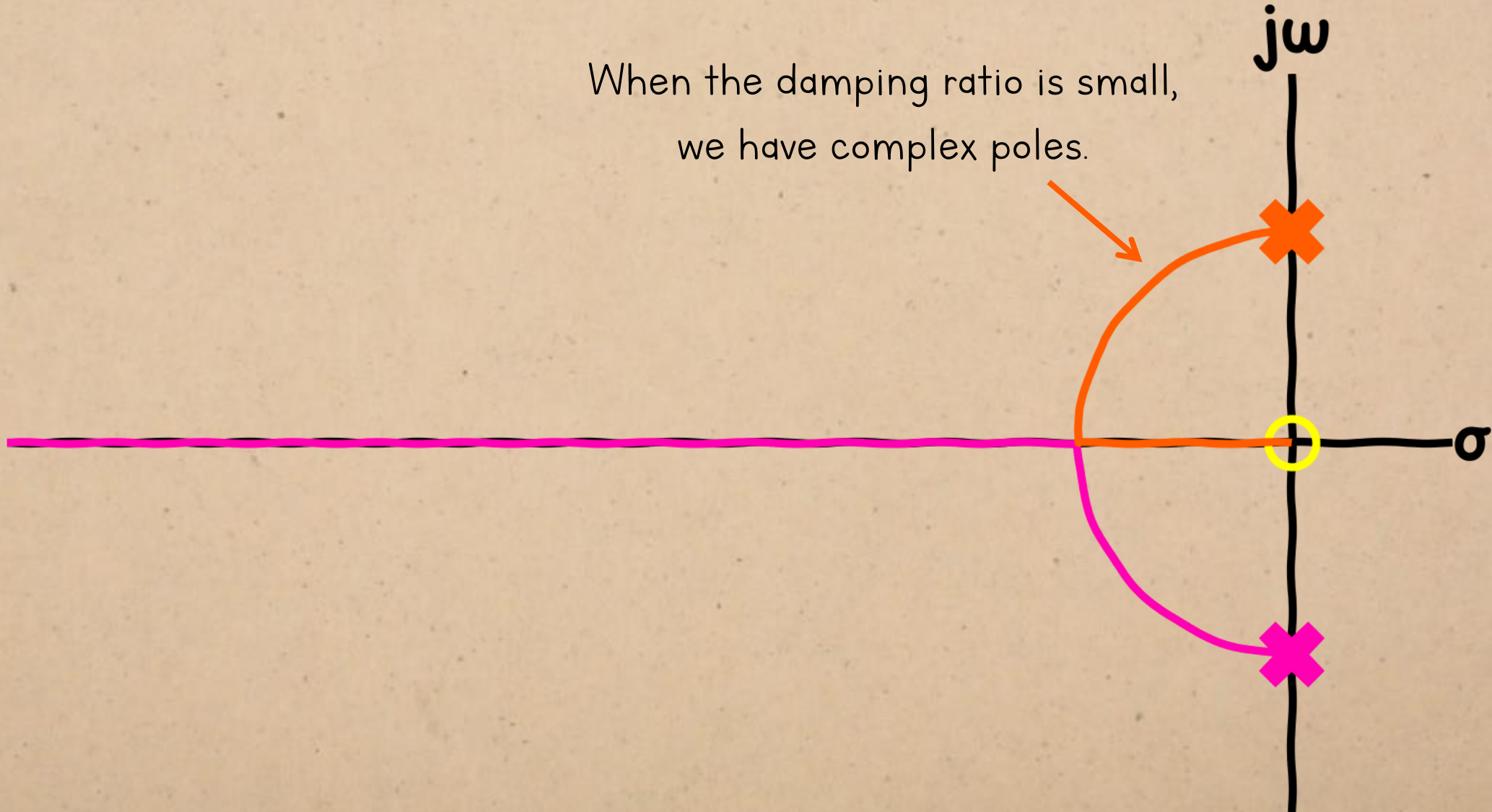
characteristic equation: $Ks + (s^2 + 1) = 0$

true solution:

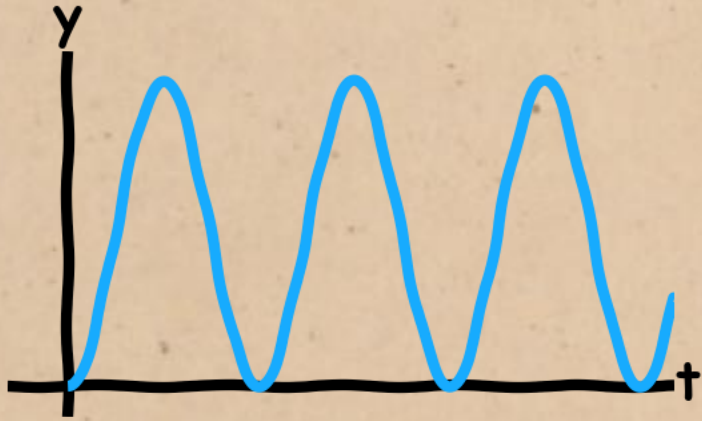


INTERPRETATION

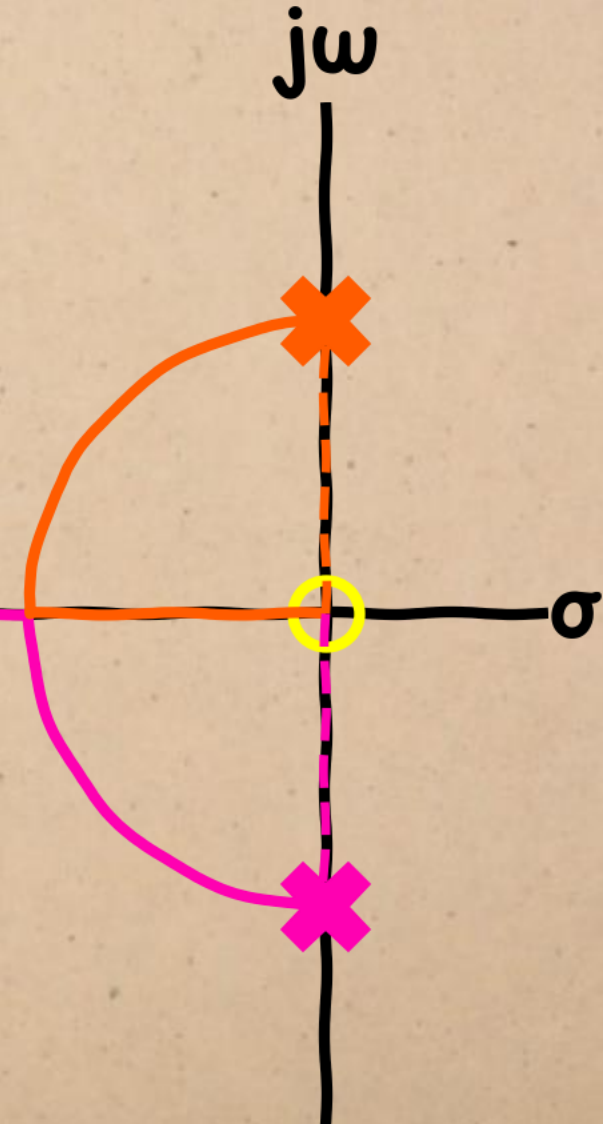
When the damping ratio is small,
we have complex poles.



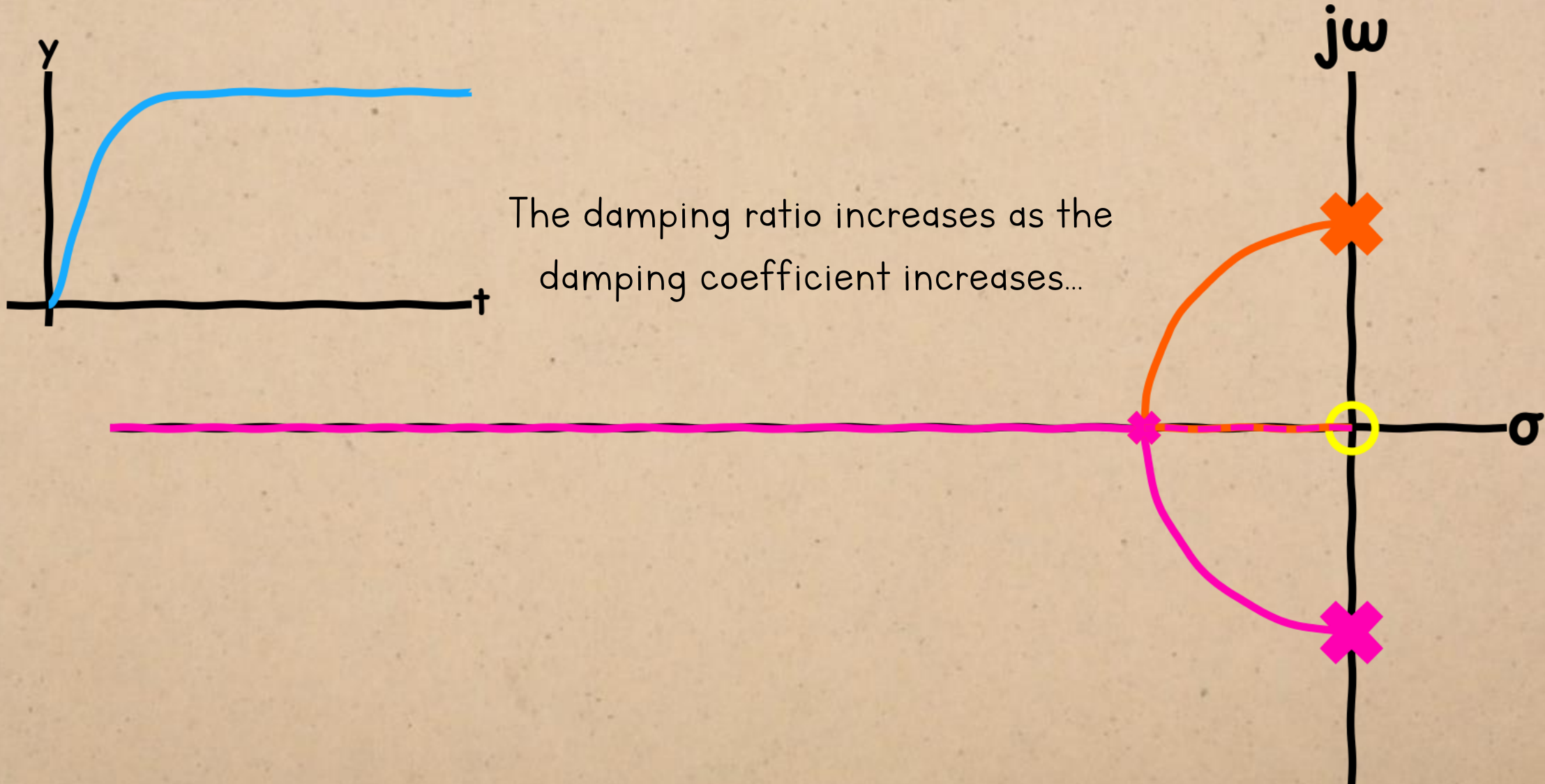
INTERPRETATION



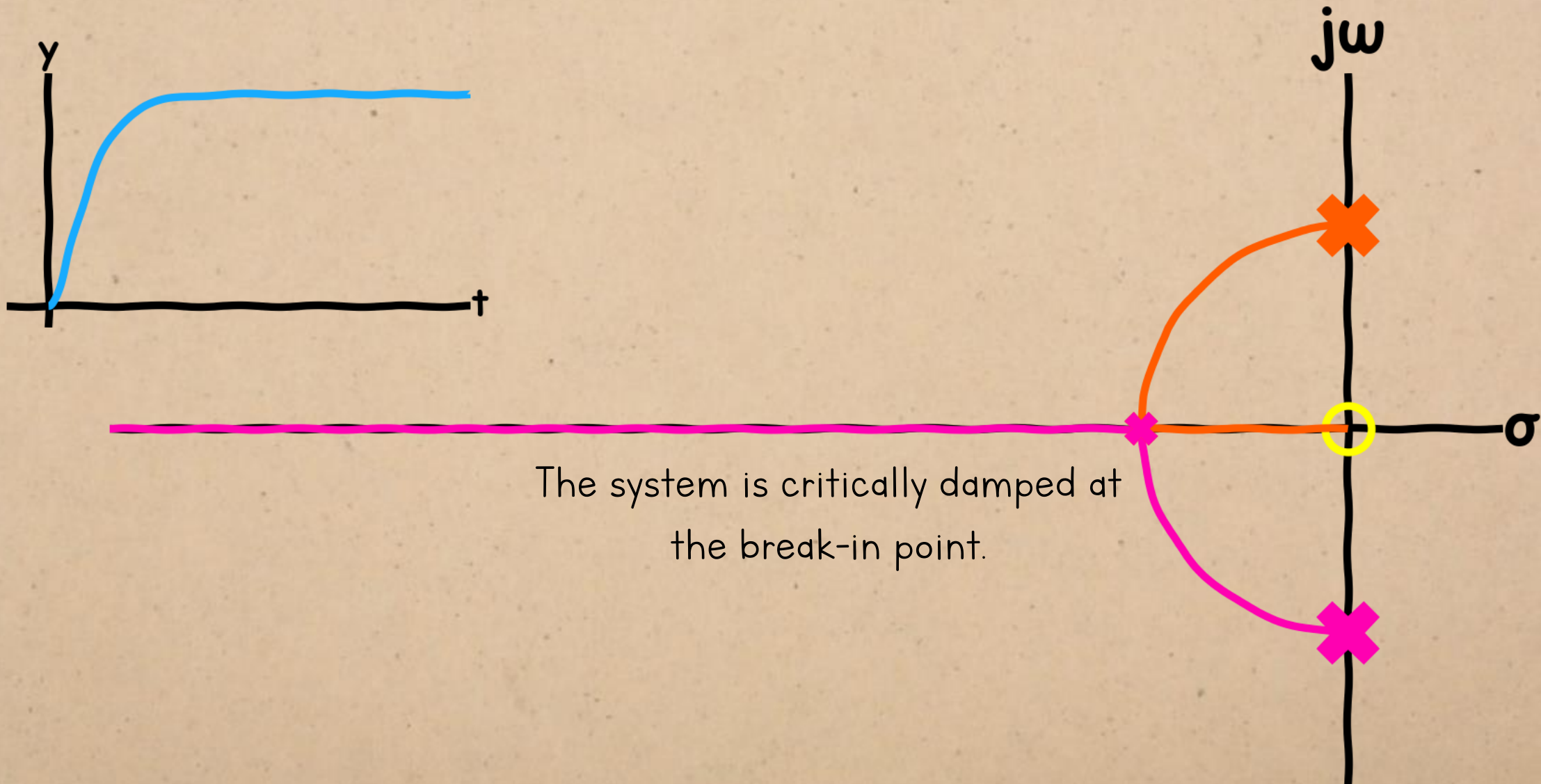
The damping ratio increases as the damping coefficient increases...



INTERPRETATION

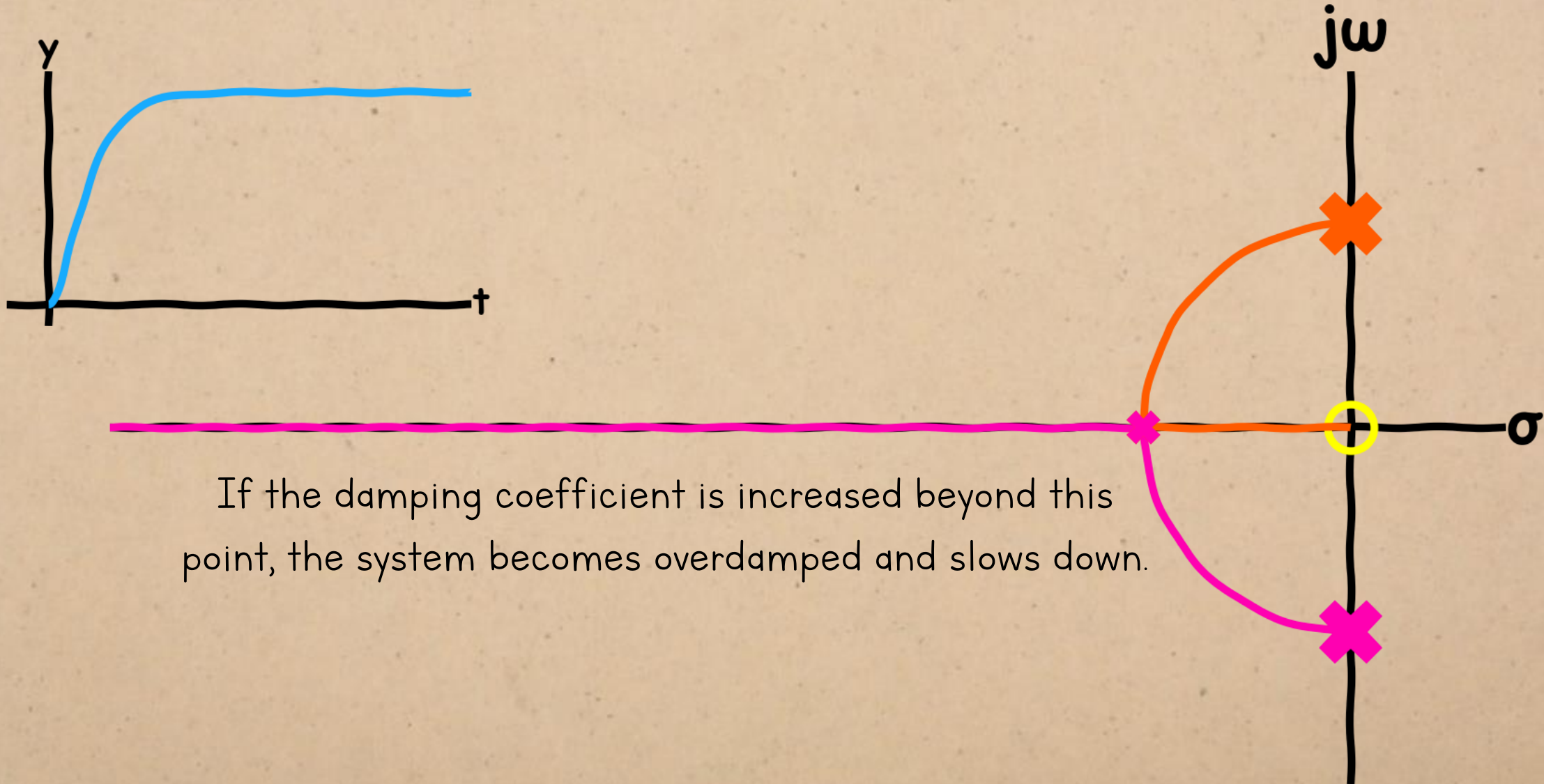


INTERPRETATION

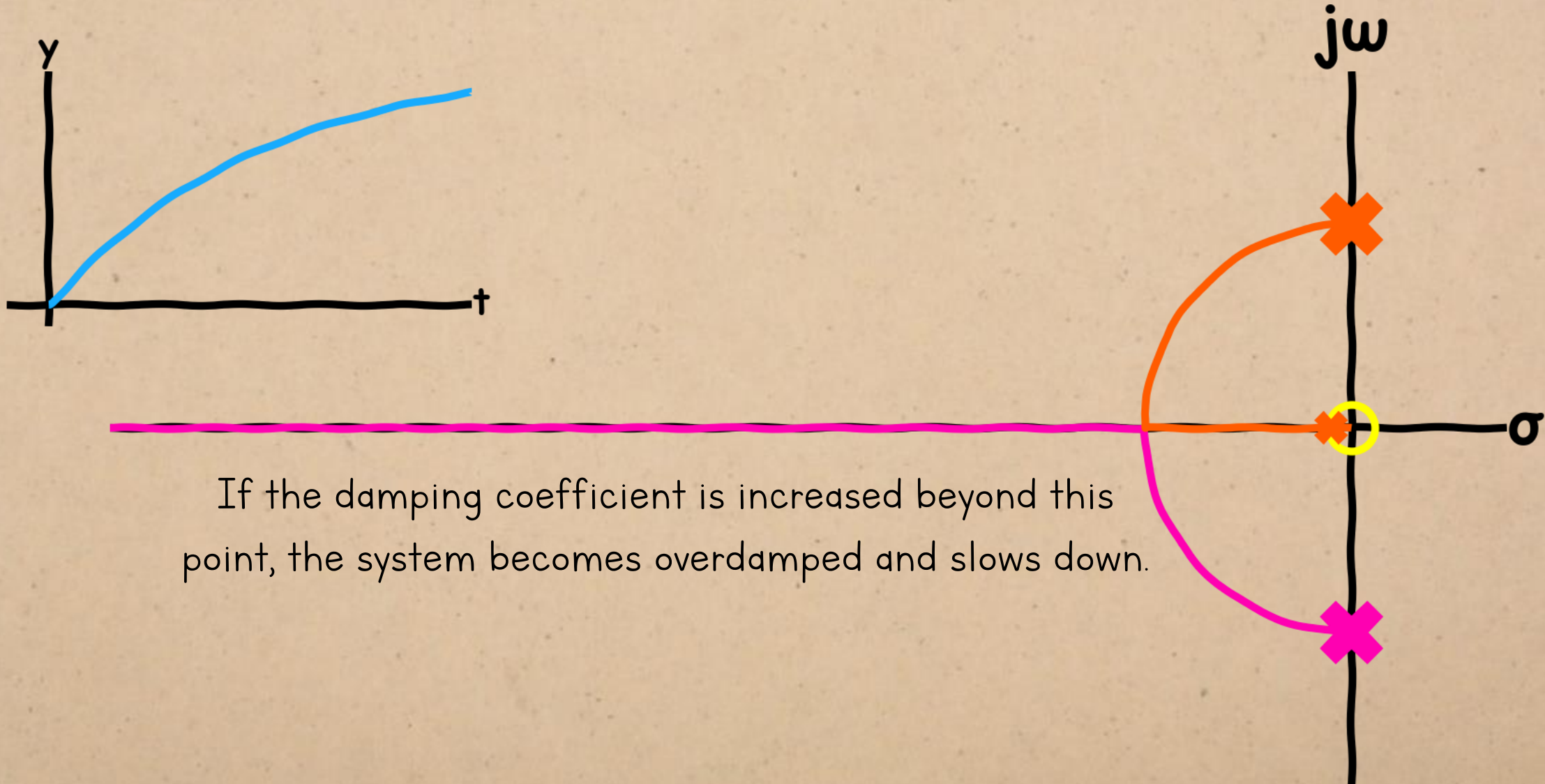


The system is critically damped at the break-in point.

INTERPRETATION



INTERPRETATION



If the damping coefficient is increased beyond this point, the system becomes overdamped and slows down.

Previously...

SENSITIVITY

The root locus can also give us information about the robustness of the system.

the sensitivity of **a**
with respect to **b**

$$S_b^a = \frac{\frac{\partial a}{\partial b}}{\frac{a}{b}}$$

proportional
change in **a**

proportional
change in **b**

Previously... **SENSITIVITY**

Often, we look at sensitivity in terms of the closed-loop system's sensitivity to the open-loop system as a whole:

sensitivity function:

$$\text{if } S = \frac{1}{1 + Q(s)} > 1$$

the system is **sensitive**

(changes to the open loop will be magnified in closed loop)

...But it can also be helpful to know the effect of specific parameters on specific poles.

ROOT SENSITIVITY

= the sensitivity of a pole's position (p) to changes in some parameter (λ)

$$S_{\lambda}^p = \frac{\frac{\partial p}{\partial \lambda}}{\frac{p}{\lambda}}$$

ROOT SENSITIVITY

= the sensitivity of a pole's position (p) to changes in some parameter (λ)

We're often interested in the root sensitivity with respect to the gain...

$$S_K^p = \frac{\frac{\partial p}{\partial K}}{\frac{p}{K}}$$

ROOT SENSITIVITY

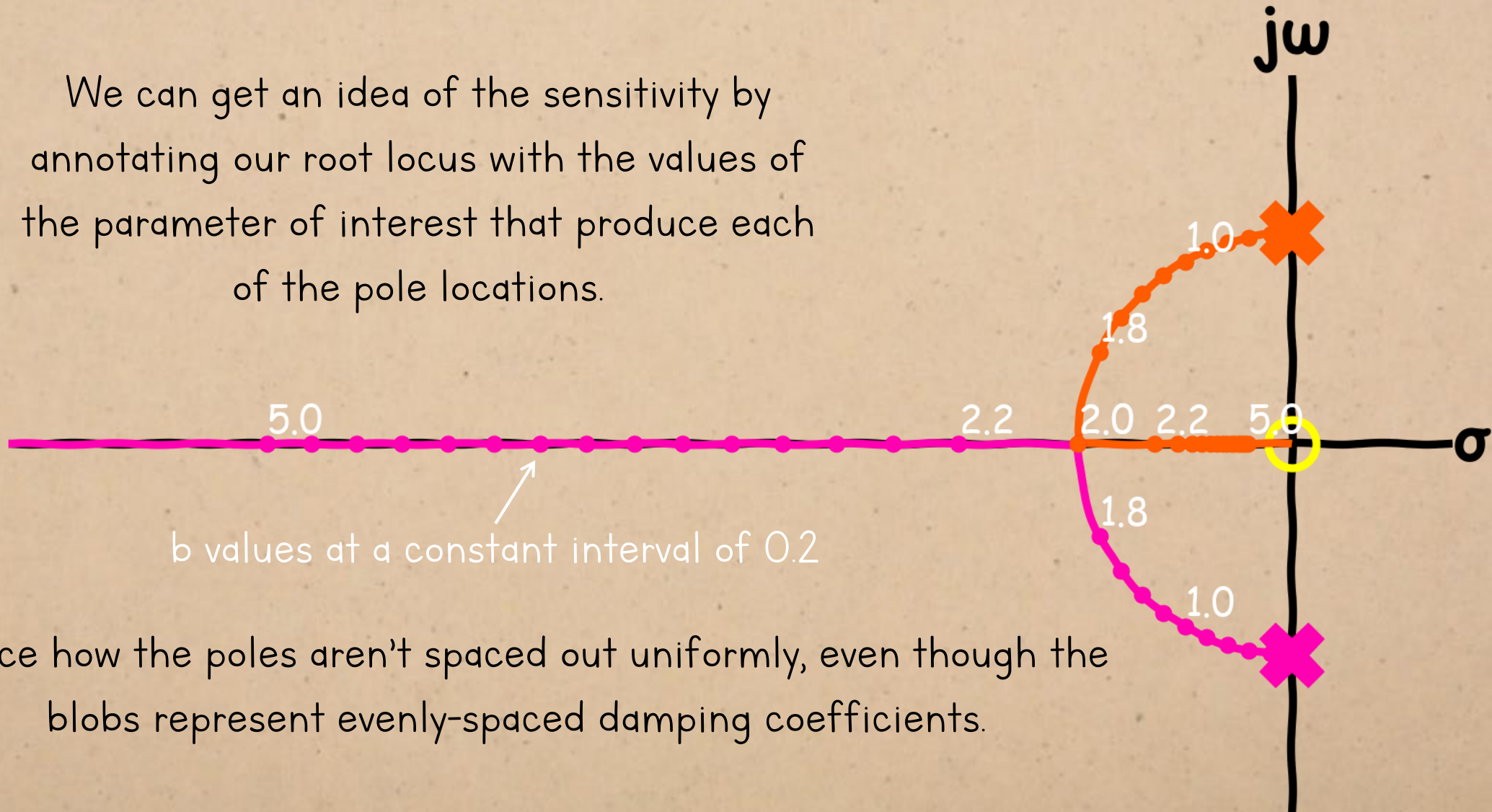
= the sensitivity of a pole's position (p) to changes in some parameter (λ)

...but we can work it out
for any parameter
(like the damping coefficient
in this example)

$$S_b^p = \frac{\frac{\partial p}{\partial b}}{\frac{p}{b}}$$

ROOT SENSITIVITY

We can get an idea of the sensitivity by annotating our root locus with the values of the parameter of interest that produce each of the pole locations.

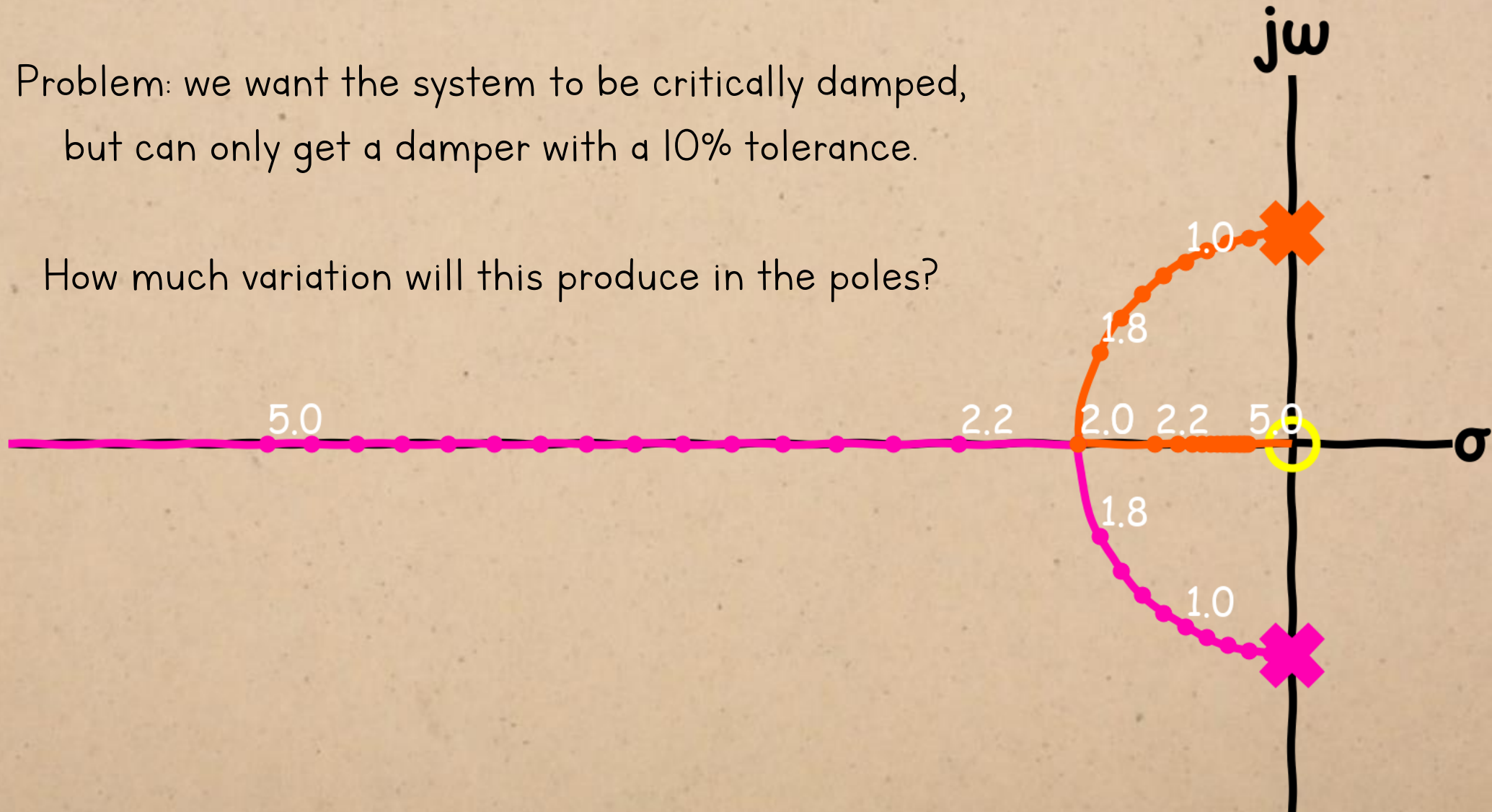


Notice how the poles aren't spaced out uniformly, even though the blobs represent evenly-spaced damping coefficients.

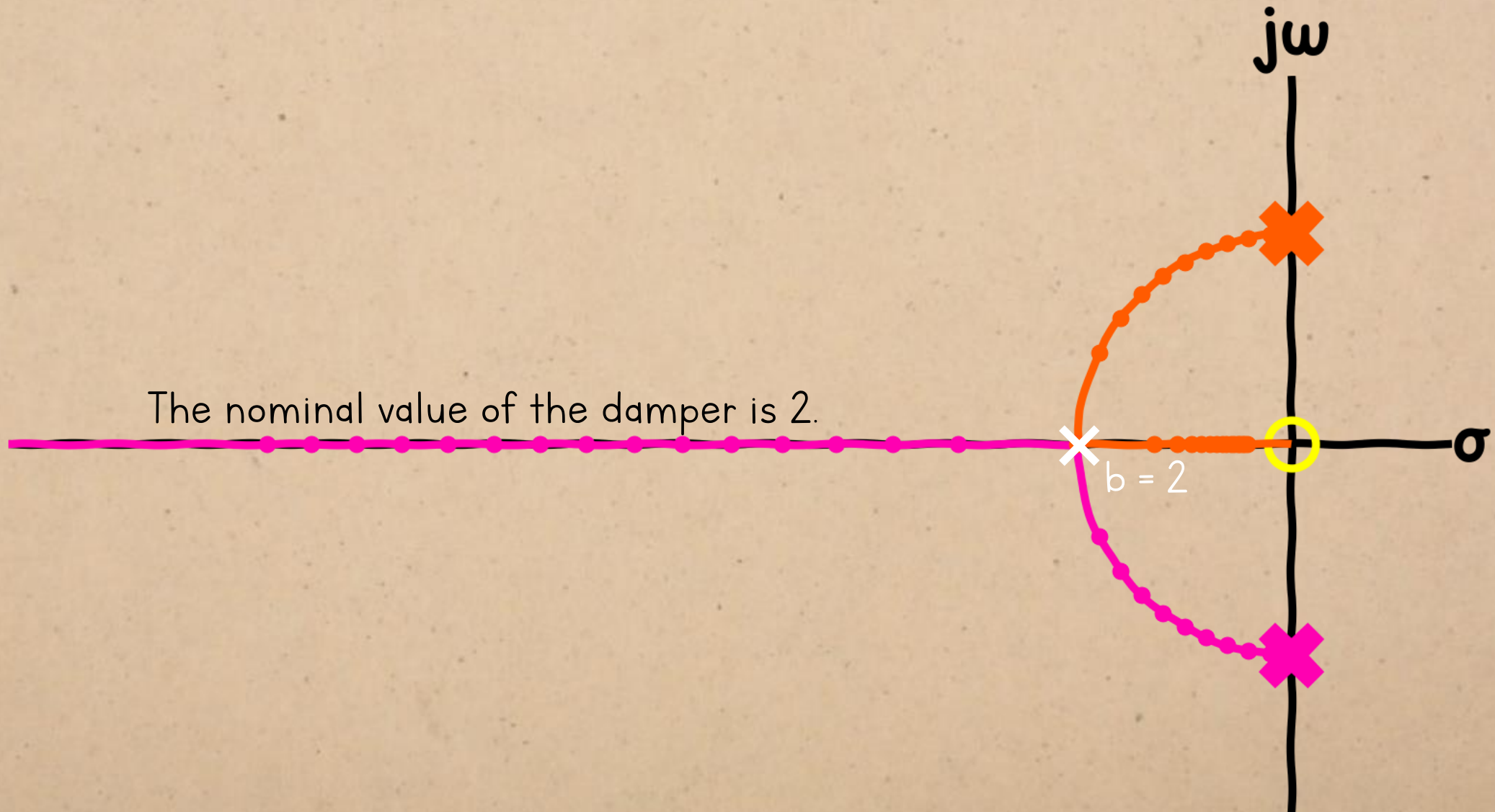
ROOT SENSITIVITY

Problem: we want the system to be critically damped,
but can only get a damper with a 10% tolerance.

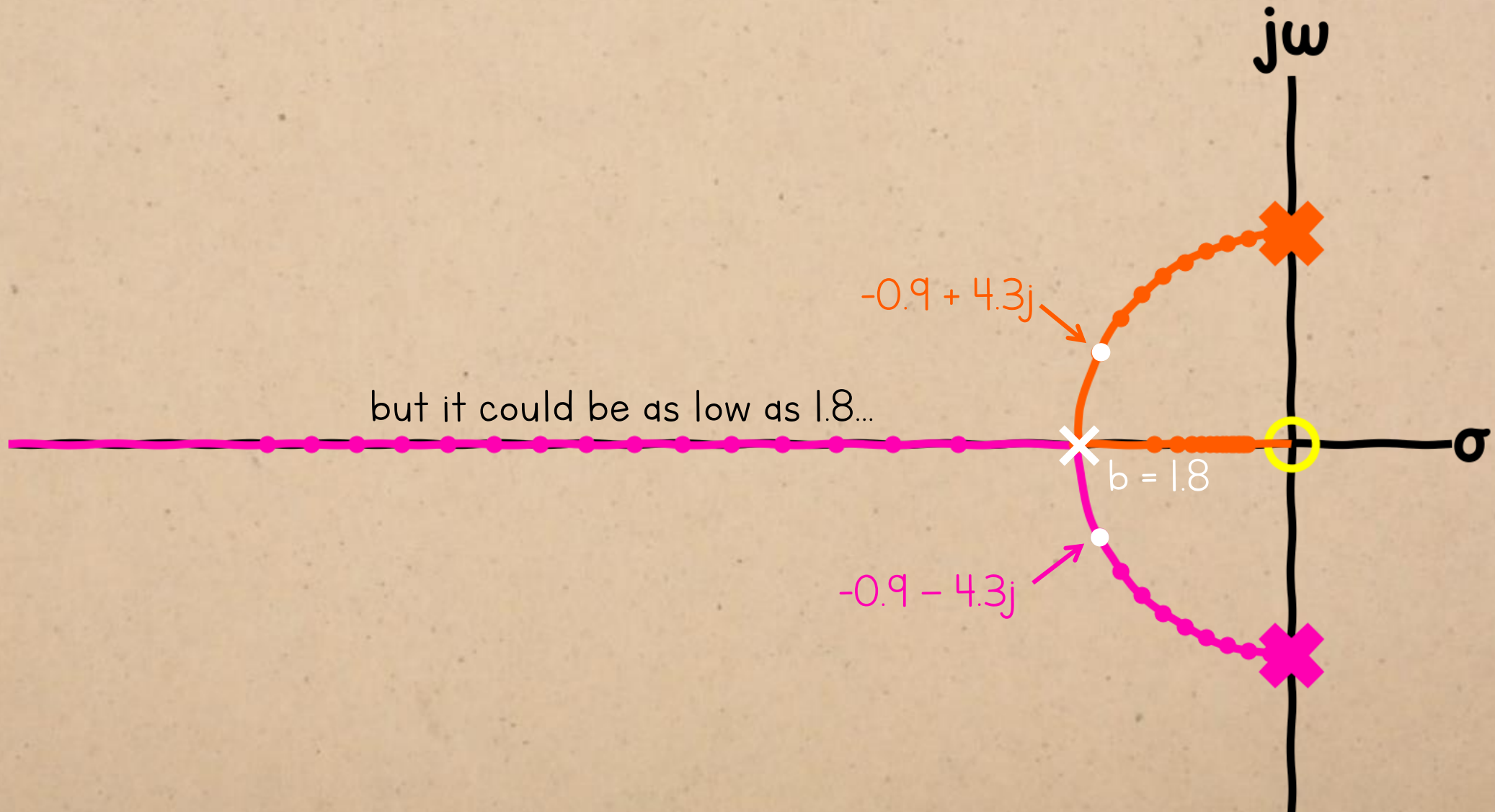
How much variation will this produce in the poles?



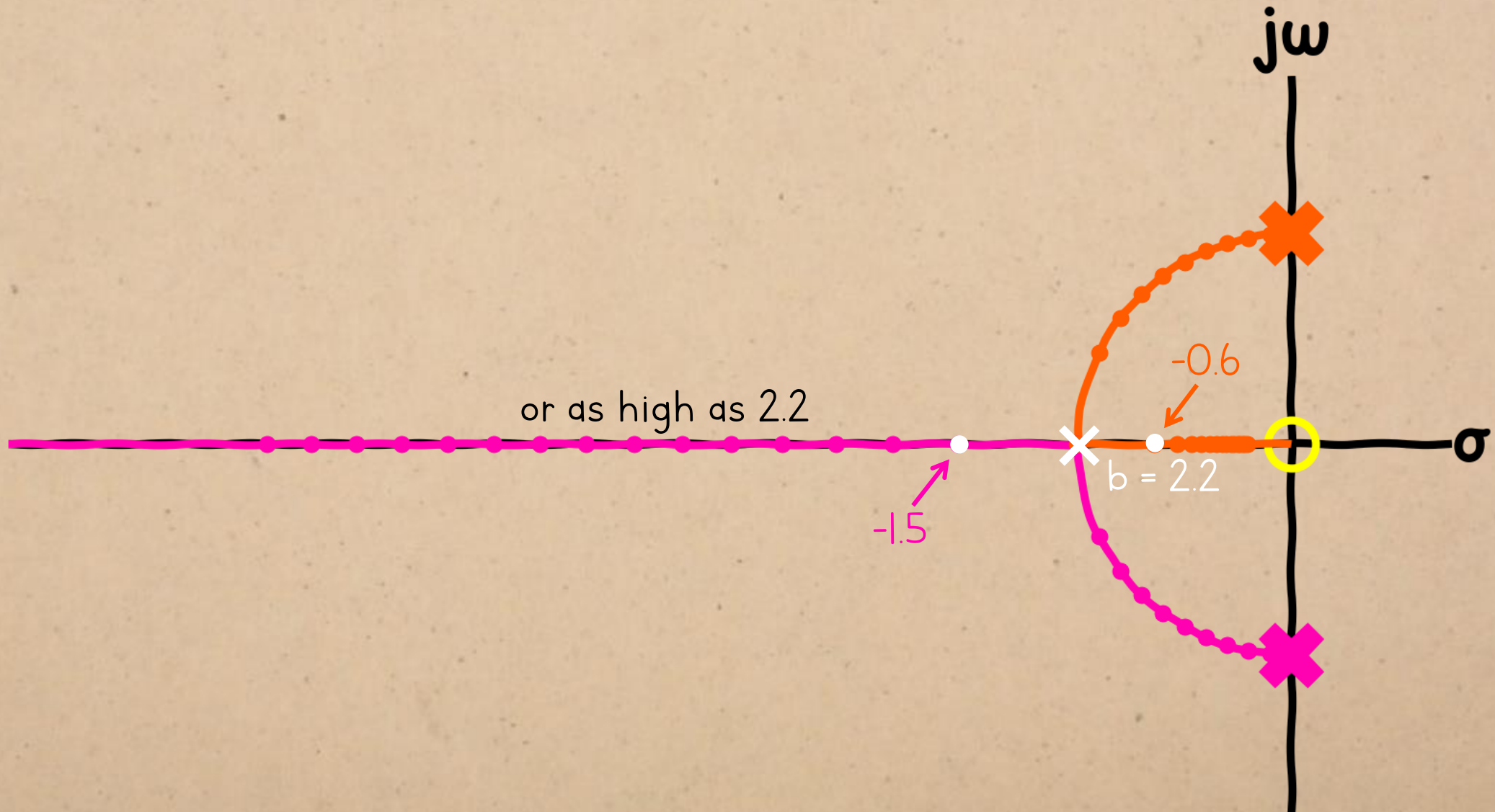
ROOT SENSITIVITY



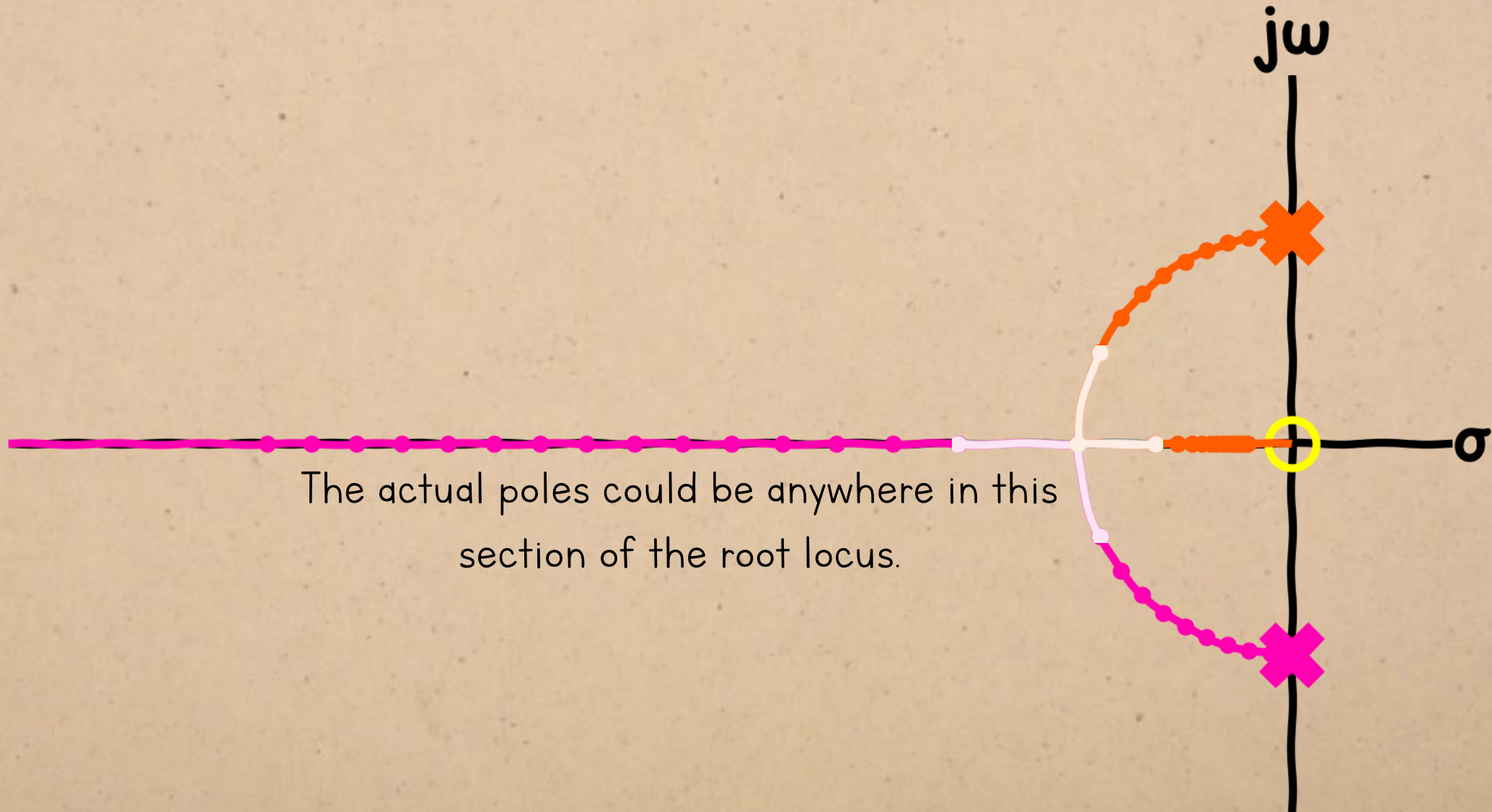
ROOT SENSITIVITY



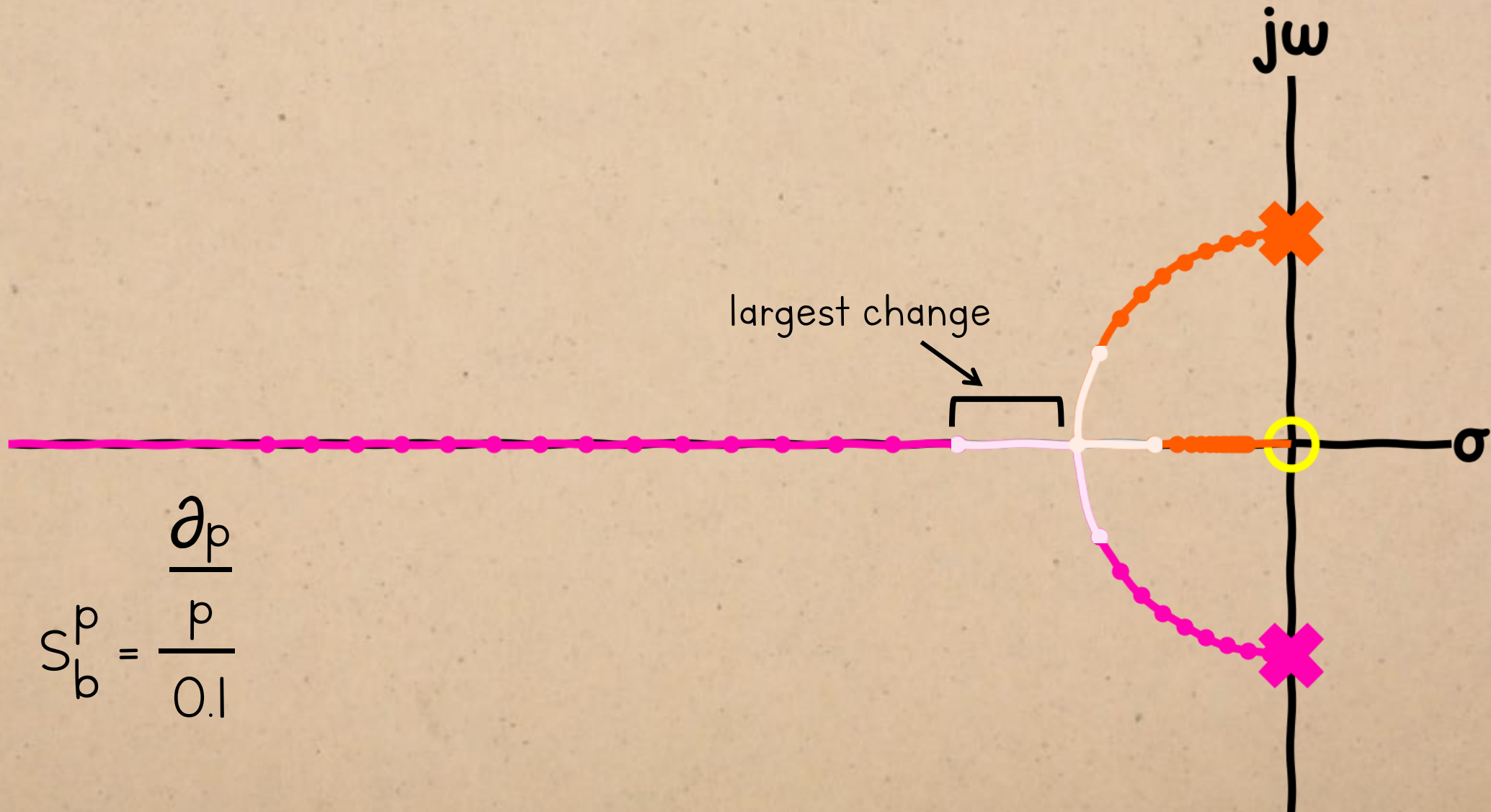
ROOT SENSITIVITY



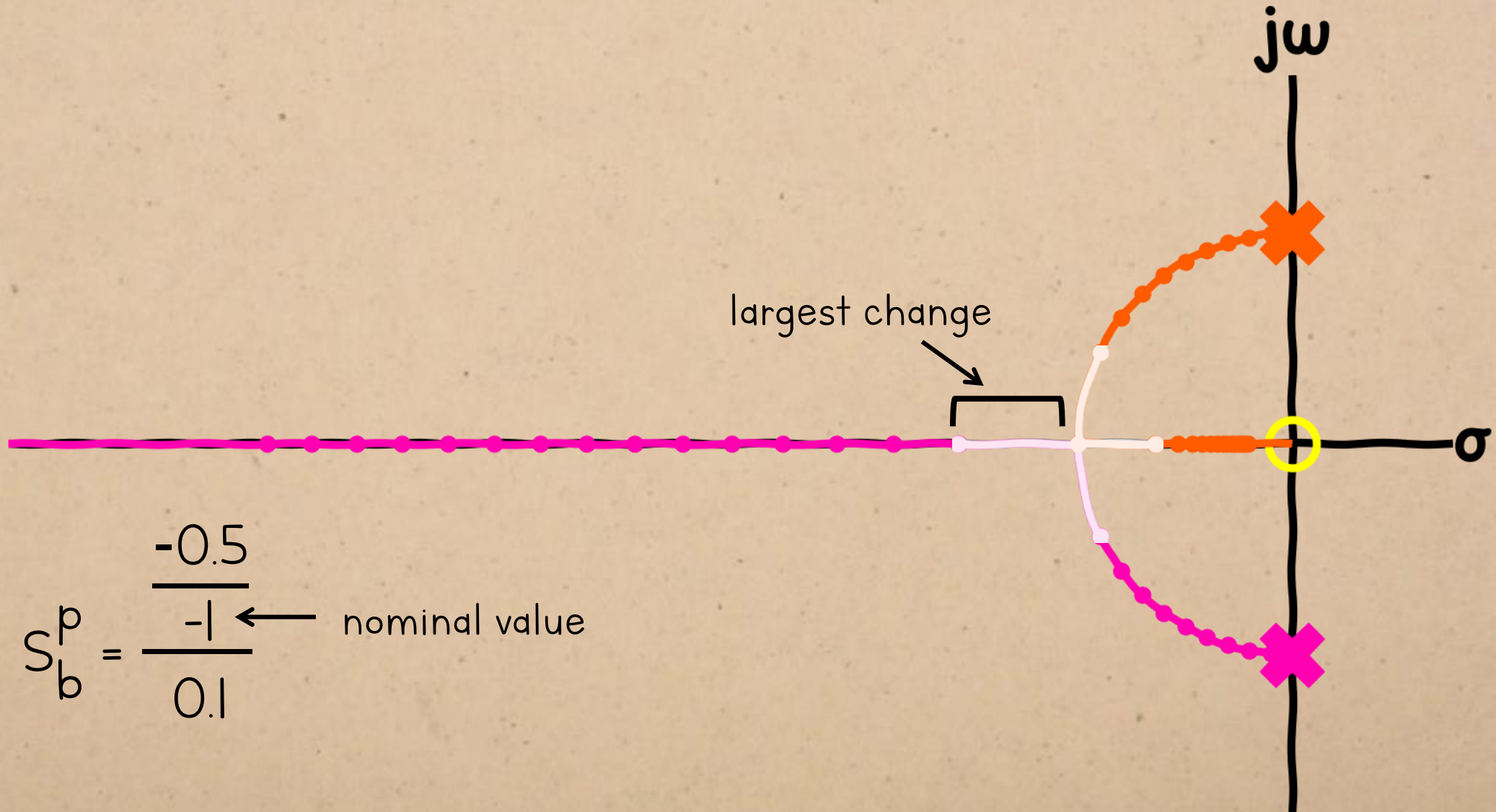
ROOT SENSITIVITY



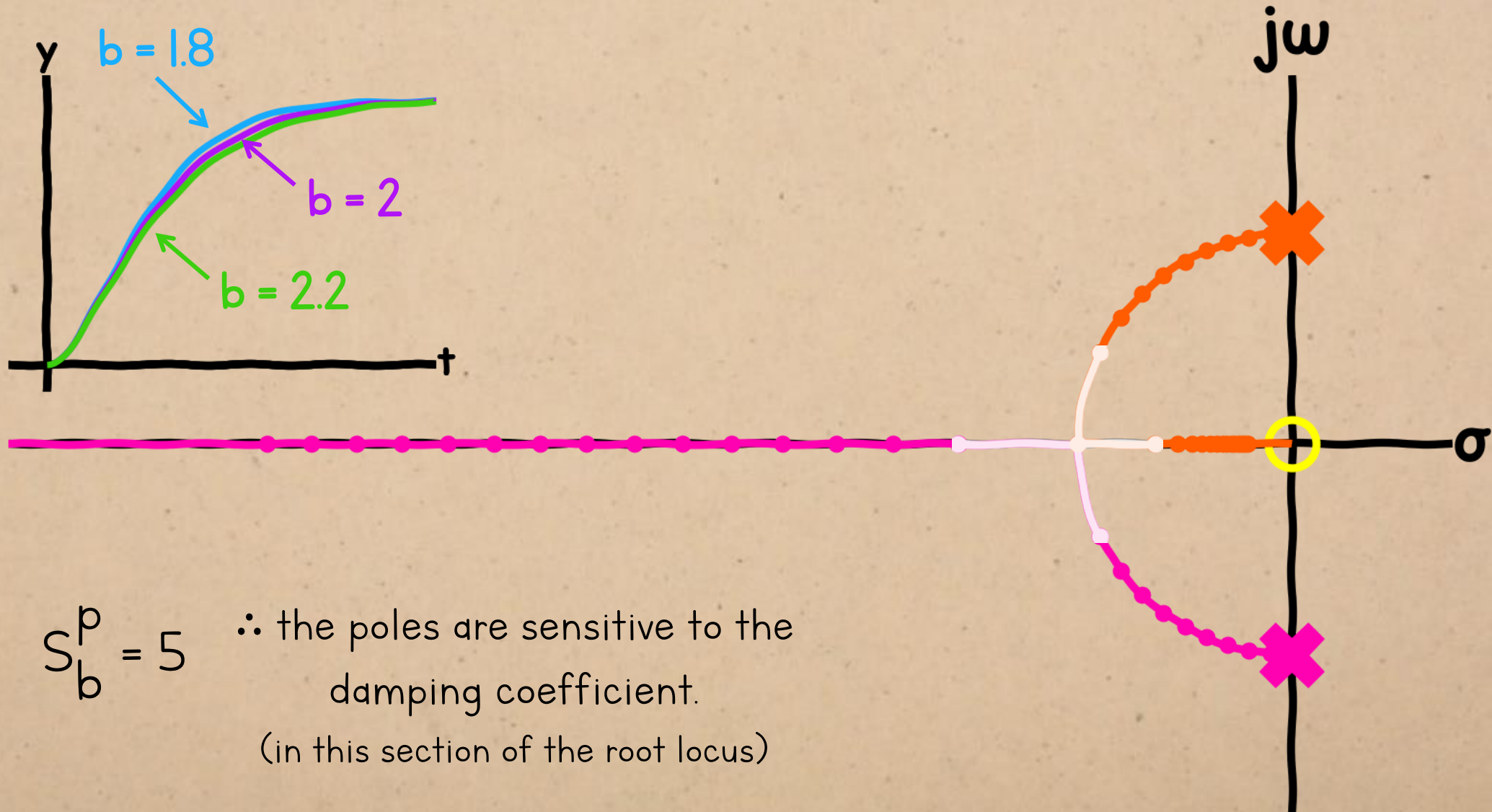
ROOT SENSITIVITY



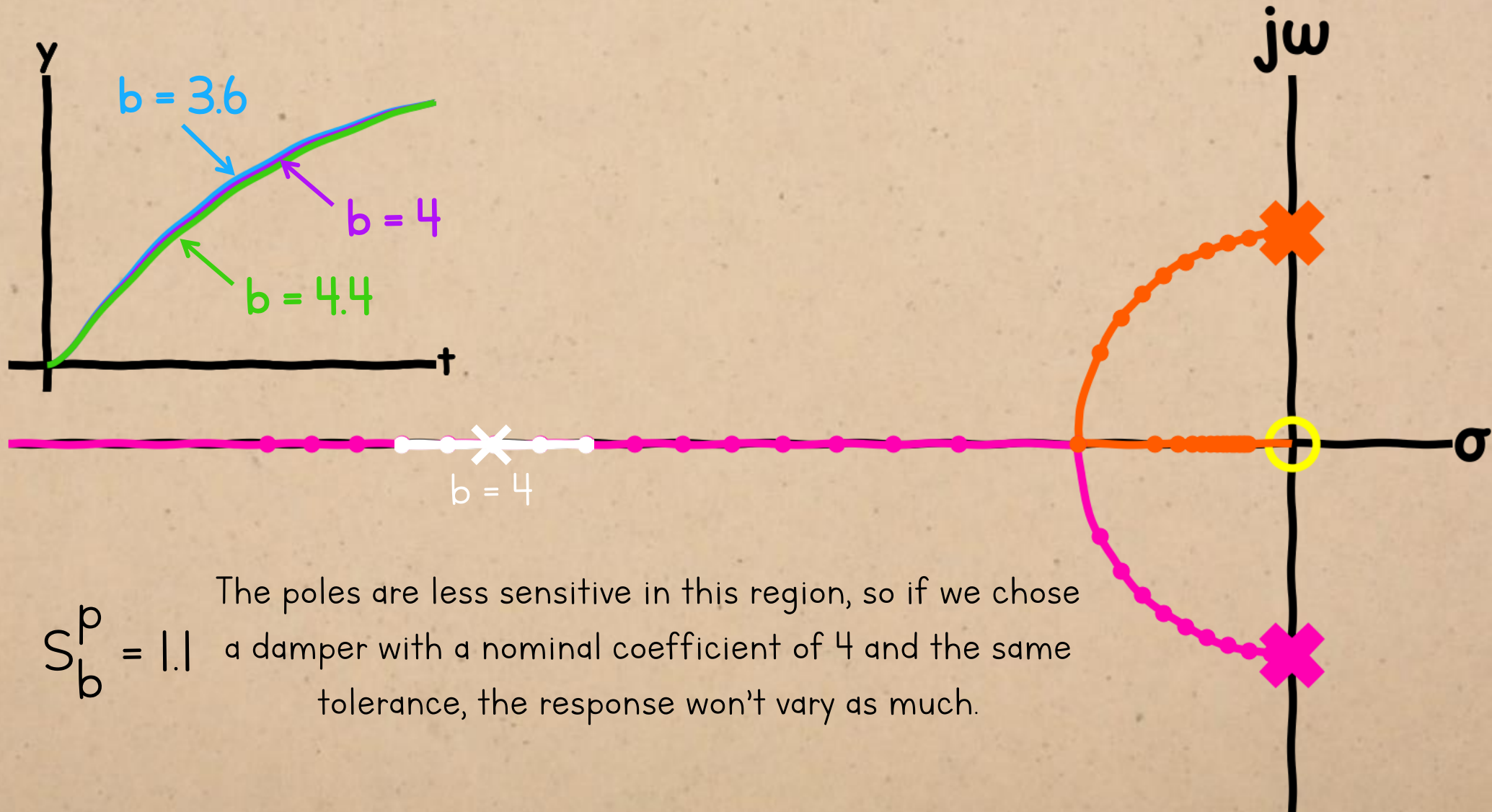
ROOT SENSITIVITY



ROOT SENSITIVITY



ROOT SENSITIVITY



$S_b^p = 1.1$
 The poles are less sensitive in this region, so if we chose a damper with a nominal coefficient of 4 and the same tolerance, the response won't vary as much.

SUMMARY

The Root Locus Plot

What does it show? The positions of the closed-loop poles for all values of the gain (or some other parameter of interest)

What can we do with it?

- Evaluate the stability margins for different parameters
- See how parameter variation affects the system's performance
- Evaluate the sensitivity of the system to a specific parameter over different ranges
- Design compensators (we'll get to this later)

